Challenger Deep

Ghost Ships and the Rubber Duck Armada

Derelict vessels, ocean drift science, and what the sea gives back

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The Ghost Ship Phenomenon

Famous derelicts and the stories attached to them

A derelict becomes famous when a story attaches to it — and the stories that last are the ones with paperwork. This paper's two best-documented ghost ships make the point from opposite ends of the earth. One is remembered because of what was in her hold; the other, because of the apparatus built to watch for her.

The first is SS Baychimo, the Hudson's Bay Company cargo steamer abandoned in the Beaufort Sea ice in 1931 and, in the words of the museum paper that reconstructed her afterlife, "the legendary abandoned vessel that floated in the Beaufort Sea icepack for over 30 years," making "occasional appearances for over 30 years in the Alaskan Chukchi and Beaufort Sea region" [1]. Her fame is usually told as a sighting story, but her documentary trail runs through a cargo: in the summer of 1930 the Canadian Department of the Interior commissioned filmmaker Richard Finnie to document Copper Inuit culture, and the ethnological collection he assembled — intended for the Canadian National Museum in Ottawa — was loaded aboard Baychimo at Fort Hearne on 5 September 1931, "eight crates in all," for shipment to Vancouver on her supply run [1]. The ship never delivered them. Trapped in ice in Peard Bay, she was abandoned in mid-October 1931, most of her people evacuated by air to Nome; after a late November storm she vanished, and the shore party assumed she had sunk [1]. She had not. In early December 1931 Iñupiat hunters spotted her near Skull Cliff, south of Barrow, and "salvaged several bales of furs from the vessel's hold" [1].

What followed was a thirty-year career as a boarded, plundered, re-sighted wreck — and each boarding left records. On 11 August 1933 the crew of MS Trader tied up to her off Wainwright, tried and failed to break her free of the ice for salvage, and "nevertheless recovered several items from Baychimo" [1]. Among Trader's people were owner Ira Rank and brothers Peter and Kari Palsson; the British traveller Isobel Wylie Hutchison, also aboard, acquired Copper Inuit parkas "from a sack of curios" on the derelict and later donated them to the Cambridge University Museum [1]. Finnie himself recognized his lost collection in print: a letter he identified as his own reached him via trader Charles Brower, handed over by "one of the local Eskimos who had been plundering the ship," and in Hutchison's book he spotted "some Copper Eskimo costumes she had acquired for the Cambridge University Museum. I knew them well, these costumes, for they were part of the ethnological collection I had entrusted to the Baychimo" [1]. Fourteen of the artifacts — a musk-ox horn blubber pounder, a copper ulu, a copper man's knife, antler-handled scissors, a skin scraper, fish rakes, bone wound pins, a leister prong [1] [1] — reached the University of Alaska Museum in

1934, accessioned as Acc. 514 under a donor recorded as “Peter Balssan,” almost certainly Trader’s Peter Palsson with his name misspelled in a ledger written years later [1]. The boxes, labelled “From Canadian Arctic taken from the ‘Beychimo,’” then sat unrecognized for 81 years until researchers rediscovered them in October 2015 [1]. Baychimo’s ghost story, in other words, is verifiable precisely where it touches paper: a commission, a manifest of crates, an accession ledger, a misspelled name.

The second famous derelict arrived with a coast guard photograph instead of a legend. The Ryou-Un Maru, “the 50-metre ‘ghost ship,’” was discovered in Canadian waters about 200 km off Haida Gwaii on 20 March 2012, a year after the Tōhoku tsunami tore it from its moorings in Japan [2]. Its story is inseparable from the tracking effort around it: the debris field it came from was visible to the NESDIS environmental satellite for about a month before dispersing beyond detection by 14 April 2011, and a December 2011 NOAA search of high-resolution imagery over a 50 × 60 km patch where models predicted debris found nothing [2]. Japan’s government catalogued 97 sightings of debris at sea from April to November 2011, most of them “larger objects such as small vessels and containers,” many capsized or partially submerged [2]. By 18 October 2012 NOAA had received 1,385 reports of suspected tsunami debris — 620 shoreline, 765 at sea — yet “only 14 objects have been definitively confirmed to originate from the Tohoku tsunami” [2]. The Russian sailing vessel STS Pallada found one of them, a Fukushima-registered fishing boat, 800 km northwest of Midway Atoll on 22 September 2011, its radiation levels below detection limits [2]. The Ryou-Un Maru was famous not because it was mysterious but because it was the largest confirmed emissary of a debris field an entire hemisphere was watching for — and because, as the review notes, “there has been no monitoring of the movement of even large pieces of debris, so there was no advanced warning” of arrivals like it, or of the concrete dock that beached on Oregon’s Agate Beach on 5 June 2012 [2].

Between these two ships sits the derelict story that created a science. In May 1991, factory-fresh Nike sneakers began washing up on Oregon beaches in such numbers that beachcombers held swap meets to match lefts with rights [3]. Oceanographer Curt Ebbesmeyer traced them through shipping lawyers to court papers: the Hansa Carrier, en route from Korea, had been hit by a storm on 27 May 1990 at 48°N, 161°W, losing 21 containers, five of them holding 80,000 Nikes — each tongue serial-numbered, so the shipwrecked shoes could be told from any others, and the serial numbers eventually showed that only four of the five containers had opened, spilling 61,280 shoes [3]. A year later came the spill this report is named for: in 1992 a container ship spilled 29,000 bath toys into the North Pacific, and after 400 were found, oceanographer James Ingraham used a computer model to track the objects’ route and extrapolate where the remainder would go [3]. The Hansa Carrier’s cargo had everything a ghost ship usually lacks — a date, a position, a serial number, a court file — and that is why it, and not a hundred anonymous wrecks, taught oceanography how things drift.

Every famous derelict casts the same legal shadow. A hull without a crew is not automatically ownerless: salvage law holds that services cannot be forced on an unwilling owner — a salvor who acts without consent is a “gratuitous intermeddler” entitled to nothing — and title holders of historic wrecks can bar salvors outright, as Spain did for its 1802 wreck in Sea Hunt [4]. The modern regulatory version is blunter: Washington’s Derelict Vessel Removal Program took custody even of a former Coast Guard 41-foot utility vessel owned by the U.S. Bureau of Indian Affairs, because “the vessel met the definition of abandoned” [5]. Fame, law, and drift are this paper’s three threads. The sections that follow pull them in order — beginning with Baychimo herself, the ship whose story Section 2 now tells in full.



Exhibit. Ghost Ships: The Japanese Ryou-Un Maru, a 164-foot shrimp vessel, was sunk by the U.S. Coast Guard in April, 2012 off the coast of Alaska in 6,000 feet of water. Simply put, similar floating vessels should be treated the. Source: S0013.

SS Baychimo: the Arctic ghost ship that kept being sighted

Baychimo's last working season already had the shape of her afterlife: late, ice-bound, and watched. Sea ice in the summer of 1931 was difficult, and travelling east she "only rounded Point Barrow on 22 August, the latest she had ever entered the Beaufort Sea on her annual run to Coronation Gulf" [1]. She reached Fort Hearne on 5 September and spent four days offloading supplies and taking on cargo — including, as Section 1 recorded, the eight crates of Richard Finnie's ethnological collection [1]. Outward-bound, she "passed Point Barrow traveling southwest on 18 September after difficult navigation through thick ice in the Beaufort Sea," her captain and crew "optimistic for a clear passage to Bering Strait" [1]. The optimism lasted a week: conditions in the Chukchi Sea worsened, and "by 25 September Baychimo was locked in ice in Peard Bay near Franklin Point" [1].

The abandonment, when it came, was orderly — which is precisely what makes the sequel strange. "By mid-October it was clear that the vessel was trapped and the decision was made to abandon ship. The majority of passengers and crew were evacuated by air to Nome. A small contingent built a shore camp and prepared to wait through the winter and keep watch on the vessel. After a late November storm, the ship vanished and the shore crew assumed she had sunk" [1]. She had instead done what she would do for three decades: moved with the ice, out of sight, intact. "In early December, the abandoned Baychimo was spotted by Iñupiat hunters and trappers near Skull Cliff, south of Barrow. They salvaged several bales of furs from the vessel's hold, and notified the shore party. By February 1932, Baychimo had once again disappeared from sight, and the remaining shore party was evacuated by air to Kotzebue" [1]. The pattern of her career was set in its first winter: appearance, boarding, salvage, disappearance.

"In subsequent years, the vessel, locked and drifting with the pack ice, was spotted and occasionally boarded. Some cargo was retrieved, although she rarely appeared close to shore and transportation was difficult" [1]. The museum paper's summary is the one this report's title trades on: an "abandoned icebound vessel said to make occasional appearances for over 30 years in the Alaskan Chukchi and Beaufort Sea region," a ship that "floated in the Beaufort Sea icepack for over 30 years" [1]. The best-documented boarding remains MS Trader's of 11 August 1933, when the derelict lay "anchored off the village of Wainwright" and Trader's crew "tied up to Baychimo and attempted to break free the abandoned vessel from the ice to tow it to a location where it could be salvaged" — failing be-

cause “Baychimo was too large for the smaller vessel to break free from the ice pack,” though they “nevertheless recovered several items” [1]. Even the transfer of those items into the historical record is a sighting-grade reconstruction: the paper surmises that on 17 July 1934, during Trader’s $200 charter carrying part of Otto Geist’s expedition from Nome to Savoonga, Palsson handed the Baychimo materials to one of Geist’s crew, who deposited them at the University of Alaska Museum [1].

Consider what kind of track this is. A modern derelict’s drift can be modelled: for the tsunami debris field, simulations using observed winds and altimeter-derived currents show that objects with very high windage (4–5%) crossed the Pacific in nine months to a year, while zero-windage debris stayed far offshore, its greatest density “northwest of Hawaii, centred about the 40°N latitude” [2]. The Ryou-Un Maru — the other ghost ship of Section 1 — “was estimated to have a windage of about 3.5%,” which is why she arrived off Haida Gwaii in the first winter while low-windage debris was still mid-ocean [2]. Baychimo had no windage coefficient and no model. Locked in the pack, she moved at the ice’s pace, and her positions survive only where human eyes met her: a hunters’ report at Skull Cliff, a trader’s boarding off Wainwright, a shore camp’s empty horizon after a November storm. Her thirty-year track is a sighting log — the reportage form of Section 3’s taxonomy, kept by the people who lived along her route.

That is also why she, and not some better-instrumented wreck, became the Arctic’s ghost ship. A derelict that sinks is an ending; a derelict that strands is a cleanup; a derelict that keeps being sighted is a serial — each appearance a new instalment, each boarding a small salvage, each disappearance a promise of return. The ice that imprisoned her also preserved her, and the coast she drifted along was inhabited by expert observers who boarded, salvaged, and reported her as a matter of course. Baychimo’s legend is what a thirty-year observational record looks like when the observers are hunters, traders, and travellers rather than satellites — accurate in its particulars, as the museum’s reconstruction of her cargo shows, and impossible to close, because the record simply stops being added to. The next section steps back from her to the category she anchors: what “ghost ship” means when legend, reportage, and documentation are kept apart.

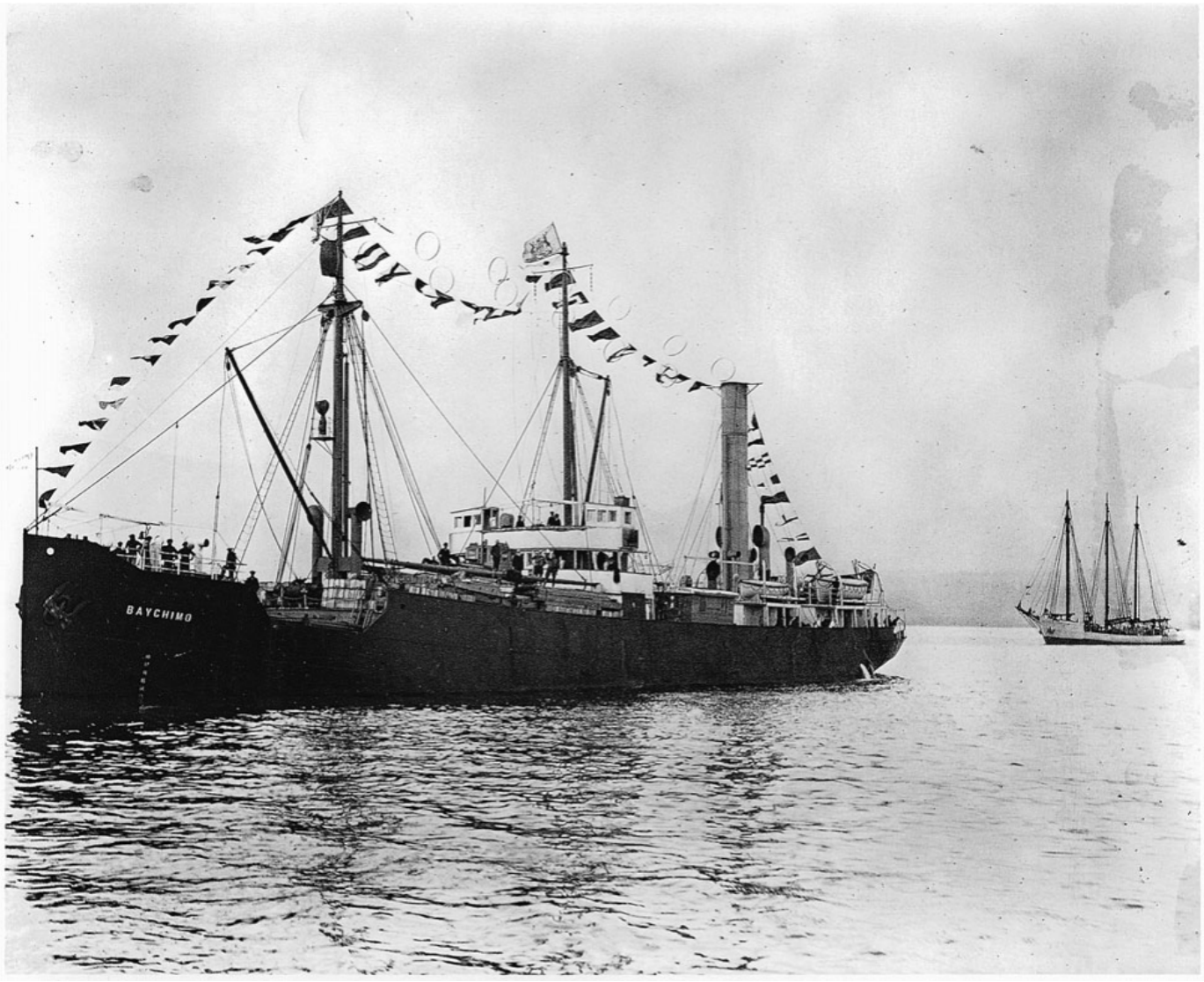


Exhibit. Fig. 1. The Hudson's Bay Company steamer, the Baychimo; photo taken by Captain George E. Source: S0010.

Ghost ships as a category: legend, reportage, and documented cases

Sections 1 and 2 used three words as if they were one. A ghost ship can be a legend — a story whose pleasure is that it cannot be checked. It can be reportage — a dated, placed claim that a vessel was seen, boarded, or found. Or it can be a documented case — a status conferred by an institution, with a file. Baychimo lived in the second sense for thirty years and entered the third only when a museum paper reconstructed her cargo's paper trail. The Ryou-Un Maru was documented from her first coast guard photograph. This section defines the category by the documented end of the spectrum, because that is where the term does work.

In the documented sense, "abandoned" is not a description but a legal act. The ADV Blue-Ribbon report defines the core problem as one of authority: "Authority refers to the legal ability of a governing agency to declare a vessel 'abandoned' and thus remove and dispose of it" [5]. A vessel becomes a documented derelict when a custodian with due-process powers says so — and the White Paper research behind the report found the category in administrative disarray: the problem "includes both commercial and recreational vessels"; "the majority of ADVs are recreational, yet commercial vessels are typically larger and on a per vessel basis, can cost several orders of magnitude more than recreational vessels to remove"; most states face "an increasing backlog of existing or 'legacy' ADVs"; "in the US, there is no comprehensive federal program"; only Washington among the five Task Force states "can be considered comprehensive"; and Canada's federal Abandoned and Wrecked Vessel Act is "comprehensive yet underfunded" [5]. A comprehensive programme, the report concludes, needs five elements — Authority, Prevention, Public Outreach and Education, Removal and Deconstruction, and Funding [5]. The category, in short, is a filing system with gaps.

The gaps produce their own kind of ghost. In 2017 the Point Estero ran aground near Cayucos State Beach, California: "The US Coast Guard removed the oil but then departed. Vessel removal costs were estimated at $70,000. The uninsured owner walked away. The vessel was not eligible for the state's recreational ADV program, and there is no commercial program. Both the county and the State Lands Commission have authority to remove the vessel, but no funding to do so. As of 2019, the vessel remains" [5]. Nothing about the Point Estero is mysterious. It is a derelict manufactured by the seams between programmes — documented at every step, removed at none. Where Baychimo's story is a crewless ship that kept moving, the Point Estero's is a crewed system's ship that could not be moved, and both belong to this paper because both are what the sea does with vessels whose paperwork fails.

Reportage, the middle sense, is where the two meet — and the documented world runs on it more than it admits. For fishing gear, the FAO review notes that direct reporting of loss by the gear's own operator "should enable more accurate location and identification of the gear, but such reporting is rare"; Norway's mandatory procedure is the exception that proves the value, with "in excess of 80 percent of losses" estimated reported [6]. Most programmes therefore cultivate sightings: a "no blame" approach to reporting, as in Washington's Northwest Straits Derelict Fishing Gear Removal Project, and public reporting channels such as the California Lost Fishing Gear Recovery Project's online form and free phone number, with GPS making reported locations precise [6]. On larger ships the sighting is bureaucratized in advance: vessels over 400 GT must carry garbage record books logging any loss of fishing gear, "subject to random inspections by the coast guard and fisheries monitoring, control and surveillance officers and port state officers" [6]. A sighting that reaches a record book, a hotline, or a coast guard photograph has crossed from reportage into documentation.

The category also scales. UNEP's Regional Seas Programme organized marine-litter work across 12 regional seas from 2005, each with a status review and action plan [6]. APEC's 2004 Honolulu seminar on derelict fishing gear called the problem "a critical problem in the marine environment," and the 2005 Bali Plan of Action extended the remit explicitly "to address derelict fishing gear and derelict vessels" [6]. The European Commission's Regulation 356/2005 laid down rules for marking passive gear in Community waters — documentation inscribed on the object itself [6]. Even outreach closes the loop: the Blue-Ribbon report's model programme prescribes "leveraging ADV stories, especially on topics such as removals, turn-in events, and problem ADVs" [5] — the documented category deliberately feeding on the first sense, story, to prevent the third.

So the taxonomy is a pipeline. Legend is reportage that never found paperwork. Documentation is reportage that found an institution — a museum, a coast guard, a record book, a removal fund. What none of the three senses yet provides is continuous observation: a derelict known not by episodes but by track. The next section takes up the in-

strument built to supply exactly that for the ocean's surface — the global drifter array — and the strange asymmetry that the sea's smallest debris is better observed than its largest derelicts.



Exhibit. Fig. 4. Above: the SS Baychimo trapped in ice in Peard Bay off Point Franklin; below: aircraft on the ice during the 1931 relief effort (Reuther & Rogers 2016) [S0010]. Source: S0010.

Reading the Ocean's Surface: Drift Science

The observational global drifter dataset

Every ghost in this paper so far has been known by episodes — a sighting here, a boarding there, a file where an institution happened to look. The Global Drifter Program is the opposite epistemology: objects built to be watched, released on purpose, reporting on schedule. Van Sebille, England and Froyland's study "Origin, dynamics and evolution of ocean garbage patches from observed surface drifters" turns that observational array into the paper's reference instrument, and its design deserves a section of its own, because everything the drift sciences claim rests on it.

The dataset is the accumulated record of the Global Drifter Program, whose buoys, "advected with the near-surface flow, have been released since the early 1980s" [7]. Each buoy carries a drogue at 15 metres depth, has "a battery life of up to 5 yr," and its "post-processed data yields geo-locations of the buoys every 6 h" [7]. Coverage is the point: "more than 85% of the ocean surface having over 100 location fixes per 1° × 1° grid box" [7]. Against Baychimo's thirty years of occasional appearances, set a buoy's five years of four fixes a day — the difference between reportage and observation, quantified.

The array even contains its own derelict problem, and the study turns it into a feature. "All buoys used here are deployed with a drogue at 15 m, but many of them lose their drogue during their lifetime, so that 48% and 52% of the data is from buoys with and without drogues, respectively" [7]. The two populations move differently: "non-drogued buoys are more sensitive to direct wind forcing and wind drift, whereas drogued buoy trajectories track the ocean flow at 15 m depth, and so are more representative of the upper ocean geostrophic flow and Ekman transport" [7]. The authors keep both, deliberately — partly for sample size, and partly because real debris is likewise split between the wind-blown surface layer and the near-neutrally buoyant mixed layer, with biofouling submerging or sinking the smallest plastics [7]. A buoy that loses its drogue is a drifter that has changed its windage — the same variable that decided the Ryou-Un Maru's fast crossing in Section 2 — and the dataset records the change instead of discarding the buoy.

Because even five-year trajectories are "too short lived to resolve the connectivity between ocean regions" on the timescales that matter, the study converts observation into statistics: all trajectories are gridded onto a 1° × 1° grid and used to build transit matrices giving "the probability that particles move to any other grid box in 60 days," with six matrices — one per two-month period — preserving the seasonal cycle [7]. Iterating the matrices advects virtual tracer for as long as one likes: in the headline experiment, "the tracer is advected for 1100 yr" [7]. The method's honesty is in its stated limits: the matrices conserve tracer, so "we do not consider the beaching of marine debris... or the abyssal sinking of particles due to heavy biofouling," only "that part of the marine debris inventory that remains in the open ocean for decades to centuries" [7].

What the instrument sees is a surface ocean that collects. The abstract states the finding plainly: debris "collects in so-called garbage patches where, due to convergence of the surface flow, the debris is trapped for decades to millennia," and from the drifter data "six major garbage patches emerge, one in each of the five subtropical basins and one previously unreported patch in the Barents Sea" [7]. The evolution is asymmetric: after 10 years, well-defined accumulation regions (tracer concentration more than twice uniform) stand in all five subtropical gyres; after 50 years the South Indian patch has shrunk, the South Atlantic grown, and the Barents patch has formed; after 1000 years "there are no longer any elevated levels of tracer in the South Atlantic and South Indian patches, while the North Pacific patch is continuing to expand" [7]. Patches leak — tracer released in the North Atlantic drifts northeast toward the Barents Sea, and Barents tracer returns southwest — so that "on centennial timescales the different basins are much better connected than previously thought," and over multi-millennial timescales much of the debris released outside the North Atlantic "will eventually end up in the North Pacific patch, the main attractor of global marine debris" [7] [7]. Even the surprise is flagged as provisional: Barents accumulation may reflect deep-water formation or "surface buoys becoming grounded in the (seasonal) sea-ice," and "observations of this area of high debris accumulation in the Arctic region are not yet available" [7].

This is the standard against which the paper's other objects are measured. A drifter is a derelict with a biography written in advance: known release, known physics, a fix every six hours, and a statistical afterlife that runs for a millennium. The tsunami-debris review shows the same data doing operational work — drifter studies fix the North Pacific Current's bifurcation into the Alaska and California Currents at "about 50°N," the fork that decided which coast

received which debris [2]. The next section maps the basins and currents this instrument revealed — the stage on which every subsequent ghost in the paper, from bath toys to tankers, performs.

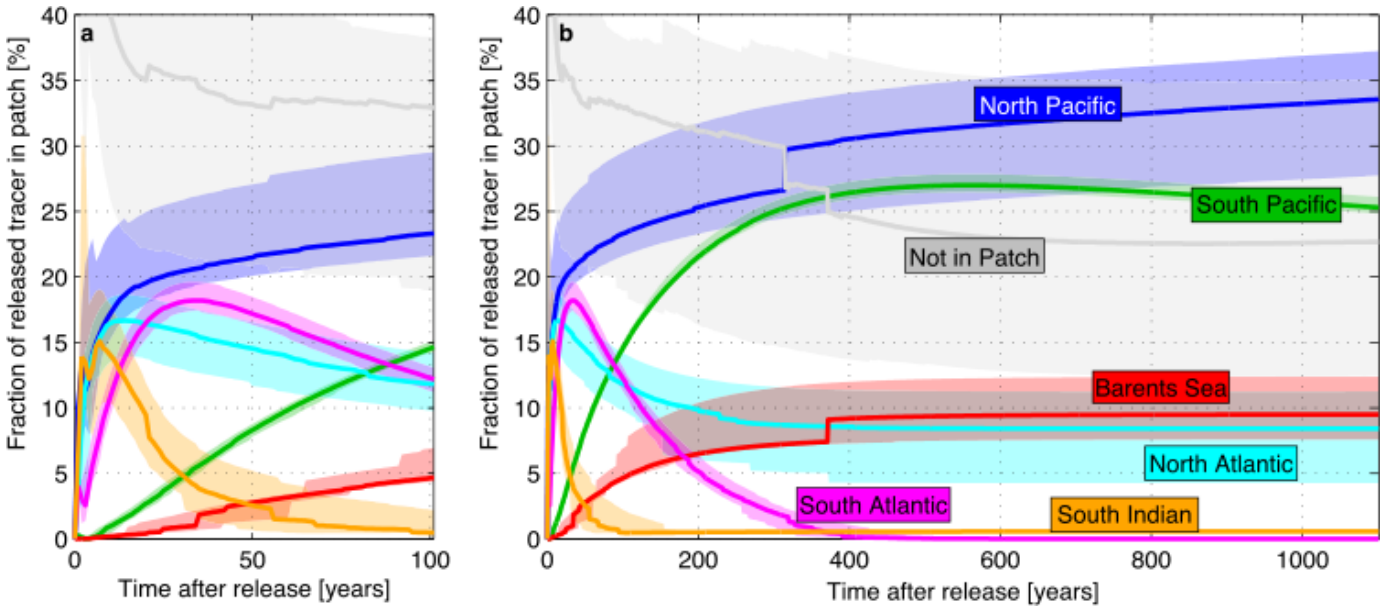


Exhibit. Figure 2. The evolution of the amount of tracer within the six garbage patches. Source: S0005.

Ocean surface-layer basins and currents

A basin, in the surface layer, is a circulation with a centre that keeps things. The mechanism is Ekman pumping: in the subtropical gyres, wind-driven convergence of the surface flow piles surface water toward the gyre's centre, and because floating debris "is generally less dense than seawater, it floats and accumulates in the regions of strongest convergence near the surface" [7]. The result is a geography of extremes — in the patches, "the concentration of plastic and other debris... can be many orders of magnitude larger than in other regions of the world ocean" [7]. But the same introduction that names the mechanism names its limits: linear Ekman theory "does not predict the timescales of patch formation from debris entering at the coastal margins, nor how eddy mixing or other non-linear processes can counteract the accumulation of debris and provide a flux out of the patch locations" [7]. A basin is not a trap with walls; it is a balance between convergence that gathers and eddies that disperse — which is why Section 4's patches leak.

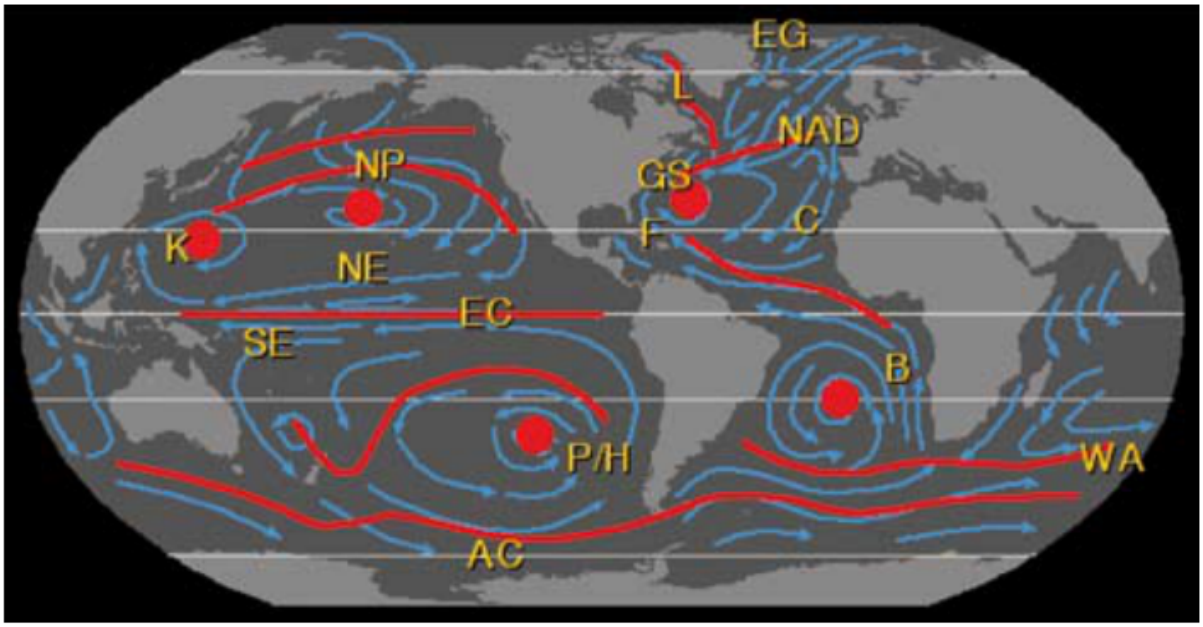
Sampling confirms the map the drifters drew. Cózar and colleagues' global compilation — 3,070 samples — found plastic in 88% of open-ocean surface samples, with concentrations "spanning over four orders of magnitude," and "the distribution pattern agreed with those predicted from ocean surface circulation models, confirming the accumulation of plastic debris in the convergence zone of each of the five large subtropical gyres" [8]. Their estimate of the open-ocean surface load is 7,000 to 35,000 tons, with the North Pacific contributing "between 33 and 35%" of it, "mainly owing to the size of this gyre" and to the densely populated Asian coast behind it; concentrations in the Kuroshio Current, "the western arm of the North Pacific Gyre, can become exceptionally high, including the highest reported for nonaccumulation regions" [8]. Eriksen and colleagues, comparing all size classes for the first time, found "plastics of all sizes... in all ocean regions, converging in accumulation zones in the subtropical gyres, including southern hemisphere gyres where coastal population density is much lower" [9].

That southern result is the basin system's central puzzle. Southern-hemisphere totals fall "within the same range as for the northern hemisphere oceans," which "is unexpected given that inputs are substantially higher in the northern than in the southern hemisphere" [9]. Two explanations compete, and both are about connection. Either plastic "is moved more easily between oceanic gyres and between hemispheres than previously assumed... such as currents from the Bay of Bengal that cross the equator south of Indonesia" [9] — the super-gyre connectivity Section 4's transit matrices show — or the losses are asymmetric: "stranding of floating plastics on local seashores seems to be more important in the northern than in the southern hemisphere," so the north sheds more of its load onto coasts [9]. A basin's budget, in other words, cannot be closed basin by basin; the surface layer is one system with regional accents.

It is also a system with a leak downward. Cózar's size distributions follow the fragmentation power law faithfully for pieces larger than 5 mm (exponent 2.93 ± 0.08, as theory predicts) and then diverge: below that size the expected ever-increasing numbers of fragments are missing, with "losses concentrated around sizes of 2.2 mm" [8]. Eriksen's discussion lists the removal processes at work — "UV degradation, biodegradation, ingestion by organisms, decreased buoyancy due to fouling organisms, entrainment in settling detritus, and beaching" — and notes the scale of the discrepancy: the entire floating load, all sizes combined, is "only 0.1% of the world annual production" of 288 million tons in 2012 [9]. The surface layer holds a small, selected residue of what enters it; the rest strands, sinks, or is eaten. Ghost ships and ghost gear are simply the residue's largest members — the pieces too big for every removal process except the shore.

The North Pacific shows the full anatomy in one basin. Its western boundary current, the Kuroshio, carries the exceptionally high concentrations noted above [8]; its extension feeds the accumulation zone "formed in subtropical waters between California and Hawaii," where Lebreton and colleagues estimate "at least 79 (45–129) thousand tonnes of ocean plastic are floating inside an area of 1.6 million km²" — a figure "four to sixteen times higher than previously reported," with the patch's connection "to substantial ocean plastic sources in Asia through the Kuroshio Extension (KE) current system" [10]. Around 60% of all plastic produced is less dense than seawater, so the basin's floating population is drawn from the majority of the material ever made [10]. Sections 9–12 take up this plastic ocean basin by basin and budget by budget. First, though, the paper returns to the accidental experiment that made these currents visible to the public — the bath toys whose drift across this very basin turned a circulation diagram into a story, told next in Section 6.

FIGURE 5
Examples of ocean convergence zones



The red dots indicate where marine litter may accumulate

Exhibit. FIGURE 5 Examples of ocean convergence zones The red dots indicate where marine litter may accumulate. Source: S0017.

Bathtub toys' drift: the Friendly Floatees as accidental oceanographers

The Friendly Floatees are this paper's control experiment: a derelict release with a logbook. In 1993 an article in the Sitka Sentinel reported that "scores of tangerine-size yellow ducks, green frogs, blue turtles and red beavers had beached in Alaska coves" — another big container spill, "bathtub toys, no less" [3]. Curt Ebbesmeyer, fresh from the Nike investigation of Section 1, "confirmed some 400 sightings along 530 miles of Alaska shoreline" [3]. Then a shipping executive directed him to a Tacoma dock, where the captain "opened the ship's log to the spill — 29,000 toys

dumped on January 10, 1992, at 44.7 degrees north, 178.1 degrees east" [3]. A known cargo, a known date, a known position: everything a ghost ship never provides, supplied by four inches of molded plastic.

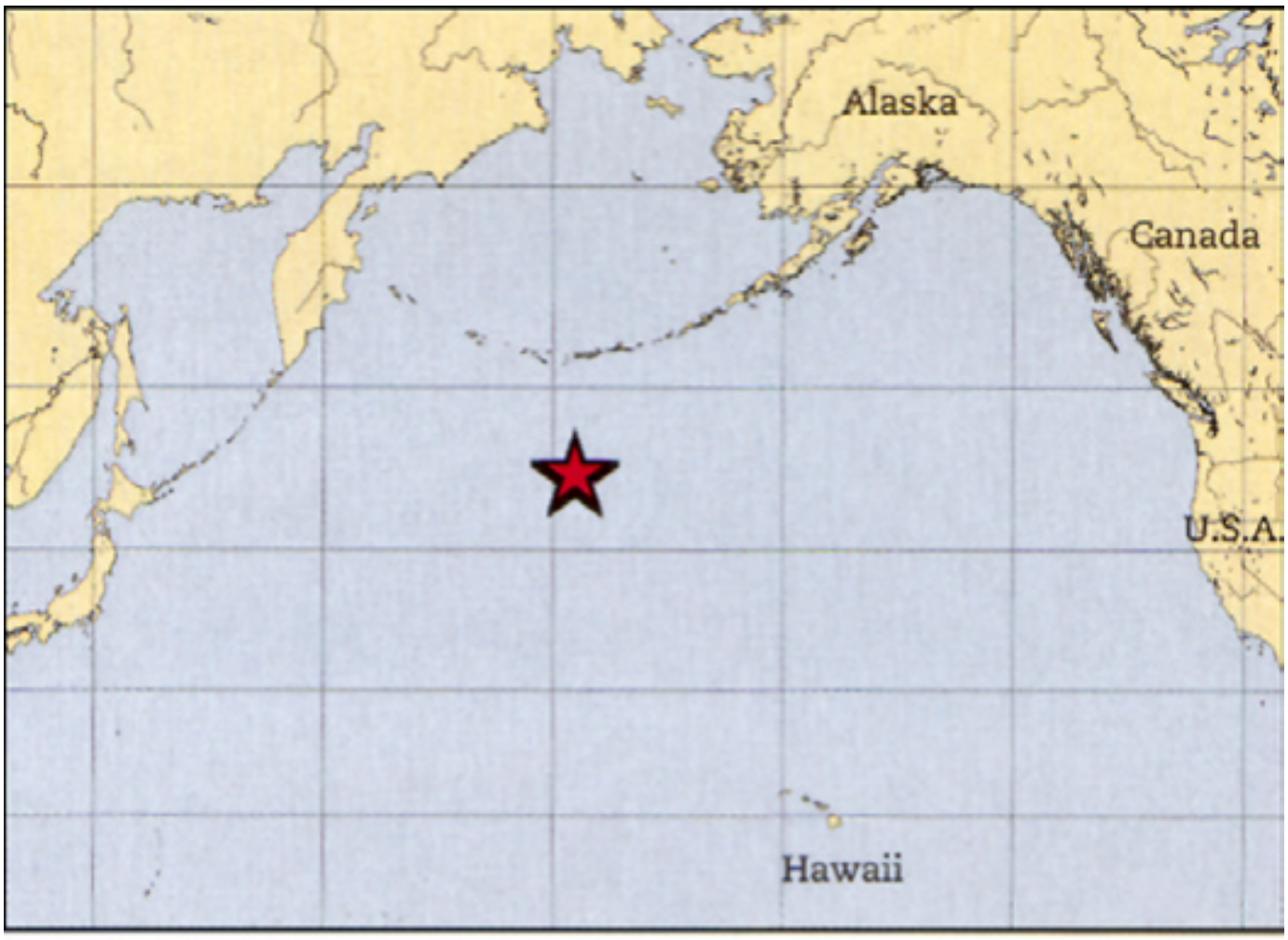
Ebbesmeyer treated the toys as instruments and first calibrated the instrument. From the U.S. distributor, Kiddie Products, he obtained samples: each four-toy set sat in a plastic shell with cardboard backing, but "soaking in seawater dissolved it, meaning the toys had probably been released to go their separate ways within 24 hours" — the experimental units were individual toys from day one [3]. Some recovered toys "appeared to have been gnawed by sea otters, so he drilled otter-tooth-size holes in his samples to judge what portion of the originals might have sunk: none. 'Very high-quality bathtub toys,' said Ebbesmeyer. 'Buoyant even when half-waterlogged.' He stuck them in his freezer to see if Arctic cold would crack them. Nope" [3]. A tracer that cannot sink, crack, or waterlog is a tracer whose strandings are pure signal — which is why this spill, and not a thousand wooden wrecks, could calibrate a model of the North Pacific.

The model was James Ingraham's OSCURS, the Ocean Surface Current Simulator, built from Navy weather data and, until the spills, "only theoretical" [3]. For the Nikes, Ebbesmeyer and Steve McLeod's beachcomber network had compiled "reliable details on 1,600 wash ups, a 2.6 percent recovery rate — very respectable," and when spill coordinates were plugged in with beaching coordinates and dates, OSCURS's predictions "matched nicely" [3] [3]. That match licensed the deeper exercise: using historical weather, they "simulated identical spills for May 27 of every year, 1946 to 1991. The Nikes went a different way each time — a tribute to the ocean's variability. 'The surface layer is a strange place,' said Ingraham" [3]. The toys then gave OSCURS "huge amounts of new data," and the result achieved an unusual form of publication: "Kiddie Products printed the duckies' calculated twisty 15,000-mile route on the back of every new bathtub-toy package. 'By far our best publication,' says Ebbesmeyer" [3].

The method generalized because the observer network did. Ebbesmeyer describes himself as a "filter feeder on floating objects," working through "his own vast, ragtag worldwide army of beachcombers" [3]; he founded the Beachcombers' and Oceanographers' International Association and its Beachcombers' Alert! newsletter, with "500 paid subscribers from as far as West Africa and New Zealand," documenting "spills of everything from onions to hockey gloves" [3]. The network's epistemology is the inverse of the drifter array's: instead of instruments reporting positions at sea, strandings are reported from shore and the sea's motion is inferred — the same inversion Section 4's transit matrices perform with buoys. And the toys taught the field its humility along with its method: "if two bathtub toys are dumped, say, in the middle of the Pacific at the same moment in the same spot, one may wash up in Hawaii while the other might end up frozen in an Arctic ice floe" [3].

The calibration outlived the toys. When the British Columbia tsunami-debris review needed a beaching rate for the debris field, it adopted Miller and Brennan's estimate that 2% to 25% would wash ashore, whose "lower bound of 2% is based on the observed recovery rate of a large number of running shoes and plastic toys accidentally released into the ocean (Ebbesmeyer and Ingraham, 1994)" [2]. The shoes' 2.6% had become, two decades on, a planning constant for a national debris emergency. Ebbesmeyer himself appears in Alaska's tsunami clean-up plan as the authority explaining why debris made landfall in December 2011, eight months after the tsunami and well ahead of model projections: debris with a high "windage" factor arrives early [11]. The accidental oceanographers had become the cited literature.

There is a darker footnote the same network recorded. On a 1999 Algalita expedition into the North Pacific gyre — the accumulation zone of Section 5 — beachcomber Steve McLeod found "piles of rope, fishing gear, liquid containers, a volleyball, even a refrigerator," and fine-net drags indicating nurdles at "129,000 per square mile — six times the mass of collected zooplankton"; a mahimahi caught for dinner "was full of nurdles" [3]. The toys were the gyre's charming emissaries; the nurdles are its permanent population. The next section turns to the oldest deliberate version of the same experiment — the message in a bottle, released on purpose, one unit at a time, by senders who understood exactly what the Floatees proved by accident.


Alaska
Canada
U.S.A.
Hawaii

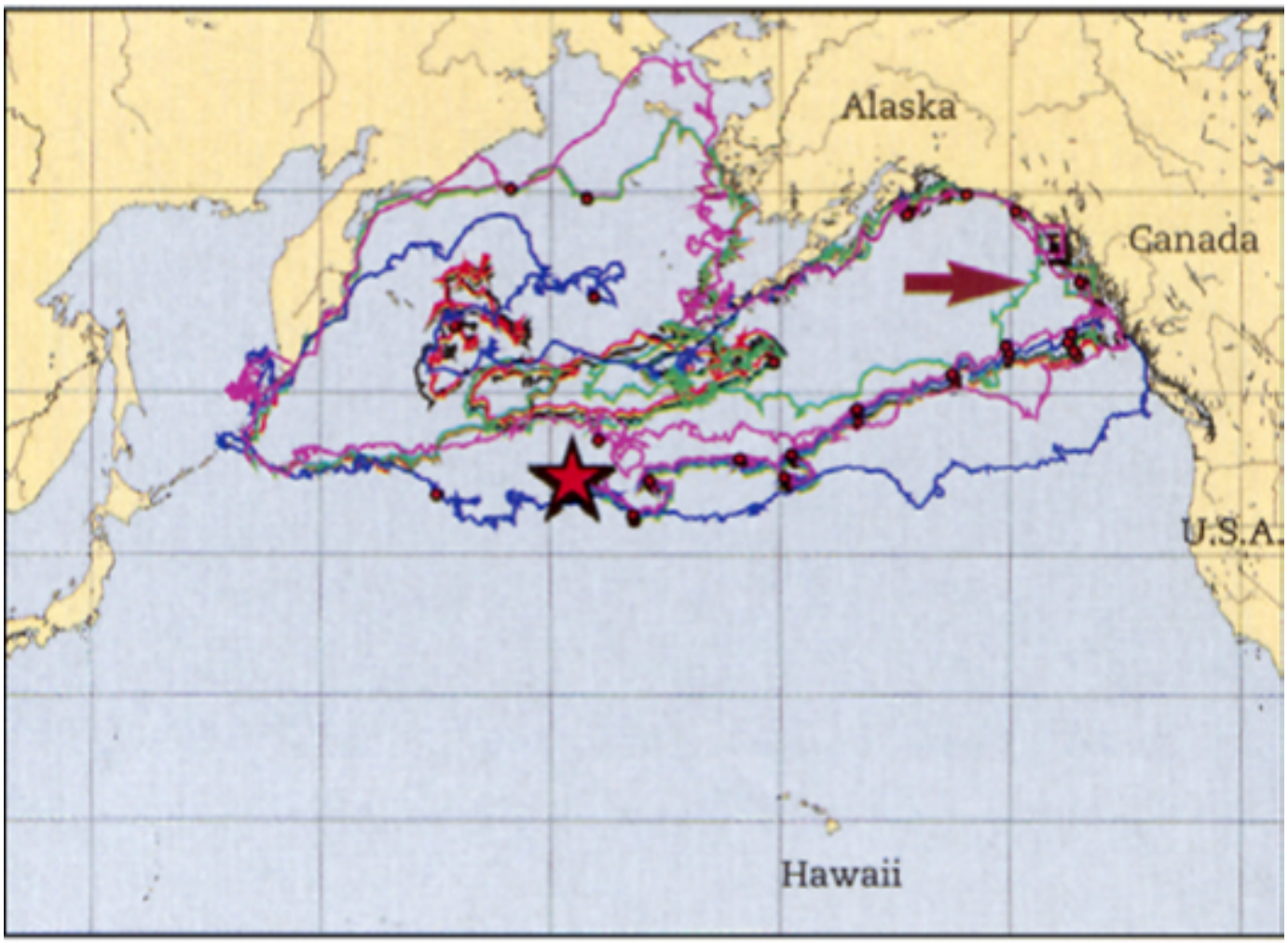

Alaska
Canada
U.S.A.
Hawaii

Exhibit. Bathtub toys' drift: OSCURS model drift tracks from the 1992 spill site (star) in the North Pacific, with beachcomber recovery points along the Alaska and British Columbia coasts (Jim Ingraham / NOAA) [S0001]. Source: S0001.

Message in a bottle and other unlikely drifters

The message in a bottle is the drifter as its sender imagined it: a known launch, a sealed text, a request for reply. It is also the only instrument in this paper whose recovery depends entirely on a stranger's curiosity — and the record shows the stranger is the least reliable component. In 1959 the Guinness brewing company dumped 150,000 bottles into the Atlantic and Caribbean, each containing a scroll of greetings from "King Neptune" and instructions for converting the bottle into a lamp, sealed with cork, wax and lead tape so well that Guinness "predicted they would go 500 years" [3]. According to one of Ebbesmeyer's sources, about 80 of them made it to Coats Island in Canada's Hudson Bay, "to be spotted by Inuit hunters. The mystified Inuit used the bottles for target practice, but then saw the scrolls and buried the rest in an unmarked grave" [3]. A 500-year message, answered with rifles and a burial: the bottle experiment's failure mode is not sinking but indifference — the same failure mode, Section 3 showed, that gear-reporting programmes fight with hotlines and no-blame rules.

The most consequential bottle in this corpus missed its target by an ocean. In December 1991 a friend sent Ebbesmeyer a bottle found on southern Vancouver Island containing dampened sheets printed in Chinese: "an appeal for the release of the famous dissident Wei Jingsheng, by then imprisoned 12 years" [3]. A year of phone calls established that "Taiwanese frogmen had been aiming propaganda bottles toward the mainland for years," and that this one had "likely been released in summer 1980 near the island of Quemoy, when coastal currents were favorable. Obviously, it had gone off course" [3]. Ingraham ran OSCURS on it and plotted the probable route — "through an island chain below Japan, then across the Pacific" — and the pair published in Eos, ending with their hope that "the magic of bottle notes" would secure Wei's release from a second prison term [3]. Wei was freed in 1997 — "no bottle connection is known" — and passing through Seattle, he met Ebbesmeyer, who gave him a copy of the paper [3]. The message failed as propaganda and succeeded as oceanography: delivered, eleven years late, to the wrong continent, and read by exactly the right person.

Most bottles are not epics. The Smithsonian author, beachcombing at Leadbetter Point, Washington, found "a semitransparent white plastic vitamin bottle with a rusted metal cap" containing "a photocopied memorandum dated Oct. 20, 1999, regarding lifeboats aboard the USS Camden" — a survival Q&A ("With water, how long could a man survive without food? A: Three weeks or longer") from a combat-support ship based near Seattle [3]. "As far as I could tell, all aboard were safe and well; I do not know why someone dropped this bottle, which probably did not drift far" [3]. That is the honest base rate of the instrument: most drift is local, most messages are mundane, and the trans-Pacific crossings are the tail of the distribution — the same distribution whose other tail Section 6's recovery rates measure.

Bottles were nonetheless the original current meter. "Eighteenth- and nineteenth-century explorers figured out the outlines of major currents simply by watching where objects traveled, including their own vessels" [3]. The instrument's modern descendants are the PALACE floats — Profiling Autonomous Lagrangian Circulation Explorers — of which a fleet of "3,000-some" was planned: drifters that "automatically sample salinity and temperature… and dive on programmed schedules as deep as a mile to catch various current layers, surfacing occasionally like U-boats to transmit data to satellites" [3]. Section 4's Global Drifter Program is the surface member of this family. The lineage runs bottle → bath toy → buoy: the same experiment, with the message progressively replaced by telemetry.

The "other unlikely drifters" widen the category past the human-made. Trees falling from Siberian and Canadian riverbanks enter the sea ice and "may go around clockwise with the Transarctic Drift" — a great Arctic gyre — "for 30 years," before being "spit out near Greenland" to circle the Atlantic or land in England or North Carolina [3]. The mechanism is old enough to be biogeography: there is evidence "even large terrestrial animals may spread and evolve by the unlikely mechanism of 'rafting' on ocean debris" [3]. And the modern bottle fleet dwarfs Guinness: "as many as 1,000 or more boxcar-size shipping containers — perhaps the most fruitful sources of intriguing objects — fall off ships annually, releasing fleets of floatable goods," while lost synthetic fishing gear alone "may run 150,000 tons a year" [3]. Every container overboard is 10,000 unaddressed bottles. What bottles and toys together give science is endpoints — a launch, a landing, and nothing between. Filling in the middle, with all the uncertainty that involves, is the work of the simulations, and it is to them that the next section turns.

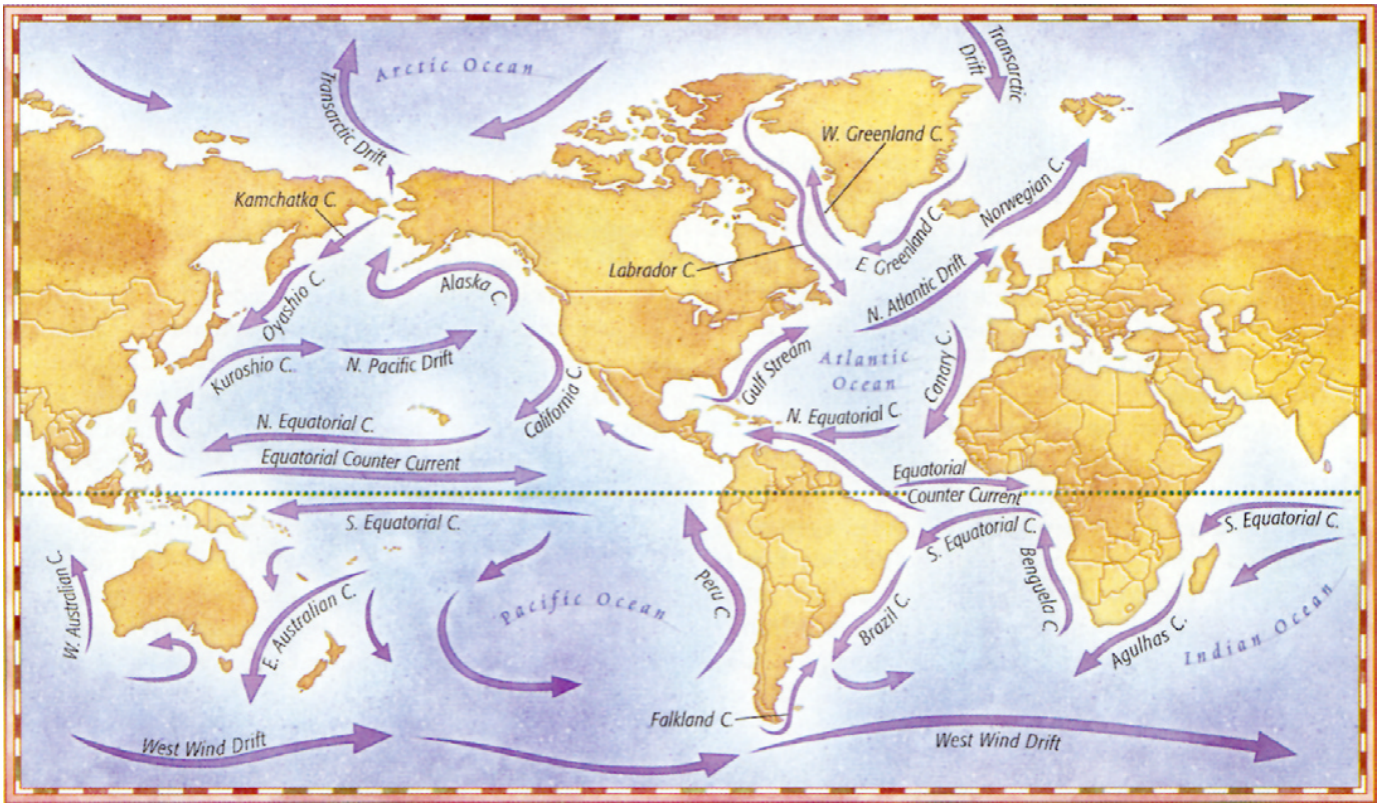


Exhibit. The world's major surface currents — the conveyor system that drift scientists read backwards from flotsam landfalls (Smithsonian Magazine) [S0001]. Source: S0001.

Model simulations of debris distribution

Simulations exist because observation fails. After April 2011 the tsunami debris field could no longer be seen from satellites, and the British Columbia review states the consequence flatly: "it is necessary to rely on models to estimate the present distribution of the debris and to forecast its distribution in the future. While models may be useful in this regard, the extent to which they can accurately simulate the transport of the tsunami debris is an open question. A major source of uncertainty is that the windage of the debris is not well established" [2]. This section surveys the model families this corpus uses — statistical, forced, and hybrid — and the single parameter that governs them all.

The first family is climatological: statistics of real drifters, projected forward. The University of Hawaii model behind the tsunami forecasts "is based on the trajectories of a large number of drifters that were deployed in the ocean over a 30-year time period," representing objects that "move with the top layer of the water column" and "have effectively zero windage" [2]. Its results are the baseline geography of slow debris: "the leading edge of the debris cloud is predicted to reach the North American continent about 2 to 2.5 years after deposition into the ocean," so low-windage tsunami debris was expected from the first half of 2013; most is carried south into the California Current, a smaller share north into the Alaska Current; and "most of the low windage debris (> 90%) remain[s] in the ocean after 5 years," with "a large fraction of the debris accumulating within the Sub-tropical Convergence Zone... referred to as the North Pacific Garbage Patch, where floating debris... has accumulated over time" and remains "trapped within the Garbage Patch for many years" [2]. Near British Columbia, simulated debris presence peaks "by about March 2014, some three years after its release" [2]. This is Section 4's transit-matrix logic in operational dress: the past behaviour of tracked objects, treated as the future behaviour of untracked ones.

The second family forces particles with actual weather. NOAA and Japanese government models drove debris with the observed winds since March 2011 and varied the windage — and the geography splits by percentage point. "Simulated debris with very high windage (4–5%) arrives at, or close to, the west coast of N. America during the winter of 2011–2012, roughly 9 months to one year after being released," and migrates farther north than zero-windage debris; by spring 2012 high-windage debris "has either already washed ashore, or is located in proximity of the coasts of Alaska, B.C, Washington and Oregon" [2]. Moderate (~2%) windage material approaches the coast in spring–summer 2012, then summer winds veer it south and winter trades carry it back west "towards Hawaii and subsequently back towards Asia" [2]. The same architecture drives the Great Pacific Garbage Patch model of Lebreton and colleagues: Lagrangian particles "advected by... sea surface currents, wave induced Stokes drift and winds," with HYCOM currents at 1/12° resolution (1993–2012), NCEP/NCAR winds, and WaveWatch3 wave fields, un-

der windage scenarios of 0%, 0.1%, 0.5%, 1%, 2% and 3%, released from coastal population hotspots, major rivers, and at-sea sources in fishing, aquaculture and shipping [10].

Calibration delivers the family's humbling result. Comparing model output against trawl-measured microplastic concentrations, "the best model fit was found for the forcing scenario with sea-surface current only (R^2 = 0.52, n = 277 cells). The regression coefficient declined as we increased the atmospheric drag term (R^2 = 0.39 to 0.21 depending on windage coefficient)" [10]. For the patch's small debris, in other words, adding the wind made the model worse — the floating population behaves as if mostly submerged, an assumption Eriksen's global census model builds in from the start: its HYCOM/NCODA circulation is coupled to the Pol3DD particle tracker, and "this model assumes that debris particles are mostly submerged in the water and extra forcing on potentially emerged parts of the debris is neglected" [9]. Yet for individual large objects the opposite holds — the Ryou-Un Maru's ~3.5% windage is precisely what the zero-windage model cannot explain [2]. Windage is negligible for the many and decisive for the few, and the few are the ones that arrive as news.

Hence the verdict the BC review keeps returning to: constructing a "nowcast" of the debris field "would require the assumption of some distribution of windage among the pieces of debris. The difficulty is that the windage of the debris is largely unknown and can only be guessed at… the results of a 'nowcast'… would depend sensitively on this assumed distribution" [2]. The models themselves are not the weak link — OSCURS was validated against 1,600 Nike strandings, and the drifter statistics rest on three decades of trajectories. The weak link is the input census: nobody can say what fraction of a debris field stands 4% out of the water. A simulation, in this field, is a sighting multiplied by an assumption. The chapters that follow use these models' outputs — patch budgets, global loads, source apportionments — with that caveat attached: the plastic ocean of Sections 9–14 is real, measured at its shores and in its trawls, but its interior geography is drawn by machines whose confidence is only as good as a guessed distribution of sails.

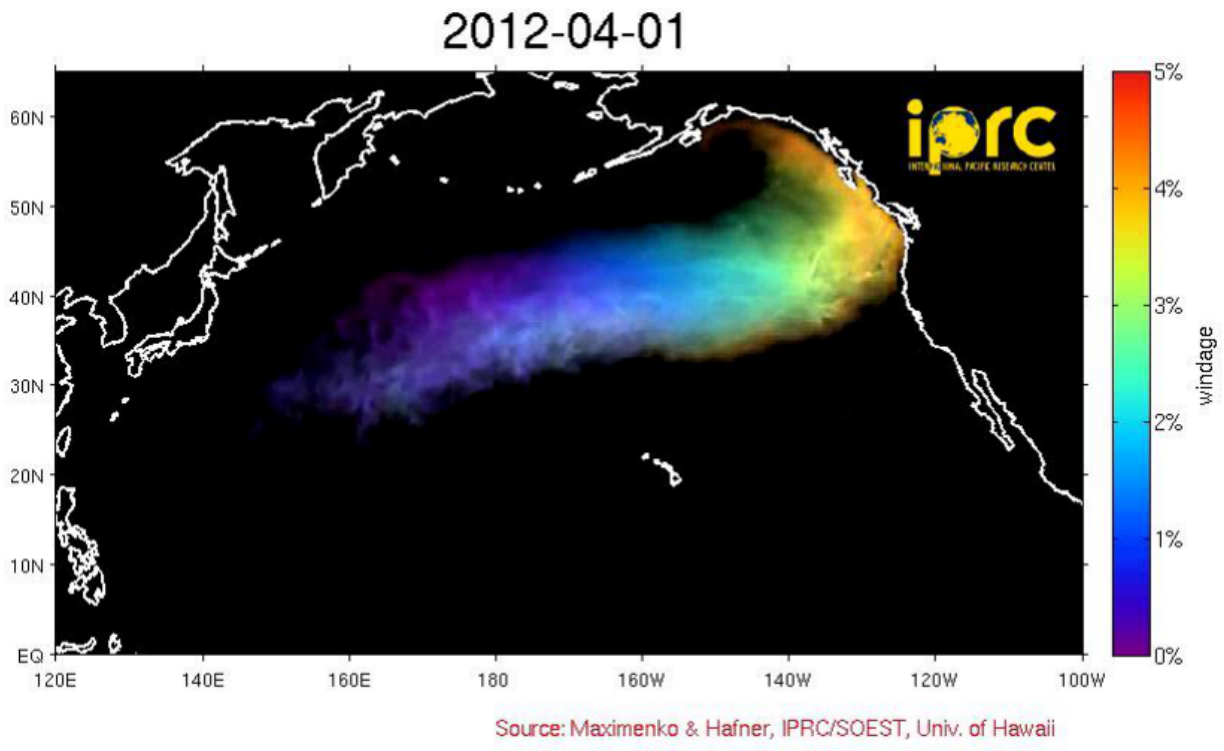


Exhibit. Figure 5. Results from model simulations of debris with varying (0-5%) windage. Source: S0012.

Where Debris Gathers: Garbage Patches and the Plastic Ocean

Five trillion pieces: plastic pollution across the world's oceans

The plastic ocean has a census, and its headline number is a floor, not a figure. Eriksen and colleagues' 2014 global estimate rests on 24 expeditions between 2007 and 2013 "across all five sub-tropical gyres, coastal Australia, Bay

of Bengal and the Mediterranean Sea," conducting 680 surface net tows and 891 visual survey transects for large debris — 1,571 locations in all [9] [9]. Feeding those observations into an oceanographic dispersal model whose inputs scale with "watershed outfalls, population density and maritime activity," and correcting for wind-driven vertical mixing, they arrived at "a minimum of 5.25 trillion particles weighing 268,940 tons" [9] [9]. The count spans four size classes — small microplastics (0.33–1.00 mm), large microplastics (1.01–4.75 mm), mesoplastics (4.76–200 mm), and macroplastics above 200 mm — making it "the first study that compares all sizes of floating plastic in the world's oceans from the largest items to small microplastics" [9] [9].

Every qualifier in that paragraph cuts the same way: downward. The authors "stress that our estimates are highly conservative, and may be considered minimum estimates," because macroplastic counts rest on "a limited inventory of ocean observations" and because the census excludes "the potentially massive amount of plastic present on shorelines, on the seabed, suspended in the water column, and within organisms" [9]. The surface total is also small against production — "only 0.1% of the world annual production" of 288 million tons in 2012 [9] — which is not reassurance but accounting: the surface layer is where plastic is easiest to count, not where most of it is. "The larger weight of macroplastic relative to meso- and microplastic, and the global estimate of floating plastic weight relative to the weight of plastic produced annually, indicates that the sea surface is likely not the ultimate sink for plastic pollution," with "sequestration in sediment the likely resting place" for what leaves it [9].

The census's real discovery is a discrepancy. Fragmentation should manufacture microplastics relentlessly — every large piece breaking into ever more small ones — yet across the size classes the study found "a tremendous loss of microplastics… from the sea surface compared to expected rates of fragmentation, suggesting there are mechanisms at play that remove <4.75 mm plastic particles from the ocean surface" [9]. Cózar's independent compilation sharpened the same anomaly: microplastic weights 100-fold below expectation, with losses concentrated near 2.2 mm (Section 5) [9]. The removal processes are enumerated — "UV degradation, biodegradation, ingestion by organisms, decreased buoyancy due to fouling organisms, entrainment in settling detritus, and beaching" — plus a subtler one: organisms ingest small particles and package them "into fecal pellets, thus enhancing sinking," while microbes biodegrade the smallest particles fastest because surface-area-to-volume rises as size falls [9]. The paper ends by reopening Thompson's 2004 question verbatim: "The question 'Where is all the Plastic?' remains unanswered" [9]. Five trillion pieces is what the surface admits to; the missing fraction is the ocean's unfiled remainder.

Cross-checks bound the uncertainty rather than removing it. For the North Pacific, an 11-year dataset estimated 21,290 metric tons of floating microplastic where Eriksen's model gives 12,100 tons for the same region; Cózar's global microplastic range of 7,000–35,000 tons brackets Eriksen's 35,500-ton microplastic component [9]. The studies agree to within their methods — and their methods differ mainly in which sizes they can see: nets catch the small, eyes catch the large, and no single instrument spans "resin pellets to large derelict nets, docks and boats" [9]. That top of the range is where this section rejoins the paper's subject. The five-trillion-piece ocean is also a transport system: floating plastics "float across oceans and transport microbial communities, algae, invertebrates, and fish to non-native regions," and the persistence that makes a dock or a derelict boat a raft is the same persistence the census counts at every scale [9]. The ghost fleet and the plastic smog are one phenomenon read at two magnifications — and the next section zooms to the patch where the large end of the distribution concentrates, the Great Pacific Garbage Patch, whose mass turns out to be carried not by the trillion small pieces but by the few big ones.

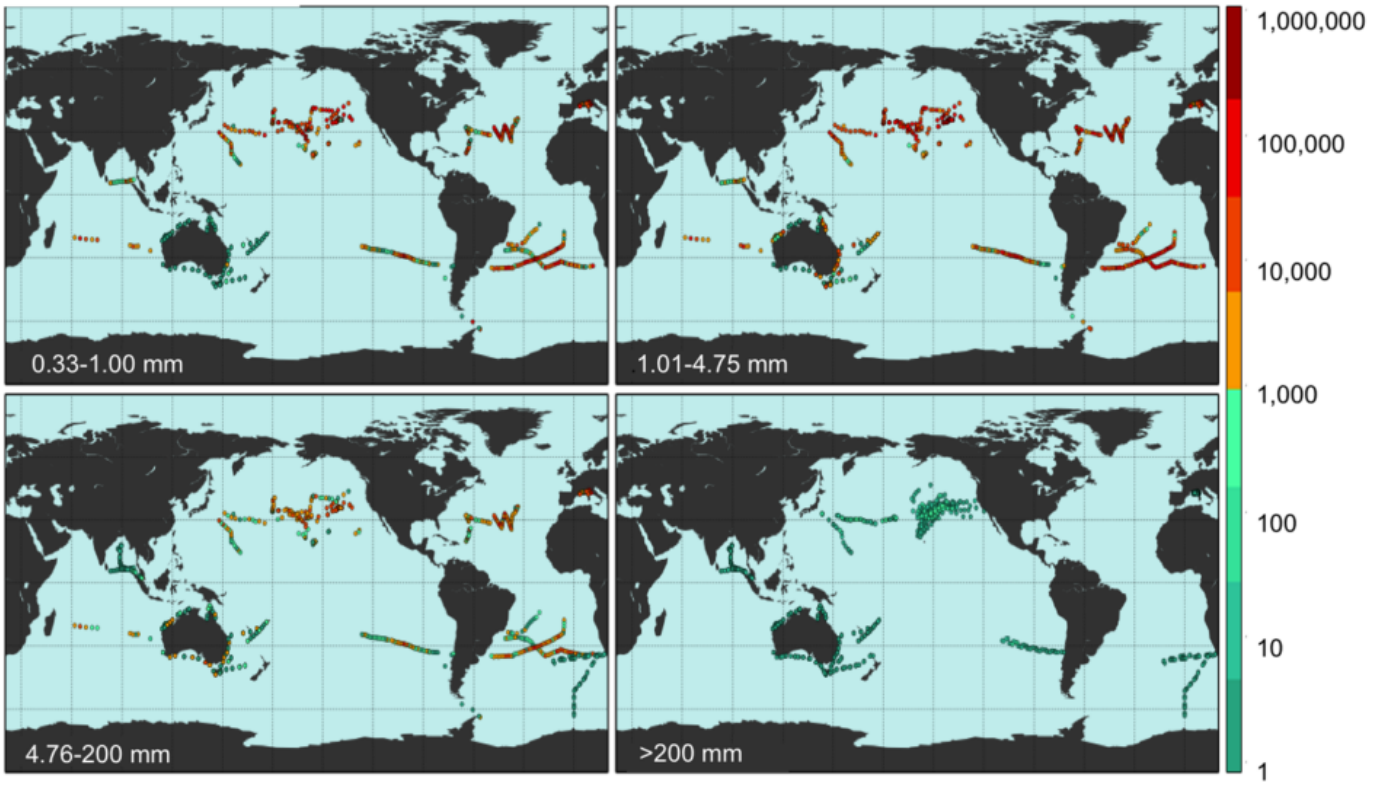


Exhibit. Figure 1. Field locations where count density was measured. Source: S0002.

Evidence the Great Pacific Garbage Patch is rapidly accumulating plastic

The Great Pacific Garbage Patch has now been weighed twice, and the second weighing changed what the patch is. Lebreton and colleagues combined two trawl types with aerial survey mosaics to quantify all debris larger than 0.05 cm inside the patch they defined — a 1.6 million km² accumulation zone in the eastern North Pacific. Their estimate: "a total of 1.8 (mid-point estimate, low: 1.1, high: 3.6) trillion plastic pieces weighing 79 k (45 k–129 k) tonnes," in an area holding concentrations "ranging from 10s to 100s kg km−2" and comprising about 87% of the plastic in their model domain [10]. Plastics were "more than 99.9% of the 1,136,145 pieces and 668 kg of floating debris collected by our trawls" [10]. The patch is not a soup with islands in it; it is a plastic-dominated sea.

Split the mass by size and the picture inverts the popular image. Microplastics dominate the count — 1.7 trillion pieces — but contribute only ~6.4 thousand tonnes; mesoplastics add ~10 kt; macroplastics (5–50 cm) ~20 kt; and megaplastics above 50 cm, a mere 3.2 million objects, carry ~42 kt [10] [10]. "More than three quarters of the GPGP plastic mass was contained in the upper size classes (>5 cm)," with hard plastics/films (type H) at 47% of mass and "plastic lines, ropes and fishing nets" (type N) at 52%; for megaplastics, "86% of their 42 k tonnes contribution was carried by fishing nets" [10]. The patch's mass, in other words, is mostly ghost gear — the largest derelicts in the census, derelict nets with "a draft of several metres" that ride below the wind's reach [10]. The polymer signature confirms the source: polyethylene and polypropylene dominate, films (37% of PE/PP waste generation) are "rarely found" — thin, high-surface items foul and sink fast — while "at least half of the collected GPGP plastics was composed of objects from marine based sources," against a model that had marine sources (fishing, aquaculture, shipping) at only 28.1% of global inputs [10] [10]. The discrepancy's explanation is partly selection: beaching strips coastal land-plastic before it can accumulate offshore, while marine gear is "purposely engineered" for durability — "strong and thick-walled nets, traps, ropes, and floats" [10].

The word "rapidly" in the title is earned twice over. First, the new total is "nearly sixteen times higher than a previous study (~4.8 k tonnes)" based on net trawls alone and four times a ~21 kt estimate that added visual surveys — a jump the authors attribute "mainly… to the use of more robust methods for quantifying macro- and megaplastics," while conceding it "could also be attributed to increasing levels of ocean plastic pollution," including the 2011 tsunami: of 4.5 million tonnes washed out, about 70% sank quickly, leaving ~1.4 million tonnes afloat, and their dispersal model puts the tsunami's contribution at "around 10–20% of the debris input into the GPGP post 2011" against global inputs of 5.32–19.3 million tonnes per year [10]. Second, decadal comparison of net-tow data shows microplastic mass concentration inside the patch rising on an exponential fit (R^2 = 0.92 within the GPGP) [10]. The debris itself testifies to accumulation over decades: of 50 objects with readable production dates, one was made in 1977,

seven in the 1980s, seventeen in the 1990s, twenty-four in the 2000s; 386 objects carried writing in nine languages, Japanese (115) and Chinese (113) leading [10]. A companion mass-budget analysis shows that this age profile is matched only by a slow degradation rate — about 3% per year — and explains the absence of post-2010 objects by "the minimum time required for plastic debris to travel to oceanic gyres (~5 years)" [12]: the patch is a lagged archive of what the Pacific rim discarded years earlier.

Two caveats keep the number honest, and both echo Section 9. The estimate is conservative — counting the outer patch at the 0.1 kg km−2 contour pushes the model "over 100k tonnes," and high-windage debris passing through is undercounted because it spreads over a larger area [10]. And the mass-balance gap persists even here: with accepted input rates, the model "predicted millions of tonnes of ocean plastic to be within the GPGP region, while we only found tens of thousands of tonnes" — a two-orders-of-magnitude shortfall implying removal mechanisms and/or fragmentation below 0.5 mm, with roughly 40% of plastic non-buoyant and sinking "soon after release" [10]. Windage decides who stays: the null-windage scenario fit best, high-windage items (the docks, boats, and buoys of Section 1) "may not accumulate within the GPGP region" but spread, beach, and exit — which is why the patch accumulates the submerged fleet of nets while the conspicuous derelicts sail through it [10] [10]. The next section asks where the accumulation zones themselves come from — the origin, dynamics, and evolution of the patches that the drifters revealed.

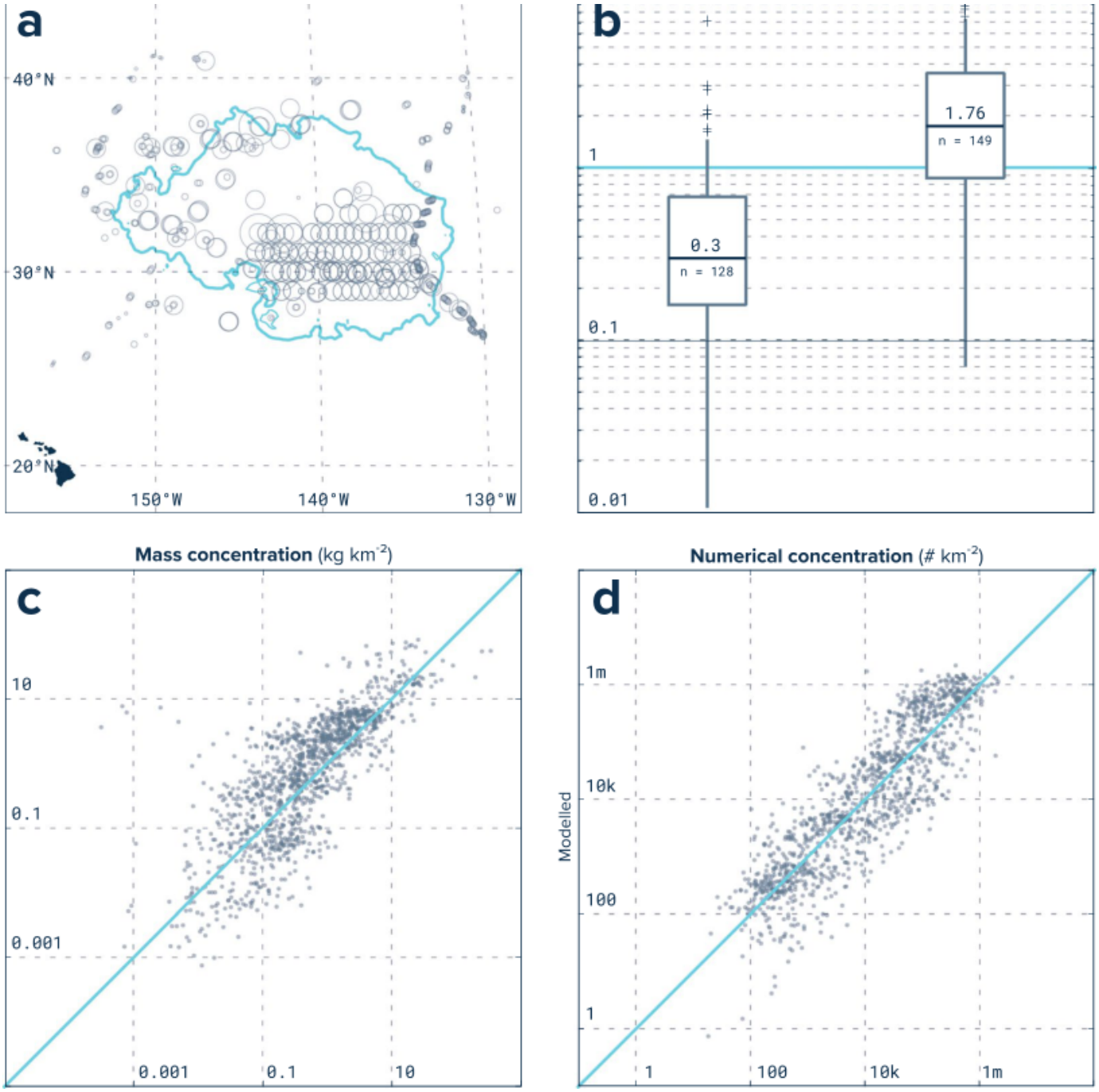


Exhibit. Figure 2. Numerical model calibration. Source: S0004.

Origin, dynamics and evolution of ocean garbage patches from observed drifters

A garbage patch is not a place. It is a statistical outcome — the place where probability piles up. Van Sebille, England, and Froyland's drifter-based experiment (Section 4) makes the origin story precise: release debris tracer from the world's coasts in proportion to human population, let the observed surface circulation move it, and watch where it gathers. The release rule itself is an admission about sources: with no global observations of actual debris fluxes, "human population is arguably the most relevant and best-constrained proxy for marine debris fluxes," so tracer was scaled to "local population density within 200 km from the coast" [7]. Where drifter coverage was too poor to build transit matrices — parts of the Indonesian Archipelago and the North Sea — no tracer could be released at all, and the Mediterranean is closed to the method entirely "since no observational drifters cross the Gibraltar Strait" [7]. The map of patch origins is thus also a map of where the observing system can see.

To find patches in the resulting flow, the authors define a tracer amplification factor, TAF = local concentration × (number of grid points / total tracer): a box with TAF 20 holds "twenty times more tracer… than if all tracer is uniformly distributed over the global ocean," and patches are "connected areas within the different ocean basins wherever TAF > 2.0" [7]. Run forward, each patch has a biography. "The North Pacific patch keeps on growing for at least 1100 yr after the release of tracer, while the South Pacific patch slowly decays after reaching a maximum size at 500 yr. The North Atlantic and Barents Sea patches reach an equilibrium size after approximately 300 yr, while the South Atlantic and South Indian patches completely disperse within 400 and 100 yr, respectively" [7]. And capture is never total: "the amount of tracer not within any of the six main patches stabilizes at approximately 22% of total tracer released," most of it — 13% of everything — residing "on the fringes of the patches" rather than inside them [7]. Patches grow by exchange: they merge ("near instantaneous jumps in the patch size are due to mergers of patches") and leak, and because the tracer is released in pulses and the system is linear, multi-year releases are just superpositions of single-year fates [7]. The long-run winner is the North Pacific — consistent with Section 4's finding that it becomes "the main attractor of global marine debris" [7] — which is why the paper's one fully weighed patch (Section 10) sits there.

Nor does a patch sit still within its own lifetime. Lebreton's GPGP modelling found "significant seasonal and inter-annual variations of the GPGP position," with dimensions "relatively consistent" but location shifting year to year; matching the expedition's 2015 conditions to ENSO and PDO climate indices picked out 2002 and 2004 as analogue years and improved the model fit (R^2 = 0.58, currents-only) — "suggesting that climatic events such as ENSO or PDO influence the average position of the GPGP" [10]. A garbage patch, dynamically, is a gyre's shadow: fixed in aggregate, wandering in detail, which is one more reason single-expedition maps disagree.

Evolution's other half is exit, and here Cózar's analysis of the size spectrum does the work. Four sinks have been proposed for the surface's missing plastic — "shore deposition, nanofragmentation, biofouling, and ingestion" — and the evidence now sketches how several operate at once [8] [8]. Fragmentation itself may be two-phase: the gap below 1 mm "could indicate a fast breaking down of the plastic fragments from millimeter scale to micrometer scale," with scanning electron micrographs showing bacterial populations colonizing particles and "potentially intervening in the fragmentation dynamics" [8]. Biofouling ballasts small, high surface-to-volume fragments until they reach seawater density and sink — but not necessarily forever: "field experiments have shown that biofouled plastic debris undergoes a rapid defouling when submerged, causing the plastic to return to the surface," so fouled plastic may oscillate vertically, sinking in lit water and resurfacing from the dark [8]. Ingestion is size-matched: the loss window (0.5–5 mm) is exactly "the most frequent plastic size ingested by fish," plastic incidence in small mesopelagic fish stomachs runs 9–35%, and the estimated plastic load standing inside mesopelagic fish "are on the same order of magnitude as our estimates of free plastic on the surface"; what fish don't retain leaves in feces sinking at "around 1,000 m·d−1," a conveyor to the seafloor [8] [8]. Large objects exit differently: bags and bottles "host a wide size range of organisms" and their cavities "facilitate their ballasting and subsequent sinking," and large plastic is "commonly observed on the seafloor" [8].

Cózar's conclusion is the honest summary of the dynamics: the open-ocean surface load is "in the order of tens of thousands of tons (10,000–40,000)," the missing fraction's "pathway and ultimate fate… are as yet unknown," and the losses "likely involve a combination of multiple sinks," including nano-fragmentation "rendering the very small pieces undetectable" and transfer to an interior where deep sediments already show "a significant abundance of microplastic particles" [8]. Patches, then, are waystations with a throttle: fed by coasts, sorted by windage, drained by fouling, ingestion, stranding, and fragmentation — accumulating only because input outruns every exit at once. The next section closes the budget formally: a global mass balance for the macroplastic that the patches are made of.

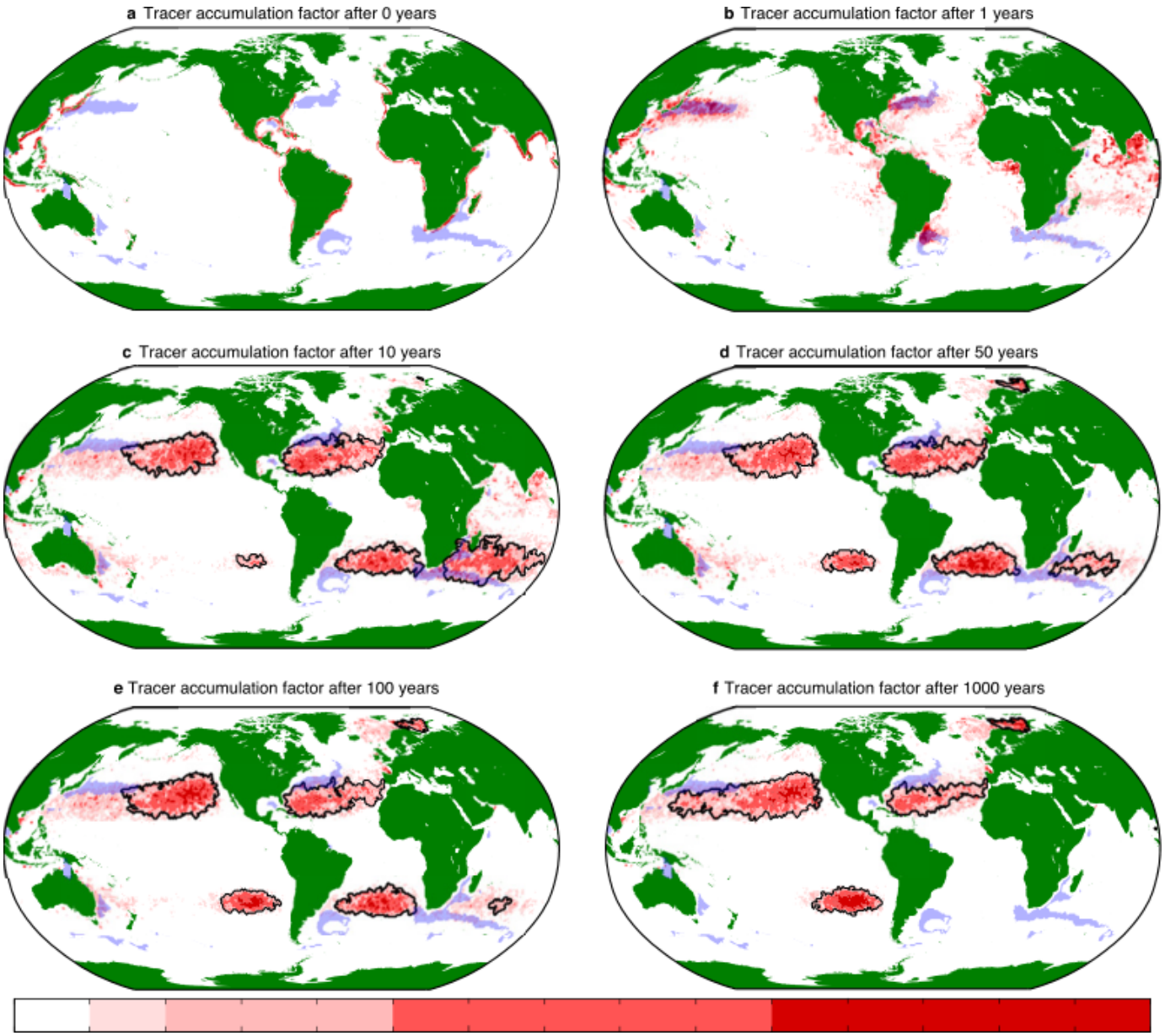


Exhibit. Figure 1. The locations and sources of the six garbage patches. Source: S0005.

A global mass budget for positively buoyant macroplastic debris

Sections 9–11 kept returning to the same hole in the ledger: accepted input rates imply far more plastic at the surface than any census finds. A global mass budget is the attempt to close that ledger formally, and Lebreton and colleagues' 2019 model does it by relocating the missing mass — not destroyed at sea, but detained on land's edge. The rival account they argue against held that degradation does the work: one earlier model proposed that "99.8% of the plastic mass that has entered the marine environment since 1950 has degraded into micro- and nanoplastics and has subsequently settled below the surface layer," predicting that under zero emissions "almost all plastic would be removed from the ocean surface layer within three years" [12]. Two field facts kill that story. Dispersal models and observations show debris "requires on average a minimum of several years to reach such remote oceanic areas" — plastic that vanishes in under three years could never stock the gyres at all — and the gyres demonstrably hold "a significant number of decades-old objects, dating as far back as the 1970s," while "the degradation of plastic initiated by solar UV radiation is severely retarded when floating in seawater" [12].

Their alternative is a box model of three surface domains — shoreline, coastal waters (out to the ~200 m photic depth), and offshore waters — with six mass compartments tracking macroplastics and their degradation products [12]. It runs from 1950 to 2015 on production data, taking positively buoyant polymers as 65.5% of production and discarding material according to market-sector lifetime distributions (packaging fast, construction slow) [12]. The input valve is the fraction of discarded plastic reaching the ocean: for 2010, 4.8–12.7 million tonnes of input against 274 million tonnes of waste gives i = 1.7–4.6% [12]. Thereafter, floating mass each year can strand or settle on the

shoreline (probability s), leak back from the shoreline into coastal water (r), escape offshore (t), or degrade out of the macroplastic class (d) in each compartment [12].

The model's referee is forensic: the age distribution of GPGP debris. The 2015 expedition landed 664 kg — 83,144 pieces, 427 with recognizable inscriptions in 11 languages, 50 with production dates [12]. Sweeping all five parameters from 0% to 100%, the authors found "only the degradation rate d significantly impacts the relative age distribution of positively buoyant macroplastics in offshore surface waters," and the best fit is d = 3% per year — with the scarcity of post-2010 objects explained by "the minimum time required for plastic debris to travel to oceanic gyres (~5 years)" [12]. A fast degradation sink is thereby ruled out by the debris's own dates: "a rapid degradation sink term of >90% per year… cannot reproduce observations of plastic age distribution at sea" [12].

What reconciles emissions with observations instead is captivity. For the model to converge, the stranding parameter "must be much greater than the release parameter" — capture dominates release — so "the landmass likely is storing a major fraction of the missing plastic debris," some of it "buried under sediments… for unknown duration" before possibly moving to deeper water [12]. The conclusion inverts the missing-plastic framing: "most of the plastic mass that has entered the marine environment since the 1950s has not disappeared from the ocean surface by degradation but is stranded or settled on its way to offshore waters, possibly slowly circulating between coastal environments with repeated episodes of beaching, fouling, defouling and resurfacing" [12]. The shoreline acts as a filter with "a delay — likely on the order of decades — for fragmented buoyant plastic to reach offshore accumulation zones," and the sorting is selective: coastal floating mass is mostly new objects, offshore mass mostly old, thick PE/PP fragments — "only certain types of plastic have the capacity to persist for a sufficient amount of time to eventually reach these accumulation zones" [12]. The budget also audits the small end: modelled secondary microplastics for 2014 (0.28–0.75 Mt) exceed surface microplastic estimates (0.093–0.236 Mt), implying "at least two-third of these microplastics have disappeared from the ocean surface layer likely by settling, ingestion, aggregation, stranding or degradation into even smaller particles" [12].

Projected forward under three emission scenarios — growth at the 2005–2015 rate, constant from 2020, or stopped from 2020 — the compartments keep accumulating or draining on the model's slow timescales, because the shoreline reservoir leaks for decades after inputs change [12]. For this paper, the budget's moral is structural: the ocean's plastic fleet is mostly not at sea. It is beached, buried, and recirculating along the coasts — the same shoreline reservoir that beachcombers sample (Section 6) and that derelict-vessel programmes clear (§§19–24) — while the famous offshore patches hold only the durable minority that escaped. The next section descends to the size class the budget treats as a sink: microplastics, and the "plastic smog" now measured in the hundreds of trillions.

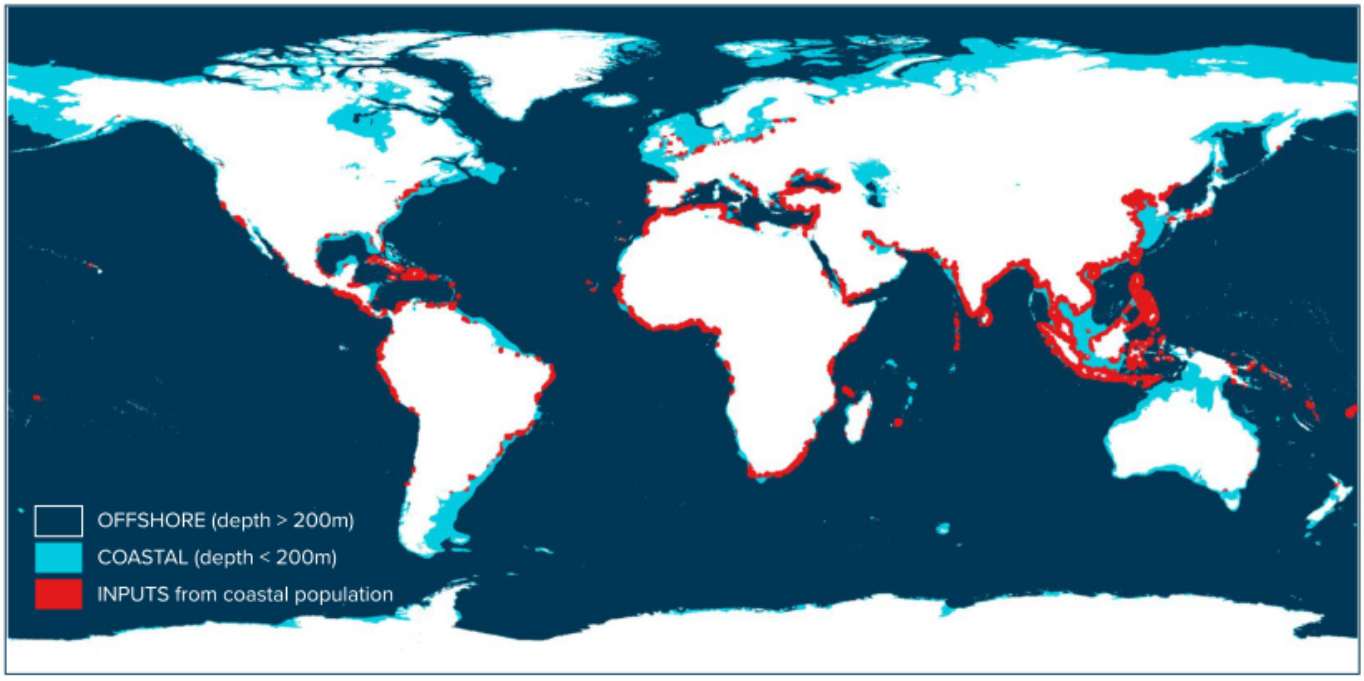


Exhibit. Figure 3. Lagrangian dispersal model source locations and global ocean surface model domains. Source: S0006.

The growing plastic smog: microplastics and primary sources

Every large derelict in this paper is on a one-way trip to becoming uncountable. Fragmentation converts objects into particles, and Eriksen and colleagues' 2023 reassessment — the census of Section 9 extended into a time series — puts the end-state in the title: "170 trillion plastic particles are afloat in the world's oceans," a "plastic smog" the authors estimate at over 25 million tonnes circulating in surface waters [13]. The dataset behind it is the largest assembled: 11,777 sampling stations compiled "from published literature and unpublished sources," mostly 5 Gyres Institute expeditions, with samples filtered to mesh sizes between 53 µm and 505 µm (94.7% at 320–350 µm), particles hand-picked under dissecting microscopes, counted individually, and converted to mass at a standard 1.36×10^{-2} g per particle [13]. The methodological spread is itself a finding: with station selection, collection, and analysis varying across four decades, "interpreting snapshots in time [is] challenging and… establishing a trend even more difficult" — which is why earlier global mass estimates for the surface layer span 93,000 to 578,000 tonnes [13].

On trend, the archive speaks in several voices. Continuous Plankton Recorder samples show "an increasing trend of microfibers since the 1960s" and "an increasing trend of macroplastic entanglement since the late 1950s"; Day and Shaw reported North Pacific microplastics increasing between 1976 and 1985; Wilcox and colleagues found the western North Atlantic increasing from 1986 to 2015 "with a rate of increase paralleling global cumulative plastic production" [13]. Against these, "other studies have found no evidence of a rise in plastic pollution over time" — a tension the authors note is partly geographic, since the positive trends come from "northern oceans surrounded by the most industrialized countries" [13]. The smog framing resolves the apparent contradiction the way Section 12's budget does: surface counts can plateau while inputs rise if export to shorelines and the seafloor rises with them — "fragmentation of large plastic results in micro- and nanoplastics leaving the OSL to shoreline and seafloor compartments" [13]. What is not in doubt is the production curve underneath: from 1.5 million tons in 1950 to "over 300 million tons" yearly, a long-term growth rate of roughly 4%, with per-capita consumption near 100 kg/yr in North America and Western Europe against ~20 kg in Asia [14]; of all plastic ever made, 8.3 billion tons, "an estimated 9% has been recycled" [14].

Half the smog, moreover, never was an object. Primary microplastics are "plastics directly released into the environment in the form of small particulates" — either deliberate additions (scrubbing agents in toiletries and cosmetics) or abrasion products: "the erosion of tyres when driving or the abrasion of synthetic textiles during washing" [14]. The IUCN's global evaluation (Boucher & Friot) quantifies seven sources — "Tyres, Synthetic Textiles, Marine Coatings, Road Markings, Personal Care Products, Plastic Pellets and City Dust" — and estimates total primary releases at 1.5 million tons per year (0.8–2.5 Mt across scenarios), concluding that "between 15 and 31% of all of the plastic in the oceans could originate from primary sources" [14] [14]. The breakdown overturns the litter-first picture: 98% of primary losses are land-based, the largest shares come from "the laundering of synthetic textiles and from the abrasion of tyres while driving," releases arise mainly from product use (49%) and maintenance (28%), and they reach the sea via road runoff (66%), wastewater systems (25%), and wind (7%) [14]. Per person, the global average is 212 grams a year — "the equivalent of one empty conventional plastic grocery bag thrown into the ocean per person/per week worldwide" — with regional per-capita releases ranging from 110 to 750 grams [14]. In countries with advanced waste treatment, the IUCN warns, primary releases "even outweigh that of secondary microplastics" — and in some countries "more plastic may be released from our driving and washing activities than from the mismanagement of our waste" [14] [14].

Crucially, these flows are additive to the familiar ones: Jambeck's 4.8–12.7 Mt/yr input estimate counts macroplastics, so primary microplastics come "in addition to the estimations made in the Jambeck report" [14]. The floating remnant of all this is comparatively small — Eriksen 2014 and van Sebille 2015 put floating microplastics at 93–268 ktons — because density sorts the smog: polypropylene floats and "eventually accumulate[s] in gyres," while denser polymers like acrylic "most probably accumulate on the ocean floor," and "a significant amount of microplastics may eventually accumulate in the deep sea… and ultimately in food chains" [14]. The era has acquired a name in the literature — the "plasticene" — and its ocean a description: a "plastic soup" [14]. For the ghost-ship narrative the smog is the background radiation of every earlier section: the medium through which derelicts drift, the fate awaiting every hull and net, and the reason the next section's question — can the inflow be turned down by 2040? — is asked about systems, not cleanups.

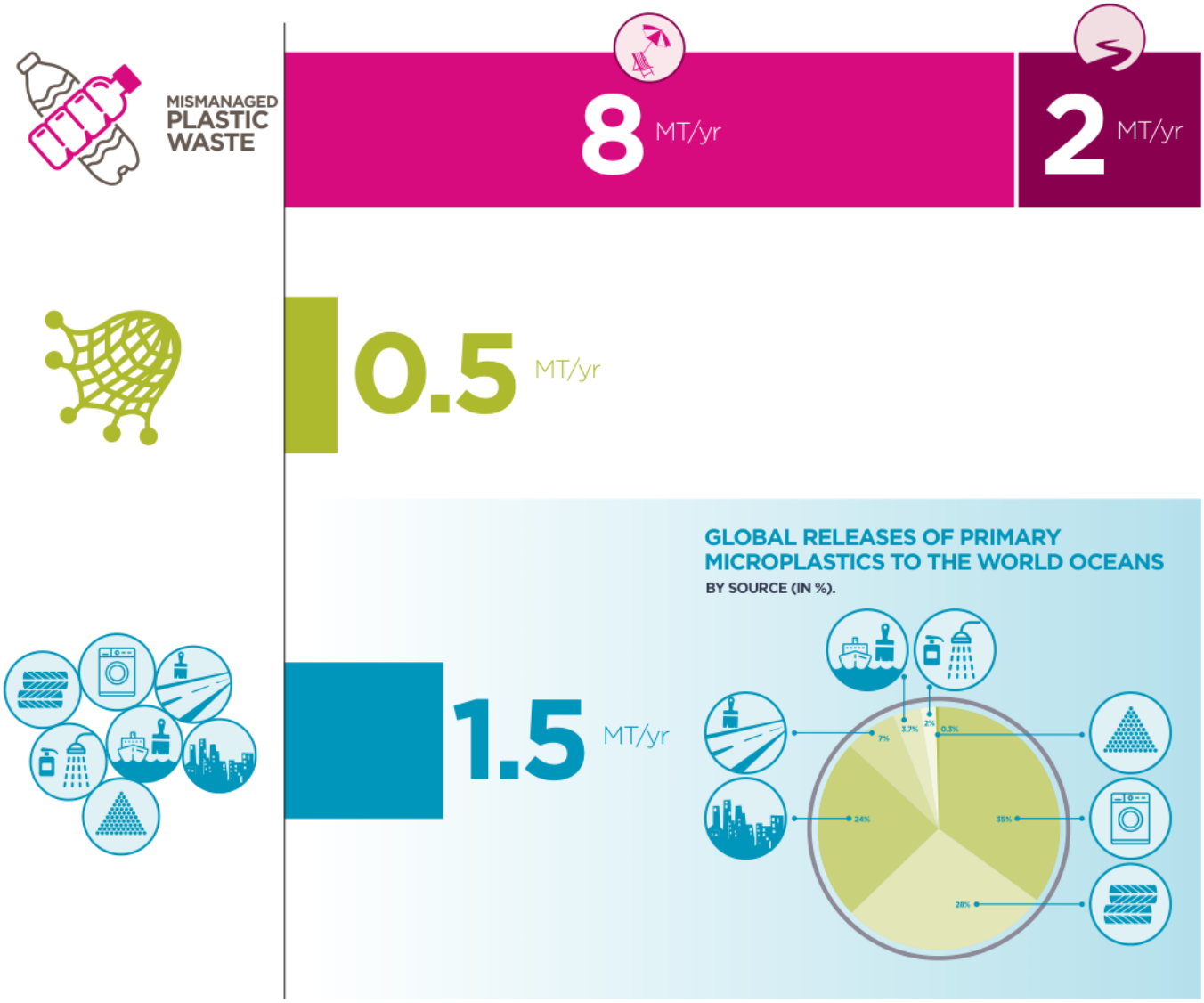


Exhibit. Global releases of primary microplastics to the world oceans by source — primary microplastics contribute 1.5 Mt/yr within total ocean plastic inputs (IUCN) [S0009]. Source: S0009.

Breaking the Plastic Wave: scenarios and system interventions to 2040

Every section so far has described a flow; this one prices the future of that flow. Breaking the Plastic Wave (Pew Charitable Trusts / SYSTEMIQ, 2020) models the global plastics system stochastically from 2016 to 2040 and asks what different futures cost. Under Business-as-Usual, annual plastic leakage to the ocean reaches 29 million metric tons by 2040 — 26 Mt of macroplastic plus 3 Mt of microplastic — with virgin plastic production at 400 million metric tons and 11 million jobs in the system [15]. Under the System Change Scenario — eight interventions applied together — leakage falls to 5 Mt/yr (4 macro + 1 micro), virgin production to 181 Mt, jobs rise to 12 million, and government costs drop [15]. The headline claim is precise: "about an 80 per cent reduction in annual leakage rates can be achieved by 2040 relative to BAU" [15].

The eight interventions span the whole chain: (1) reduce growth in plastic consumption; (2) substitute suitable alternative materials; (3) design products and packaging for recycling; (4) expand waste collection in the Global South; (5) increase mechanical recycling capacity globally; (6) scale up chemical conversion; (7) build safe disposal capacity; (8) reduce plastic waste exports [15]. They are interdependent — "collection precedes recycling, landfilling and incineration; and design for recycling helps improve the economic viability and scalability of mechanical recycling" — and the 80% figure is conditional: it "depends on the condition that all system interventions are applied ambitiously and concurrently; deviation from the levels set for any modelled intervention could yield a different outcome" [15] [15]. In the 2040 System Change "wedges," each ton of plastic entering the system has exactly one fate: Reduce (eliminate/reuse) takes 130 Mt (30%), Substitute (paper, coated paper, compostables) 71 Mt (17%), Recycle 84 Mt mechanically plus chemical routes, Dispose (including chemical conversion to fuel) 101 Mt (23%), and 44 Mt (10%) remains mismanaged — open burning, terrestrial and ocean pollution [15].

The economics invert the usual assumption that prevention is the expensive part. In the cost stack, "Reduce levers are often the most economical to implement while plastic substitutes are typically more expensive," with several reduce options showing negative net costs — savings relative to BAU — and the whole comparison excluding the unpriced externalities of mismanaged waste [15]. The report's prioritization follows: "a reduction in plastic production — through elimination, the expansion of consumer reuse options or new delivery models — is the most attractive solution from an environmental, economic, and social perspective. It offers the biggest reduction in plastic pollution, often represents a net savings, and provides the highest mitigation opportunity in GHG emissions" [15]. Mechanical recycling ranks next — "more attractive than chemical conversion or substitute materials from an economic, climate, technology readiness and regulatory point of view," each recycled tonne offsetting 48% of GHG — while plastic-to-plastic chemical conversion "has not been proved at scale," carries "higher costs, energy requirements and GHG emissions," and should expand only as energy decarbonizes; controlled disposal is "a last resort" [15]. Substitution is judged case-by-case: substitutes "are typically more expensive than plastics and their carbon impact could be better or worse" [15]. The macro result: today 95% of plastic demand is met by virgin plastic; under System Change, 2040 virgin demand falls 11% below 2016 levels even as plastic utility grows — "essentially decoupling plastic growth from economic growth" [15] [15].

The co-benefits are quantified with the same care: GHG emissions 25% lower annually by 2040 (14% lower cumulatively 2021–2040); open burning cut from 133 Mt/yr under BAU to 23 Mt/yr; 700,000 net new formal jobs in middle- and low-income countries; governments saving US$70 billion in present value over 2021–2040; and formal recognition due to "the global community of 11 million waste pickers who are currently responsible for 60 per cent of global plastic recycling" [15] [15]. But the scenario's honesty matters as much as its optimism: even under System Change, "5 million metric tons of plastic waste will still flow into the ocean in 2040 and a cumulative 248 million metric tons of plastic will have entered the ocean between 2016 and 2040" [15]. The conclusion is blunt about the cornerstone: recycling and waste management scale-up "alone will not be enough to avoid plastic pollution within budgetary and political constraints at the current levels of plastic production — let alone the expected growth," and near-zero leakage requires innovation beyond the maximum levels modelled [15]. For this paper, the scenario work closes the plastic arc with the same lesson the derelict-vessel sections will repeat for ships: end-of-pipe removal cannot outrun an unmanaged inflow. The next part turns to the sea's own demonstration of that principle — the day in 2011 when a coastline's worth of debris entered the Pacific at once.

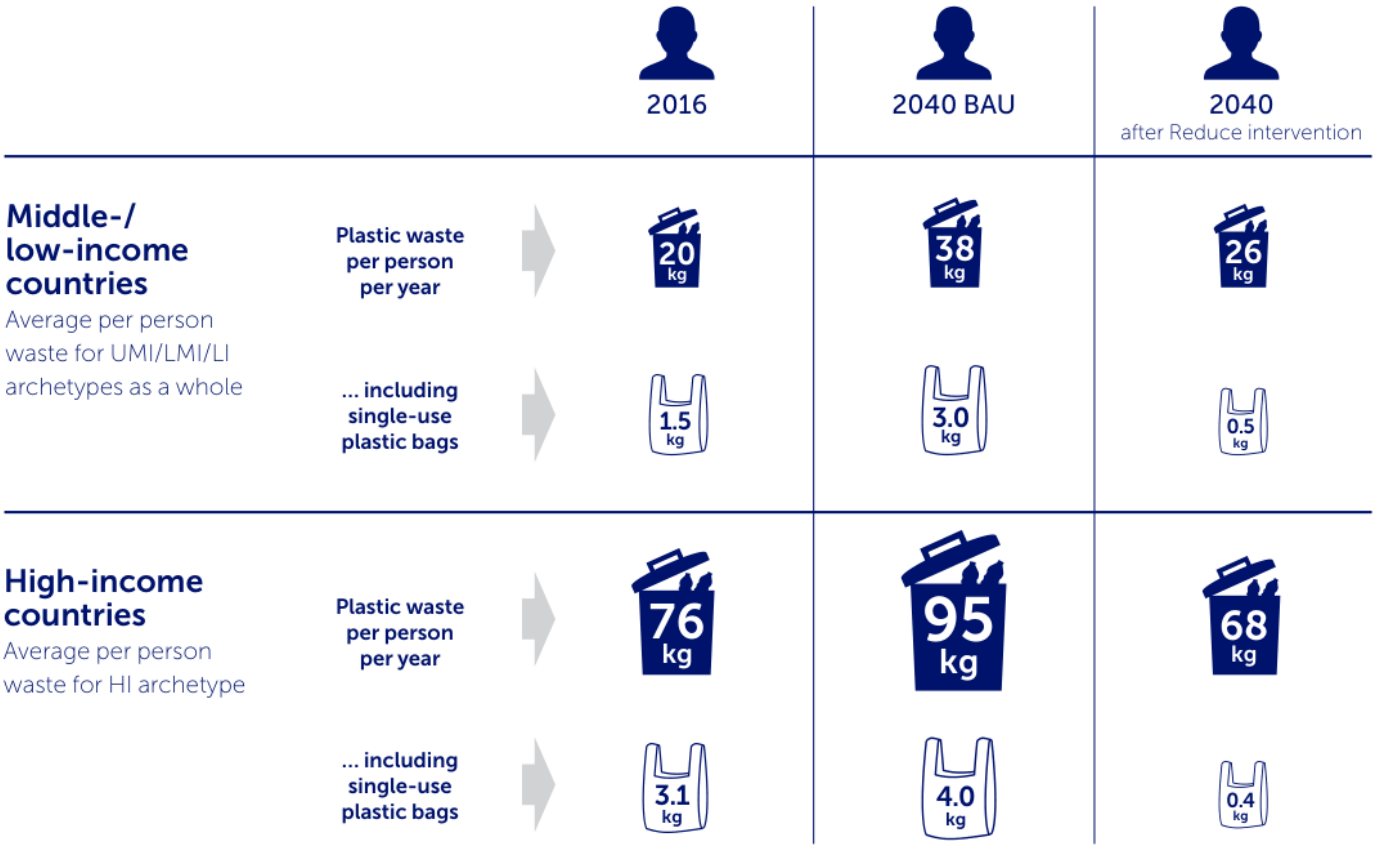


Exhibit. Figure 24: Change in total plastic waste generation and plastic bag consumption per capita in the Business-as-Usual Scenario and after Reduce interventions are applied Most plastic reductions in the System Change Scenario are in hi. Source: S0008.

The 2011 Tsunami Debris Field

A severe marine debris event: what the sea took from Japan

On 11 March 2011, the Tōhoku earthquake and tsunami "inundated 217 square miles of coastline," killed nearly 16,000 people, and destroyed much of the Sendai coast; when the wave retreated, "it dragged rubble from buildings, people's belongings, and infrastructure into the ocean" [16]. NOAA's Severe Marine Debris Event Report treats what followed as a distinct category of pollution incident — not a spill from a vessel but the partial relocation of a coastline into the sea. The Government of Japan estimated "that the tsunami swept about 5 million tons of debris into the ocean," of which "70% likely sank right away"; the remaining ~30%, roughly 1.5 million tonnes, dispersed, with "no estimate of how much is still floating" [16]. The BC science review reaches the same figure and calls the event "probably one of the largest single-source loadings of marine debris to the North Pacific Ocean" — while cautioning that "the significance of this addition is uncertain because the existing load of marine debris is not well quantified" [2].

What the sea took was, at first, visible. The debris field showed in satellite imagery for about a month, then dispersed below detectability by 14 April 2011; NOAA's December 2011 high-resolution search of the modelled 50 × 60 km patch found nothing, and the review concludes the field's "overall distribution... is unobservable and effectively unknown" (Sections 1, 8) [2]. Confirmation therefore proceeded object by object. Japan catalogued 97 at-sea sightings, all west of the dateline, mostly "small vessels and containers," many capsized or partially submerged [2]. The Russian training vessel STS Pallada made the first confirmed find — a Fukushima-registered fishing boat 800 km northwest of Midway on 22 September 2011, its radiation "below detection limits" [2]. By 18 October 2012 NOAA held 1,385 reports (620 shoreline, 765 at-sea) from BC to Hawaii, "of all these sightings, to date only 14 objects have been definitively confirmed to originate from the Tohoku tsunami" [2]. The confirmed set is famous because it is short: "vessels, buoys, sports balls, a floating pier, and a motorcycle in a container" — a soccer ball on Middleton Island, Alaska (April 2012), the Harley-Davidson in its container on Graham Island, BC (April 2012), the concrete dock at Newport, Oregon (June 2012) [16] [16]. NOAA's outreach had to teach the public windage triage — items riding high (upright vessels, Styrofoam, buoys) move FAST; half-submerged items MEDIUM; lumber, low in the water, SLOW [16] — and to repeat that "not every item found on our shorelines is from the Japan tsunami" [16].

The second dock made the event's scale physical. In December 2012 a dock "over 65 feet long and 20 feet wide and weighed more than 185 tons" grounded on Mosquito Beach in Olympic National Park, inside a designated wilderness overlapping the Olympic Coast National Marine Sanctuary; a multi-agency team with the NSF JTMD Invasive Species Team reached the site to sample non-native species and attach tracking beacons before removal [16]. The response architecture around such arrivals was improvised but real: NOAA's Marine Debris Program led, with a public reporting address (DisasterDebris@noaa.gov) feeding a tracking database visualized in ERMA, bi-weekly coordination calls whose participant list "exceeds 200 representatives," and standing liaison with Japan and Canada through the State Department [16] [16]. A December 2011 Subject Matter Expert group united the SCUD, OSCURS, and GNOME modelling leads; GNOME, fed with real weather and Navy HYCOM currents and a windage range from the USCG leeway library, produced a "hindcast, not a forecast" — with NOAA stressing that "a model is not reality" [16]. Removal authority was a patchwork: the Army Corps of Engineers under Section 19 of the Rivers and Harbors Act for obstructions, USCG and EPA under CERCLA/CWA/OPA for hazardous debris, and everything else to "state, local, or tribal jurisdictions" [16] — the same authority-without-funding seam that Section 3 found for derelict vessels.

The risk ledger, as the BC review keeps it, is dominated by uncertainty rather than alarm. Miller and Brennan's beaching estimate — 2% to 25% of the debris over a few years, the 2% floor being the Nike/Floatee recovery rate — spread over 15,000 km of coast gives "2–25 tonnes of debris per kilometre" against a Washington baseline deposition of "about 0.7 tonnes per km per year": potentially years of elevated loading, with "very wide uncertainty" [2]. Radioactivity risk is "believed to be very low": iodine-131's half-life is ~8 days, cesium-137 salts are water-soluble and wash off, and tested debris read below detection [2]. The greatest navigation risk is large objects — "shipping containers, floating or partially submerged vessels" — in coastal shipping lanes; houses were "believed to be very low" probability survivors of the crossing; invasive-species transport is a real but unquantified vector [2]. NOAA's public guidance said the same in plainer language: harmful radiation is "highly unlikely," tested items "known to be from the tsunami" showed "no radioactive contamination above normal levels" [16]. A severe debris event, the reports conclude, is a chronic problem delivered all at once — and the next section follows the debris field's modelled road to one coastline in detail: British Columbia.

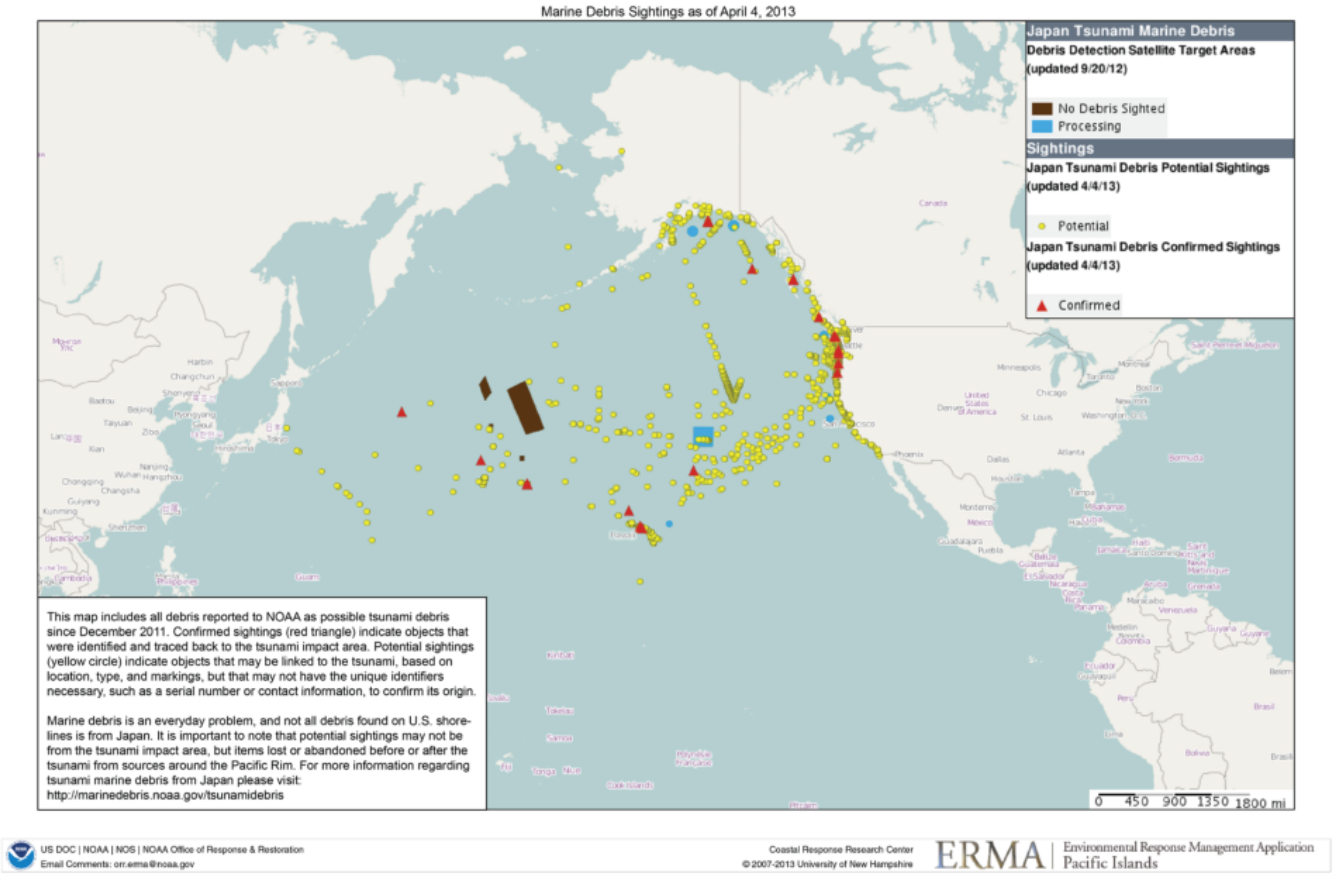


Exhibit. Marine debris sightings as of April 4, 2013: potential Japan-tsunami debris sightings (yellow) and confirmed, traced objects (red) along the North American coast (NOAA ERMA) [S0011]. Source: S0011.

Transport of tsunami debris to the coast of British Columbia

British Columbia is where the tsunami debris story becomes a forecasting problem with a coastline attached. The Canadian Science Response review (updated March 2013, superseding the original SSRP) exists because the first advice went stale: the original simulations assumed "objects with essentially zero windage" and climatological winds, while the update uses "the actual wind forcing over the Pacific from March 2011 through August 2012" and models windage explicitly [2]. Its framing quantities set the scale: initial Japanese estimates of debris swept to sea ran "between 20 and 25 million tonnes," but the updated Government of Japan figure is "about 1.5 million tonnes of tsunami-generated debris remains afloat as of March 2012," the preponderance of it "construction material from houses that were swept to sea" — lumber above all [2] [2]. The review's first conclusion is compositional: "only the most buoyant and durable objects will survive the trans-Pacific crossing and reach North America," individual pieces ranging to "hundreds of tonnes or more," most of them far smaller [2].

The road itself is a three-current relay. Debris entered the sea along northern Honshu where "the Oyashio and Kuroshio Currents merge together... to form the Kuroshio Extension, a swift eastward-flowing current"; in the central Pacific the flow becomes "very broad and sluggish" as the North Pacific Current (West Wind Drift), which carries the field across [2]. Near North America the current "splits (bifurcates) into northeastward and southeastward flowing current systems... the Alaska Current and California Current," with drifter studies placing "the latitude of bifurcation is about 50°N," varying year to year [2]. The split is destiny: north-branch debris "is more likely to enter Canadian waters," south-branch debris the US coast and Hawaii [2]. But currents alone don't beach anything — "currents can only bring debris close to the shore. Debris is then deposited on the shoreline by the action of retreating tides, onshore winds, and waves," and the seasons gate the delivery: summer's northerly winds drive upwelling and offshore transport that "will help to protect the coast," while winter's southerlies create "downwelling conditions and onshore transport of surface waters which favours deposition" [2]. BC, in effect, receives its debris in winter.

Dispersion guarantees the field arrives as weather, not as a fleet. Eddy diffusivity in the Kuroshio Extension runs 5,000–10,000 m^2/s; at the low end, a cluster's dispersion scale reaches ~400 km in six months — "a circle with a radius of 400 km encompasses an area of about 500,000 km^2, which is comparable to the area of the province of Alberta (660,000 km^2) or the state of California (424,000 km^2)" — and within that expanse debris concentrates in

"patchy clumps and elongated filaments," not uniform spread [2]. Windage then sorts the cloud (an object's velocity is the sum of water velocity and a wind-driven slip, U = U_water + U_wind [2]). The review's conclusions divide the arrivals by class: "debris with essentially zero windage will begin to arrive along the west coast of North America during the first half of 2013," most of it turning south with the California Current, "a smaller portion… transported north into the Alaska Current, and it is from this load that low windage debris is most likely to wash ashore in British Columbia… for a number of years" — while "the great bulk of low windage tsunami debris" stays at sea, "collecting in the North Pacific Garbage Patch" [2]. Debris "caught in the Garbage Patch" is "unlikely… subsequently [to] reach the coast of British Columbia" [2].

BC's early, dramatic share was the other class. High-windage debris "tends to follow a more northerly course than does low windage debris," so "the coast of British Columbia [received] considerable amounts of relatively high windage debris" in the first year — "although such debris may not represent a large fraction of the total debris load, it is nevertheless potentially significant since very large objects that are potentially hazardous to navigation tend to have appreciable windage," naming the Haida Gwaii container (April 2012) and the ghost ship found off Haida Gwaii on 20 March 2012 (windage ~3.5%) [2] [2]. The forecast's practical limits are stated plainly: models "are not sufficiently accurate to identify specific shoreline locations in British Columbia of tsunami debris deposition," so the guidance is climatological — "shorelines that receive debris at present (e.g., collector beaches), particularly those found along the west coast and around Haida Gwaii are also likely to receive tsunami debris," existing deposition patterns "not expected to change greatly," and the Strait of Georgia protected by "seaward flowing near-surface currents at the mouth of Juan de Fuca Strait and the barrier presented by the Gulf and San Juan Islands" [2] [2]. Since most arriving debris is unmarked small pieces, "the only indicator of tsunami debris may be an increase in the quantity of debris (by weight) washing ashore relative to the long-term average" [2]. The next section zooms out from BC to the full arrival record — what actually came ashore, state by state.

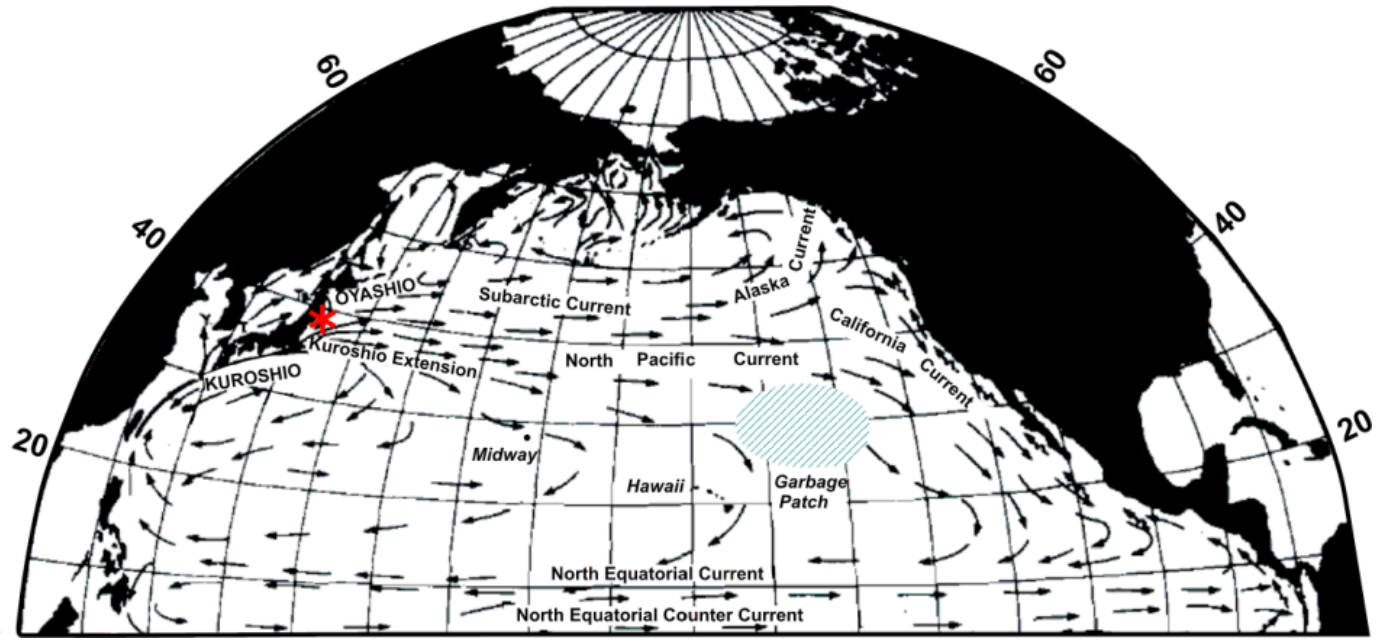


Exhibit. Figure 2. Schematic diagram showing major currents of the North Pacific. Source: S0012.

Tsunami debris arrivals and sightings by state

Read the arrival record as a map and it draws the circulation of Section 16 in objects. NOAA's confirmed and notable items list — 26 entries in the Severe Marine Debris Event Report's appendix — is dominated by derelict vessels (ten of them, scattered from the open North Pacific to Midway Atoll), two floating docks, and a scatter of personal-scale artifacts: "soccer ball found in Alaska," "volleyball found in Alaska," "soccer ball found in Washington," "Harley-Davidson motorcycle in a container, found in British Columbia," "steel tank found in British Columbia," "blue plastic bin found in Hawaii," "large yellow buoy found in Hawaii," "plastic sign found on Midway Atoll" [16]. Confirmed tsunami debris has been found "on the shores of Alaska, British Columbia, Washington State, Oregon and Hawaii" [2], while reports — 1,385 by October 2012 — also came from California, which contributes reports but no headline confirmed objects [2]. The distribution, the BC review warns, is "likely more representative of the frequency of human passage rather than the actual distribution of the tsunami debris" [2]: states appear in the record in proportion to their beachcombers as much as their beaches.

Alaska got the first wave and the strangest cargo. The Alaska cleanup plan (Marine Conservation Alliance Foundation, which ran 83 cleanup projects removing "more than 2.1 million pounds of debris" from Alaska's coast in 2003–2011) records the chronology problem plainly: currents move "at approximately six miles per day," so models projected landfall "in the spring of 2013. However, debris identified as possibly being generated by the tsunami made landfall on the west coast in December 2011, eight months after the tsunami," with sightings following in Washington and BC; Ebbesmeyer attributed the early arrivals to high windage, and Gulf of Alaska winter storms "coincided well with the movement of the debris… to send it towards Alaska" [11]. What Alaska actually received was foam: "A large amount of the debris appears to be Styrofoam. Some of it is in the form of buoys while the rest appears to be insulation from buildings. This is a particularly bad type of debris as it will easily break apart on the beaches and be ingested by animals" [11] — with listed species at added risk, "the Steller sea-lion and short-tailed albatross" [11]. The GPGP study supplies the matching negative: foam is negligible inside the patch while Alaska cleanups remove it in abundance, confirming that high-windage material exits to shorelines instead of accumulating [10]. Alaska also hosted the event's only sinking: the plan's "Ghost Ships" passage records that "the Japanese Ryou-Un Maru, a 164-foot shrimp vessel, was sunk by the U.S. Coast Guard in April, 2012 off the coast of Alaska in 6,000 feet of water," and recommends the same treatment for similar vessels — "the alternative is to allow it to shipwreck on shore, causing environmental damage followed by a costly removal" [11]. The ghost ship of Section 1 thus ends not as a wreck but as a deliberate deep-water disposal, a hazard removed before it could become a derelict on a beach.

British Columbia's entries are the high-windage vanguard of Section 16: the motorcycle container at Rose Spit (18 April 2012), a derelict vessel, and the steel tank [16] [2]. Washington's record mixes the intimate and the monumental — soccer ball, blue plastic bin, derelict vessels, and the second floating dock (the Misawa dock, in the Olympic Coast National Marine Sanctuary/Olympic National Park) [16] — and Washington is the state whose paperwork survives best: a state response contingency plan developed with NOAA and tribal input, "17 presentations at community public forums including Indian tribal meetings," more than $100,000 spent by February 2013 "on expenses related to JTMD including outreach, a dedicated response phone line, JTMD cleanup and removal operations, and biofouling or Aquatic Invasive Species assessment," plus NOAA-led shoreline monitoring at 11 sites [16]. Oregon contributed the Agate Beach dock (5 June 2012) and two derelict vessels [16] [2]. Hawaii and Midway, at the gyre's southern reach, collected the long tail — blue plastic bins, a large yellow buoy, multiple derelict vessels (one "broken into pieces"), a plastic sign on Midway [16] — consistent with the models' southward veer of moderate-windage debris toward the archipelago (Section 8) [2].

The record's real lesson is bookkeeping. With only 14 objects definitively confirmed against 1,385 reports, and most arriving material unmarked, each state's "tsunami debris" total is an artifact of its monitoring: Alaska had MCAF's monitoring programme and long-term baseline sites; Washington had 11 NOAA monitoring sites and a contingency plan; BC, the BC review notes, "has not had a systematic shoreline monitoring program" at all [2] [11] [16]. The arrivals by state are therefore also a census of readiness — which is the subject of the next section: the cleanup planning and public outreach built, state by state, while the debris was still in transit.



Exhibit. Tsunami debris arrivals: a barnacle-encrusted Japanese fish bin (Y.K. Suisan Co.) recovered in Hawaii, and the Harley-Davidson motorcycle that drifted across the Pacific from Japan (NOAA) [S0011]. Source: S0011.

Clean-up planning and public outreach

The tsunami debris response was planned in the interval between the models' two arrival waves — after the high-windage vanguard of 2011–12, before the current-driven main body expected from spring 2013 — and its documents are unusually candid about method. Alaska's plan (Marine Conservation Alliance Foundation, 2012) starts from the state's hard constraint: a coastline too "vast and inaccessible" for ground survey, so "aerial surveys are essential to help prioritize areas by amount, location and type of marine debris," flown in small aircraft with GPS-referenced cameras, the pilot marking priority stretches and a technician logging photography and position data [11]. The State of Alaska began initiating such a survey in June 2012 [11]. Prioritization converts directly into procurement: the coast is divided into segments let by Request for Proposals — "proposers should be required to submit bids for entire segments" — with contractors responsible for removing all debris, "weighing and categorizing the debris as to type and if it is tsunami debris (if possible)," and disposing of it properly, under HazMat standards where fuel and chemical containers appear [11]. The RFP "should not specify means of cleanup": the plan buys outcomes, not methods [11].

Urgency was material-specific. Because "much of the high-windage debris is Styrofoam, it is critical for the environment to clean the shores as soon as possible before the winter storms break them to pieces making them nearly impossible to remove" [11] — the fragmentation clock of Sections 5 and 11 running on a beach, on a deadline of one storm season. For the current-driven wave, the plan splits the work by size. Small and mid-sized objects — "refrigerators, commercial fishing nets, and even bowling balls are just a few of the small and medium-sized debris items expected to make landfall" — are a manpower problem: "primarily low-impact human power... most sites on rocky beaches are limited to human power and hand tools," ATVs and trucks only where possible, executed by local partners who are "experts in their region who understand the land, water, protected species, weather, and necessary safety precautions," inside a working season that runs only "from May to mid-September due to wind, sea, and sea-ice conditions" [11]. Project selection runs on two tracks — open solicitation in the impact zone (local knowledge, community priorities, cheaper) and pre-identified segments from the aerial survey and historic catchment knowledge, ranked by concentration, environmental harm, habitat sensitivity (rookeries, breeding grounds, endangered species), and accessibility [11]. Large items get their own doctrine — "Large-Item Strike Teams" of salvers, helicopters, and landing craft, because off Alaska's road system nothing else moves a house, car, or the white transport container that washed ashore at Naikoon Provincial Park, Canada — in three steps: "1) surveying to catalog GPS coordinates; 2) prioritize and map a collection route; 3) contract with salvers, landing craft, helicopters," with the economical play being to chart landings for years and mount the big removal in 2015 or 2016, resurveyed against the 2012 baseline [11] [11]. Current-driven debris, the plan notes, "may stay with the current and pass by Alaska shores and continue to circulate until it lands at a later date. It may also follow currents into the Bering Sea" [11].

Public outreach was the other half of the system, because the public was the sensor network. NOAA's trifold brochure reduces beachcomber doctrine to four verbs — "GRAB IT" (small items, watch sharp edges), "TOSS IT" (dispose/recycle properly), "BE CAREFUL" (large items "may contain hazardous material such as oil, which requires an expert to remove"; contact lifeguards or rangers), "REPORT IT" — reports, with location, date, and description, to DisasterDebris@noaa.gov [16]. The same brochure performs expectation management: "Not all debris found on U.S. shorelines or at sea is from Japan, so please use your discretion when reporting items"; on radiation, "it is highly unlikely that any tsunami-generated marine debris will hold harmful levels of radiation," and tested items, "including items known to be from the tsunami," showed "no radioactive contamination above normal levels" [16]. The infographic adds the protocols with human weight: items with marine life attached should be reported immediately (invasive species), and it is "highly unlikely that any human remains from the 2011 disaster will appear on U.S. shores" [16]; personal items, handled through the State Department, went to Japanese consulates to try to identify owners [16]. Detection itself was crowdsourced statistics: since tsunami debris is indistinguishable from everyday debris without identifying marks, "one of the few ways we will know when tsunami debris arrives is if the amount and type changes," tracked against monitored baseline beaches [16].

Underneath the planning sits the legal reality of Section 3, documented in the report's own authority appendix: removal authority rests with the Army Corps of Engineers (Rivers and Harbors Act §19), USCG, and EPA for hazardous debris under CERCLA/CWA/OPA and the National Contingency Plan, but the coastal/inland boundary "in practice... has been case-by-case," and where debris is non-hazardous solid waste, CERCLA authority is limited — "it may be most appropriate that debris determined to be non-radioactive solid waste, be addressed through state solid waste management authorities" [16]. The Marine Debris Act's contribution is programmatic: it "authorizes EPA and NOAA to develop a public education program for outreach and volunteer monitoring" [16]. The report's summary (May 2013) concedes the shape of the whole effort: "the majority of direct federal agency efforts have focused on data collection and monitoring work conducted as part of the agencies' existing activities and responsibilities," with historical

baseline data “especially valuable” precisely because tsunami debris hides inside the normal debris stream [16]. Planning, in other words, mostly meant watching well, asking the public to watch too, and pre-positioning contracts for what the watching found. The paper now turns from an event with too much debris at once to the chronic version of the same problem — the derelict vessels that accumulate one abandonment at a time.



Exhibit. Marine debris notice used in public outreach: pick up small debris, never touch large or hazardous items, and report finds to authorities (NOAA) [S0011]. Source: S0011.

Abandoned and Derelict Vessels: The Modern Ghost Fleet

Abandoned and derelict vessels defined

Section 3 introduced the category; the Blue-Ribbon Program report (Pacific States/British Columbia Oil Spill Task Force, January 2020) is where the category gets its working machinery. The report's premise is that "abandoned and derelict vessels (ADVs) are a growing global problem that harm aquatic health and the health of humans and wildlife; threaten commerce and navigation safety; and deplete resources that communities depend upon" [5]. Through "deliberate action or negligence, ADVs break up, sink, or block navigation channels," leaking "oil, lubricant, and other toxic substances" — including persistent organic pollutants that "accumulate in animal fat, becoming concentrated in top predators like orca whales and otters" — while sunken hulls "scour or crush sensitive habitats like eelgrass beds and kelp meadows" [5]. The definitions themselves live in the report's Appendix A ("For a complete list of definitions used in this report, including 'abandoned' and 'derelict', see Appendix A"), but the operative definition is functional and appears in the authority discussion: authority is "the legal ability of a governing agency to declare a vessel 'abandoned' and thus remove and dispose of it" [5] [5]. A vessel is abandoned, in the only sense that frees anyone to act, when an institution with jurisdiction says so through a legal process — and the report's central finding is that "a legal process for an aquatic land custodian to accomplish seizure, impoundment, removal or custody of an ADV through due process is currently lacking across most west coast jurisdictions," with authority needing to be "exacting in procedure but flexible in its application" [5].

The White Paper conclusions the definition must serve are by now familiar (Section 3): the problem spans commercial and recreational vessels; "the majority of ADVs are recreational, yet commercial vessels… can cost several orders of magnitude more than recreational vessels to remove"; a "legacy" backlog grows alongside new abandonments; cars and vessels are treated inconsistently; "in the US, there is no comprehensive federal program"; only Washington's state program is comprehensive; Canada's federal Act is "comprehensive yet underfunded"; and "no jurisdiction has a comprehensive outreach and education program" [5]. The five programme elements follow: Authority, Prevention, Public Outreach and Education, Removal and Deconstruction, Funding — the skeleton for Sections 20–24 [5].

Where definitions bite is at their edges, and the report catalogues five gaps in state and local authority, each a failure of a definition to cover a real case [5]. First, a "narrow focus on recreational vessels only" leaves commercial ADVs — fewer, but costlier and with "a larger single point of impact on the environment" — outside removal authority. Second, a "narrow focus on reasons for removal": statutes limited to emergencies or threats to human safety leave "no room for flexibility" or for removing nuisance vessels before they become emergencies. Third, geography: authority limited to "public aquatic lands only" abandons private landowners to "lost property or trespass laws." Fourth, local authority: state-level approval requirements mean "even if a local agency wanted to fund and remove an ADV, they need state approval if the vessel is on state-owned aquatic lands." Fifth, private owners "usually cannot access state funds" and "do not actually have the legal authority to remove the vessels" on their own property — with one exception: "Alaska… authorizes private property owners to declare vessels abandoned or derelict, and therefore subject to removal" [5]. At the federal level, five agencies hold pieces of ADV authority — NOAA, USACE, USCG, EPA, FEMA — but as Point Estero showed, "federal authorities and responsibilities frequently end once contamination is" removed, leaving the hull behind [5].

Two neighbouring definitions sharpen the boundary from either side. On the near side is the "at-risk" vessel: Florida's Vessel Turn-In Program (created July 2022 under its Derelict Vessel Prevention Program) accepts only vessels that are "floating on waters of the state," in "an At-Risk condition," with clear title and no liens, whose owners have received a citation or warning — and states the boundary explicitly: "Derelict vessels are not eligible for this program" [17] [17]. The West Coast turn-in programmes make the same cut: they dismantle vessels "that do not yet satisfy the definition of 'derelict' or 'abandoned', but are likely to become derelict or abandoned in the near future" [5]. On the far side is admiralty's abandonment doctrine for wrecks, where courts refuse to let time do the declaring: in Sea Hunt, the appeals court "significantly stressed that '[t]he mere passage of time since a shipwreck is not enough to constitute abandonment,'" the Fairport court treated lapse of time as "one factor among several," and US courts have held the United States has not abandoned warships sunk in the Civil War more than a century ago [18]. Between the at-risk boat that is not yet derelict and the warship that is never abandoned, "derelict" is less a description of a hull than a decision about one — which is why the next section treats the Blue-Ribbon Program itself: the attempt to make that decision systematic across five states.



Exhibit. A decommissioned Coast Guard vessel in public custody on stands — abandoned and derelict vessels pass into public hands for disposal (ADV Blue-Ribbon Program report) [S0014]. Source: S0014.

The Blue-Ribbon Program for the western states (AK, CA, HI, OR, WA)

The Blue-Ribbon Program is the White Paper's diagnosis converted into a blueprint: 33 recommendations across the five elements (Authority, Prevention, Outreach/Education, Removal and Deconstruction, Funding), drafted by the ADV Workgroup of programme leads from Alaska, California, Hawaii, Oregon, and Washington, and aimed at state legislatures — Canada is excluded from the recommendations because ADVs there "are addressed primarily at a federal, and not a provincial, level" [5]. Adoption "of each recommended element… would result in a comprehensive ADV program that would successfully remove legacy ADVs and prevent and remove future ADVs," with implementation varying by state [5]. The programme begins by fixing the vocabulary (its Appendix A glossary): "**Abandoned** — the legal condition of the vessel, in terms of ownership. **Derelict** — the physical condition of the vessel, in terms of seaworthiness" [5]. The distinction does real work throughout: a vessel can be derelict but owned, abandoned but sound; "commercial/recreational" classifies "the original intent of the vessel usage as built (e.g. an old tug converted to a houseboat would be a commercial vessel)," while "seaworthy" means "physically fit and in full working order, able to encounter and withstand the ordinary perils of the sea during its contemplated use" [5].

On authority, the recommendations generalize Section 19's gap list into powers. Agencies should be able to dispose of removed vessels for best public benefit — sell, deconstruct, recycle — with statutes that do "not encourage or prioritize the sale of removed vessels regardless of condition," because resold wrecks are "abandoned or derelict multiple times under multiple owners"; proceeds reimburse the removing agency, the remainder funding future removals [5]. Removal authority must cover "any vessel, whether commercial or recreational," judged on "current condition and situation, not on its as-built intended use and/or ownership status" — the programme's exhibit is Washington's DVRP taking custody of a former USCG 41-foot utility vessel legally owned by the Bureau of Indian Affairs, selling it, and depositing the funds in the Derelict Vessel Removal Account: the federal vessel "posed the same threat to human safety and the environment that a similarly sized recreational vessel posed" [5]. Private property owners should be empowered to declare vessels abandoned or derelict (the Alaska model), because "many con-

taminants from ADVs migrate by ocean currents and sediment movement or via the food chain" beyond the property line [5]. And ticketing authority for registration enforcement should extend to state agency personnel [5].

Prevention is designed around why vessels actually become ADVs: "aging and weathering; vulnerability to neglect; technological changes; owner inability to keep up with maintenance costs; maintenance costs exceeding the commercial value of the vessel; damage following an incident exceeding the value of the vessel; federal or state sponsored fishery reduction and fishery disaster relief programs that render a vessel's original purpose obsolete" [5]. The cost asymmetry of delay is priced by the F/V Western: a 69.9-foot, 78-gross-ton wooden crabber from 1934, denied moorage at Coos Bay in 2014 for its condition, anchored unauthorized, beached twice in storms, and "finally sank in 2015 near a busy navigation channel in sensitive fish and invertebrate habitat. While floating… [it] would have cost $30,000 to remove; once sank, it cost $95,000" [5]. Against the White Paper's prevention gaps — no registration requirements, no insurance requirements "especially for wreck removal," no turn-in programmes — the model programme specifies four elements: a registration system, an ADV database, insurance requirements, and a vessel turn-in program [5]. Registration is modelled on motor vehicles (and on oil/tire disposal fees): annual registration and renewal for recreational and commercial vessels alike, fees funding the programme; Washington's is "the most established program for registration revenue collection," though its commercial criteria should expand beyond vessels that linger — it currently misses "large container ships that do one port of call per year" [5].

Prevention also reaches the marina lease. States should shorten lease terms on state-owned aquatic lands (workgroup suggestion: five years): Oregon's 15-year leases make compliance checks so infrequent the state had to mandate site visits every three years, while WA DNR "has recently switched to shorter lease terms," and non-renewal is "less legally problematic" than terminating an active contract [5]. States should also stop feeding the problem: agencies auctioning their own dilapidated surplus vessels "at extremely low prices" contribute future ADVs — California's OSPR was required to auction an older vessel; Washington's restrictions (RCW 47.01.470) are the counter-model [5]. Outreach, the third element, starts from a null baseline: "none of the ADV programs within the Task Force jurisdictions have a comprehensive public outreach/education component"; programmes rely on websites, social media, and partners, do little target-audience research, and lean on NOAA's Marine Debris Program as "a go-to resource for funding ADV outreach and education efforts" [5]. Implementation is itself planned: after finalization, the Task Force will meet federal agencies individually, host an expert round-table, build a tailorable outreach suite ("Spills 911"-type materials distributed through networks such as Dockwalkers, plus materials for legislators), and "report on implementation/results" [5]. The next three sections take the elements in order: authority's remaining gaps, prevention through registration, fees, and enforcement, and the removal machinery itself.



Exhibit. The derelict F/V Western at anchor — the class of abandoned vessel the Blue-Ribbon Program was created to remove (ADV Blue-Ribbon Program report) [S0014]. Source: S0014.

Gaps in authority: no legal process for seizing and removing derelicts

Section 19 named the defect; this section maps it agency by agency. The White Paper's finding, carried into the Blue-Ribbon report, is that "a legal process for an aquatic land custodian to accomplish seizure, impoundment, removal or custody of an ADV through due process is currently lacking across most west coast jurisdictions" [5]. What exists instead is a mosaic of partial mandates, tabulated in the report's federal-authorities survey [5]. NOAA's primary ADV focus is "within National Marine Sanctuaries," and while its Marine Debris Program funds removal grants on public and private lands, ADVs are only one kind of debris to it: the grant programme "cannot be considered a true ADV-removal program because it includes all marine debris, not just ADVs" [5]. The Army Corps of Engineers "addresses ADVs only in federally recognized navigation channels, and only if the vessel impacts the maintenance or navigation of the channel" — and, in the table's bleakest cell, is "authorized to remove the vessel, but has no funding to do so" [5]. USCG (coastal) and EPA (inland) "remove pollution threat from vessel where there is an environmental or a public health threat": petroleum removal draws on the Oil Spill Liability Trust Fund, but "vessel removal funds would have to come from the agency's budget," and "neither agency has dedicated ADV removal funding" [5]. FEMA's authority "arise[s] from [the] Robert T. Stafford Disaster Relief and Emergency Assistance Act" and it "can only fund ADV removals under declared emergencies" [5]. Every federal door, in other words, opens onto a different room: sanctuary, channel, pollution threat, declared disaster. A derelict that is merely rotting — the common case — fits none of them.

The state-level gaps (Section 19) compound the federal ones: recreational-only scope, emergency-only triggers, public-lands-only geography, local governments blocked by state approval, and private owners without authority or funds [5]. The report's first authority recommendation is therefore structural: give the aquatic land custodian power to "remove a hazard, nuisance or threat while protecting the vessel owner's rights and due process," in a programme "strict in process and unrestricted in capability" — "legal authority should be free of gaps in jurisdiction, clear in process, and have the ability to adapt to changing circumstances" [5]. Process matters as much as power: "the process by which state agencies gain the legal right to remove, deconstruct, sell or use a vessel should be clearly spelled out in state statute in straightforward language comprehensible to any agency staff member" [5]. And prioritization must not become a veto: because resource-limited states rank removals, "local authorities should have a voice in removal prioritization and, when appropriate, should have authority to seize, impound, remove or gain custody of either recreational or commercial vessels" — letting a city or owner act "based on local needs and not compete with other localities for state or federal approval" [5].

Washington — the model state — exhibits both the fix and the residual gap in one example. Its DVRP prioritizes by threat, but "Washington State's statutes allow any authorized public entity to remove vessels within their jurisdiction. This gives local entities, such as ports, the ability to remove a vessel that is a low priority for the state" [5]. Yet "Washington is also an example of limiting authorities of private property owners. Counties typically have ADV removal authority on private property, but if the county refuses to exercise its authority, the property owner cannot use the State program to remove the vessel" [5]. A county's refusal thus strands the owner with the derelict — the Point Estero pattern (authority present, action absent) reproduced at kitchen-table scale [5]. The JTMD appendix tells the same story federally: boundary disputes between EPA and USCG resolved "case-by-case," and non-hazardous debris defaulting to state solid-waste authorities because the federal trigger is pollution, not dereliction [16].

The extreme end of the authority problem is worth naming here because later sections live in it: for ancient sunken warships, even ownership can be indeterminate — "What is the modern day flag State of Etruscan vessels? Greek, Italian?... what about the Phoenician fleet? Must it be considered Lebanese, Syrian, Tunisian or, even, Spanish?" — which is why the UNESCO Convention works through an "interested State" with a "verifiable link, especially a cultural, historical or archaeological link," entitled to be informed or consulted rather than to seize [18]. Between a Phoenician wreck no state can claim and a fiberglass sloop no agency can touch, the legal problem is identical in shape: dereliction is common, but the power to end it is rare, fragmented, and usually unfunded. The next section takes up the cheaper half of the cure — preventing vessels from entering the gap at all, through registration, fees, and enforcement.

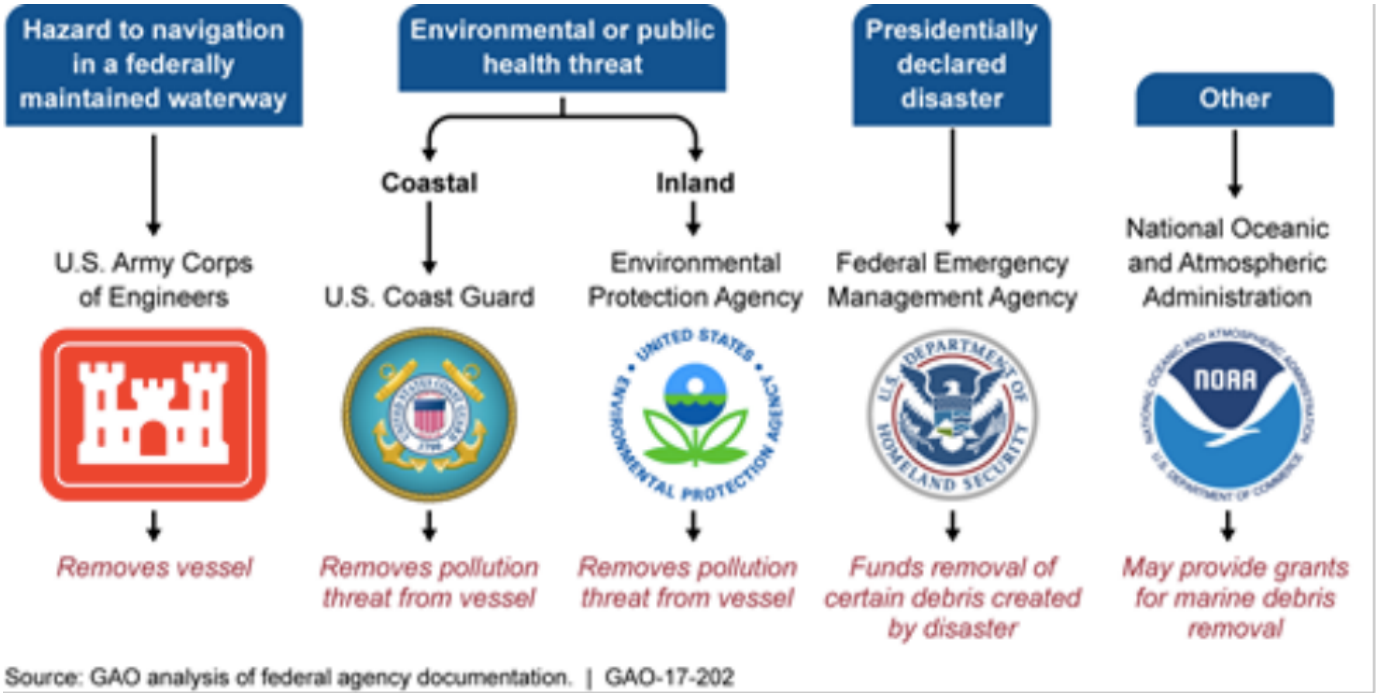


Exhibit. Figure 1. U.S. Federal Agency ADV Authorities State and Local Authorities. Source: S0014.

Prevention through registration, fees, and enforcement

Prevention is the Blue-Ribbon Program's cheapest element, because it attacks the information problem underneath everything else: most jurisdictions cannot say which vessels they have, who owns them, or which are sliding toward dereliction. The first recommendation is a registration system "including fees and adequate enforcement, for both recreational and commercial vessels" — modelled on motor vehicles and on oil and tire disposal fees, with annual renewal, so that every vessel has a current owner of record and the fees themselves fund the programme [5]. The second is surveillance with a human face: "a comprehensive database to track and (potentially) prioritize ADVs," holding "location, condition assessment, and (to help with disposal decisions) prioritization" for vessels of concern [5]. Reporting should be "seamless and allow for immediate reporting from the field" — a free mobile app or hotline with a consistent question set — with two cautions built in: the process should never encourage "the public to board or enter ADVs," and "while a vessel may appear abandoned to the reporting party, the vessel owner may not agree. Tracking ADVs is a related but separate subject from vessel ownership and seizure" [5]. Marinas close the information loop: moorage owners should be required to "collect and maintain vessel and owner information for annual submittal to the state titling and registration authority" [5]. Prioritization itself is a resource doctrine — vessels ranked "based on risk, impact, and ease of removal," locals involved in the ranking (state lists can otherwise block local action), and an explicit efficiency rule: "it is not unreasonable to prioritize removal of multiple lower risk vessels if it can be done for the same cost as removal of a single mid or higher risk vessel" [5]. The economic engine of the whole system is one ratio: "removal costs are often three times more for sunken vessels than removal of floating vessels," so every vessel caught early is a two-thirds discount [5] (Section 20's F/V Western — $30,000 afloat, $95,000 sunk — is the case in point [5]).

Insurance converts an owner's future default into a priced, private obligation. The headline recommendation: "require wreck removal insurance above the value of the vessel for both recreational and commercial vessels" [5]. At the marina level, the instruments already exist in pockets: surety or performance bonds for vessel removal and repair (the Port of Astoria requires them for repair work, costs passed to slip renters), the Port of Port Townsend requiring insurance or a bond before haul-out, the Port of Bremerton requiring tenants to name the Port on the policy "so the Port will be notified if the insurance is canceled. This in turn would cancel the vessel owner's tenancy" [5]. For industrial lessees (fish processing, vessel repair, barges and tugs) bonds should reflect cargo reality — bunker C fuel, lead, PCBs, asbestos [5]. Two structural proposals attack the end-of-life cascade in which commercial vessels are "sold or given to progressively smaller and smaller companies... until eventually the owner is unable to pay for vessel upkeep": a bond for disposal costs "during initial construction and registration," charged to "the initial owner, who profits from the vessel," and secondary-liability laws with pre-sale seaworthiness surveys for larger, older vessels (suggested limits: 65 feet, 40 years), so that an unseaworthy vessel "may only be scrapped or repaired," and a seller who transfers it anyway "may be liable if the vessel becomes abandoned or derelict in the future" [5]. Washing-

ton is “the only West Coast state with secondary liability requirements” (vessels 65 feet and over; DNR proposes lowering to 35 feet, the size beyond which vessels can’t be trailered) [5]. The cautionary tale is San Francisco Bay’s Black Kite: after the USCG spent $90,000 responding to the derelict 122-foot tugs Black Kite and Black Hawk, the Black Kite reverted to the City of Richmond, was auctioned by the US Marshals, and “was sold for $1 to a local artist. Within months of the sale, both California and the USCG were responding to the threat of an oil spill from the vessel” [5] — a dollar sale transferring a six-figure liability to the party least able to carry it, exactly what survey-plus-secondary-liability is designed to prevent.

Enforcement is where prevention budgets go to die, so the report funds it directly: states should “establish a mechanism for local law enforcement agencies to be funded or compensated for time and equipment needed to enforce vessel registration and aquatic laws, and to issue civil penalties,” and may issue grants to local law enforcement for ADV cleanup time [5]. Cost recovery completes the loop — agencies need authority “to recover the costs of government-funded vessel removal efforts from the current, and potentially past, vessel owners” [5]. Washington’s plumbing shows what a mature version collects: the derelict-vessel fee brings in “$800,000 annually from their approximate 250,000 recreational vessels and 1,700 commercial vessels” — $3 per recreational vessel per year, $1 per linear foot for commercial [5] — inside a boating-revenue system that raised $108 million in the 2021–23 biennium, 75% of it from taxes and registration fees [19]. California’s alternative apportions by use rather than ownership: a marine fuel surcharge raising “about $2 million per year” that helps fund its turn-in programme [5]. Prevention, assembled, is a paper trail with teeth: know the fleet, watch the fleet, price the risk onto owners and builders, and pay the people who enforce it. The next section covers what happens when prevention fails on schedule — removal, deconstruction, and the turn-in programmes that intercept vessels just before the line.

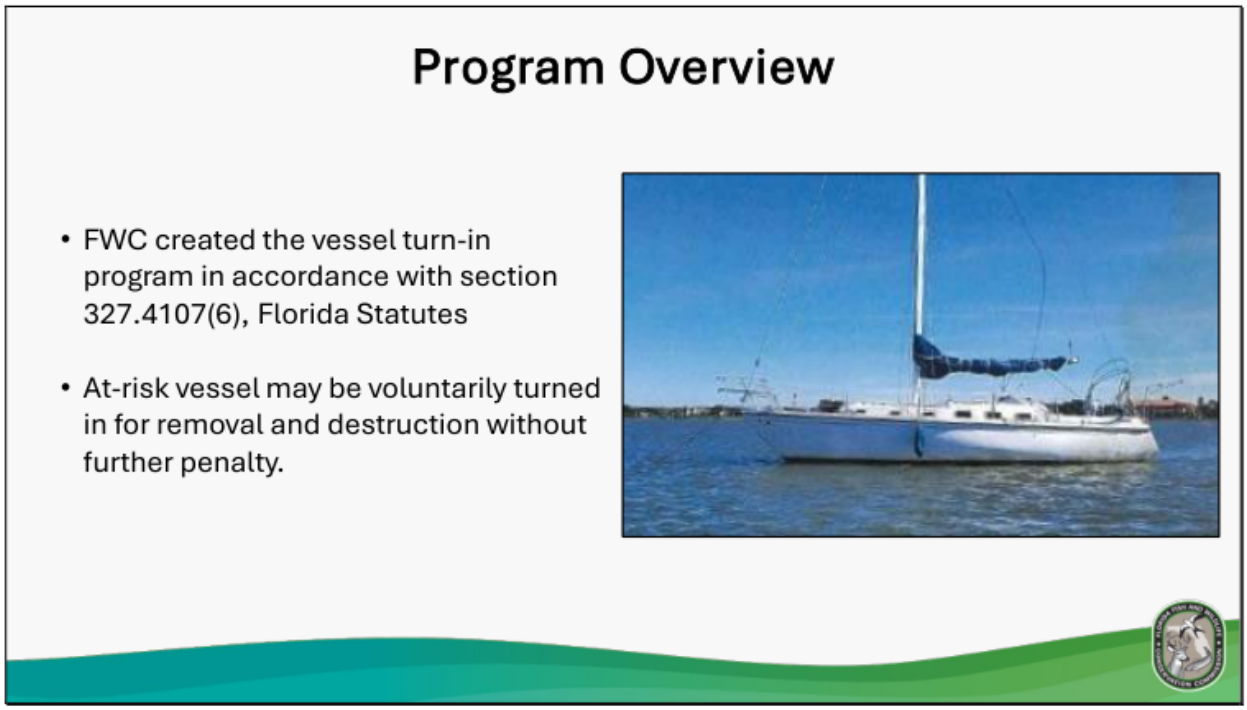


Exhibit. Florida’s Vessel Turn-In Program: an at-risk vessel may be voluntarily turned in for removal and destruction without further penalty (FWC) [S0016]. Source: S0016.

Removal, deconstruction, and vessel turn-in programs

Removal is where the ADV problem becomes a logistics industry, and the Blue-Ribbon report’s Section VI reads like an operations manual written by people tired of improvising. First, the contractor market: states should “publish and maintain a list of qualified (licensed, bonded, insured) vessel removal/recycling contractors” with statewide price agreements — Washington’s list is the model — because the default alternative, a formal RFP per vessel, is “labor-intensive and often yields no bids and meanwhile worsening condition for the vessel of concern,” while California’s emergency contracting procedures show the faster path [5]. Responders need liability protection; deconstruction needs permitted facilities, including for large commercial vessels, because ad hoc shipbreaking sites delay approvals and weaken hazardous-materials control [5]. The report’s most operational idea is a hand-off: “coordinate with USCG to establish a practice of ‘passing’ contractors from USCG employment during the pollution control phase of a response, to state or local control for the vessel salvage and deconstruction phase” — the equipment

already mobilized with federal money for oil removal stays on site for the state's hull work, cutting remobilization costs [5]. Geography should be exploited, not endured: "plan targeted local ADV removal events to leverage the cost of a single mobilization," and establish "safe and secure shore-side vessel storage" so hulls can be stockpiled for bulk deconstruction and waste shipped on "empty return barges or trucks… often available at lower rates" [5]. Disposal is the binding constraint: with EPA ocean-dumping permits restricted, "ADVs are most often deconstructed and disposed of in landfills," and "the nearest landfill [may be] too small to accept the volume of waste associated with vessel deconstruction, even for relatively small vessels" — worst in Alaska, where roadless communities make waste shipment "extremely expensive," arguing for negotiated community-landfill rates for vessels removed before they sink [5].

The cost curve explains the urgency. Removal costs are "highly variable, with a distribution heavily skewed toward relatively inexpensive removals": a small floating boat that fits a trailer "may cost as little as $100 to remove," the average recreational removal (Washington and California data) is "approximately $3,000," and "the cost to remove a large, sunken commercial vessel can reach well over $1 million" [5]. California's Sacramento–San Joaquin Delta inventory prices a regional backlog: "$30 million is needed to remove the 52 existing, known commercial ADVs," with 1–2 new commercial abandonments a year implying "approximately $4 million… on an annual basis" [5]. And the money mechanics exclude the poor: Washington, Oregon, and California fund removals but typically require "the lead agency pay for the removal up front and seek reimbursement later. This limits participation, excluding communities that cannot afford the initial costs" — hence the funding recommendation to move state money quickly and upfront (Section 24) [5] [5].

Vessel turn-in programs (VTiPs) are the deconstruction pipeline's intake valve — the state dismantles vessels that "do not yet satisfy the definition of 'derelict' or 'abandoned', but are likely to become derelict or abandoned in the near future," at "a small fraction of the cost compared to when they are abandoned" [5]. On the West Coast only Washington and California run them, recreational vessels only: Washington's, established in 2014, lets owners and marina operators apply for free disposal; it "received over 100 applications in its first two years," rehomed vessels where appropriate, and "destroyed broken-down vessels, avoiding thousands of dollars in future emergency response costs" [5]. Florida's VTIP shows the model at scale and in mid-repair. Run by the Fish and Wildlife Conservation Commission since July 2022 under the Derelict Vessel Prevention Program, it takes at-risk vessels from titled owners who have been cited or warned, walks them through junking the title, and contracts the removal and destruction — "voluntary and free to the at-risk vessel owner," sparing them fines, compliance costs, derelict-vessel criminal charges, and "thousands of dollars in removal cost reimbursement" if the vessel later has to be removed as derelict [17] [17]. The throughput is real: "501 applications and approved 295 vessels for removal… 261 vessels have been removed," with 90 more applications in the 2026 contract year [17]. The programme's failure mode is also instructive: boats kept sinking between approval and pickup, so proposed 2026 rule amendments clarify that "vessel owners remain responsible for maintaining vessels until the removal contractor takes custody," that "vessels that sink prior to removal become ineligible for the VTIP program" even if approved, and that a surrendered vessel that becomes derelict may still bring enforcement against its owner [17]. A turn-in programme, in other words, is a race between paperwork and buoyancy — the same race as the F/V Western, run at fleet scale. What funds the race is the next section's subject: Washington's boating accounts, the one place on the coast where the money side of the model programme actually exists.

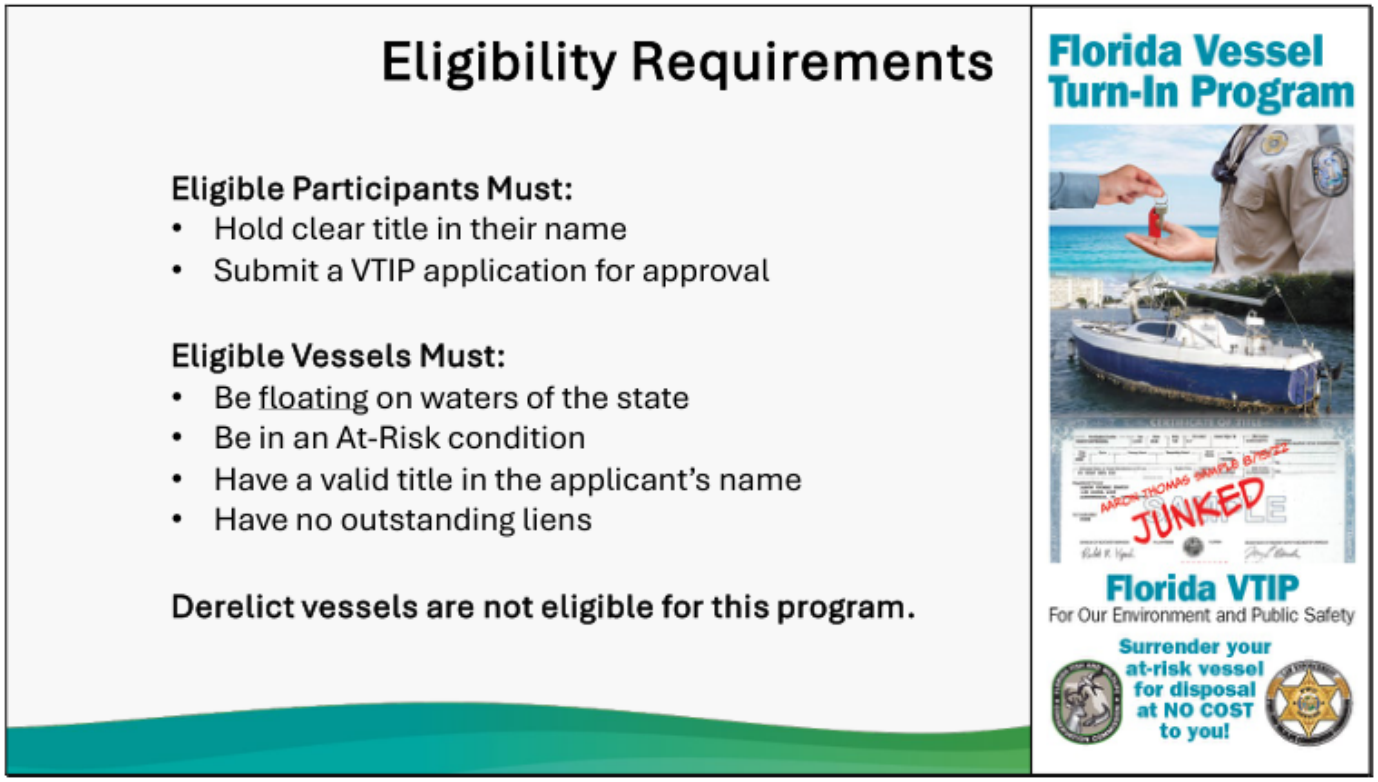


Exhibit. Staff will assist vessel owners in completion of the application and help them through the process of junking their title. Eligibility Requirements Eligible Participants Must: • Hold clear title in their name • Submit a. Source: S0016.

Funding the work: Washington State's boating accounts

Every prior section's gap reduces, in the end, to this one. The Blue-Ribbon report's funding chapter opens flatly: "Funding is a major obstacle for every Task Force jurisdiction," split into two distinct problems — money for the legacy backlog and money for the annual inflow — and "most jurisdictions have no dedicated ADV funding at all, and very few have funding to address both recreational and commercial ADVs" [5]. Washington is the exception that shows what a working system looks like, and the state's Joint Legislative Audit and Review Committee (JLARC) audited it in 2025. The architecture is statutory: derelict vessels have their own code chapter (RCW 79.100), boat registration its own (Chapter 88.02 RCW), and the watercraft excise tax its own (RCW 82.49) [19]. Around those anchors, Washington collected $108 million in boating-related revenue in the 2021–23 biennium from fifteen sources — 75% of it taxes and registration fees — deposited across 47 accounts, with 84% of revenue concentrated in just six accounts [19] [19]. Total recreational boating expenditures for the biennium were $85.89 million, 87% drawn from seven accounts [19].

The derelict-vessel money runs through three channels. First, the dedicated fee stream the Blue-Ribbon report cites: the derelict vessel removal fee yields "$800,000 annually from their approximate 250,000 recreational vessels and 1,700 commercial vessels" at $3 per recreational vessel and $1 per linear foot for commercial vessels [5]. Second, registration economics: statute directs "86% of boat titling and registration fees to support programs and activities related to recreational boating, such as derelict vessel removal and invasive species control," the remaining 14% to general purposes and administration [19]. Third — the step-change — the watercraft excise tax: 0.5% of a boat's fair market value (minimum $5), a rate "not changed since it was established in 1983," which until 2022 went entirely to the general fund; in 2022 "the Legislature changed the distribution. Beginning in fiscal year 2023, 25% goes to the derelict vessel removal account" [19]. The Blue-Ribbon appendix had modelled the end-state years earlier: DVRP funding options combining an 11–14% excise-tax share with $3–$4 recreational fees and $1–$3 per-foot commercial fees each land near $6 million per biennium [5]. A fourth, quieter channel is fuel: 1% of Washington's 49.4¢/gallon gas tax flows into the Marine Fuel Tax Refund account; boaters using at least 41 gallons can claim refunds, and the unrefunded remainder transfers to the Recreation Resources account, whose funds "must be used for recreational boating" [19].

The output side is measurable. In 2021–23, four agencies spent $21 million on environmental protection; DNR's $10.6 million share ran the Derelict Vessel Removal Program, and "program expenditures increased 367% after this increase in funding" from the excise-tax allocation; "during the 2021-23 biennium, DNR removed 226 boats from

Washington's waters and owners turned in 91 voluntarily" [19]. The same accounts fund the adjacent machinery: DFW's derelict fishing gear removal ($554,000) — officers "recovered 1,283 crab pots in Puget Sound, releasing 2,104 crabs trapped in the pots" in 2023, with contracted retrieval by the Northwest Straits Foundation — plus aquatic invasive species work ($4.7 million) and Parks' Clean Vessel sewage programme ($3.6 million) [19]. Roles are divided cleanly: DOL registers boats and collects the excise tax; DNR leases state aquatic lands and runs the DVRP; DFW handles derelict fishing gear and marine law enforcement; Parks administers the federal boating-safety and sewage grants; RCO runs grants and the boat.wa.gov portal [19].

Washington is thus the control case for the report's funding recommendations, which the other states mostly lack the plumbing to execute: move money to local agencies "quickly and upfront" rather than by reimbursement; build builder-pays disposal financing (best done federally, else by reciprocal state agreements); maintain "a reliable annual funding mechanism"; authorize cost recovery from current and past owners; and compensate local enforcement [5]. The contrast cases are California's use-based model (a marine fuel surcharge, ~$2 million a year, apportioned "based on vessel use, not vessel ownership") and the default elsewhere — "annual appropriations of general funds, or funds associated with clean water programs," i.e., no dedicated stream at all [5]. The lesson Washington teaches is boring and decisive: derelict removal happens at the speed of its revenue line. A $3 fee and a redirected excise tax — unglamorous, permanent, hypothecated — removed 226 boats in two years, while the unfunded authorities of Section 21 removed none. The paper now leaves vessels whole and turns to the gear they shed: abandoned, lost, or discarded fishing gear, the derelicts' smaller and far more numerous kin.

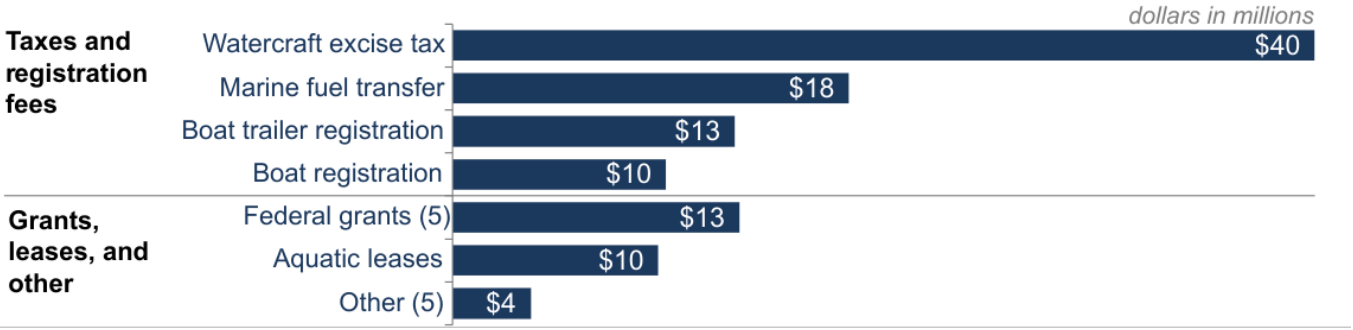


Exhibit. Figure 4: State agencies collected $108 million in revenues from seven primary sources (2021-23 biennium) dollars in millions Taxes and Watercraft excise tax $40 registration Marine fuel transfer $18 fees Boat trailer registration. Source: S0015.

Gear That Keeps Fishing: Abandoned, Lost and Discarded Fishing Gear

Magnitude and composition of ALDFG

Derelict vessels are counted in hundreds; derelict gear is counted in tonnes and kilometres. Breaking the Plastic Wave puts the global leakage of abandoned, lost or otherwise discarded fishing gear at "estimates ranging from 640,000 to 1,150,000 metric tons" per year, "expected to increase as a result of growth in fishing effort and aquaculture," and flags ALDFG as "among the most damaging" of all ocean plastic sources — damaging precisely because it was "designed to trap and kill fish," and because it leaks most intensely "in areas with high densities of marine wildlife, as this is where fishing is concentrated" [15]. The FAO's regional review supplies the texture under that headline. Loss is gear-specific: an estimated "29 per cent of lines are lost each year, 8.6 per cent of all traps and pots, and 5.7 per cent of nets," with gillnets "the highest risk of being lost," bottom trawls low, and purse seines and midwater trawls lowest — though no comparable data exist on how much of each gear type is produced or used, so tonnage shares remain estimates built on estimates [15].

The Baltic offers the best-instrumented small case. Under FANTARED 2, Swedish open-sea gillnetters targeting cod and turbot in 1998 lost about 1,500 nets a year — 155–165 km of netting, 3.6–3.8 nets per vessel — yet that was "less than 0.1 percent of nets lost per year," with fishers recovering only about 10% themselves; the main cause was gear conflict, and the eventual "ghost nets" surfaced in trawl hauls as long nets near the conflict area or small remnants scattered widely [6]. Norway's regulated fishery looks similar on paper — under 0.1% loss, recovery of 80–100% — "despite these reportedly low loss rates, between 1983 and 1997 the Norwegian net retrieval programmes recovered 6,759 gillnets targeting Greenland halibut," a reminder that small percentages of enormous fleets are still

thousands of nets [6]. Britain's tangle-net fleet lost 263 nets a year from 18 vessels (a third recovered); the hake fleet 62 from 12 (half recovered) [6]. Spain's Cantabrian fleet averaged 13.3 nets per vessel per year; the offshore rasco/monkfish métier lost 27.9 per vessel and 2,065 nets fleet-wide, driven by trawl interactions [6].

Proportional loss peaks in specific métiers: Brittany's wreck gillnetting lost "just under 3 percent of nets" — the largest share — while the flatfish and monkfish métier lost the most length, "just under 5 km of net per vessel per year"; on Portugal's Algarve, losses ran 3.2, 6.0 and 7.4 sheets per boat per year across local, coastal and hake fisheries [6]. The deepwater northeast Atlantic is the darkest corner: DeepNet found vessels that physically "are often not capable of carrying their nets back to port (the net stores are used to hold fish)," so "only the headline and footropes are brought ashore while the net sheets are discarded, being either bagged on board, burnt or dumped at sea"; anecdotal evidence from one shark vessel suggests "approximately 600 x 50 m sheets of net (30 km) are routinely discarded" per 45-day trip, extrapolating to roughly 1,254 km of sheet netting a year regionally, plus some 15 nets (750 m) lost per day in the deep-slope fisheries [6]. The Mediterranean, fished by vast numbers of small vessels, has one basin-wide estimate of 2,637–3,342 tonnes of gear lost per year, extrapolated from Turkish losses; France's 65-vessel offshore hake fleet loses about 0.2% of its nets annually (36–73 nets) [6].

The Pacific numbers are larger and stranger. A Korean survey found 18.9 kg of litter per hectare in fishing grounds, "83 percent of which was composed of fishing nets and related materials"; a six-month Incheon survey located 194,000 m^3 of debris weighing 97,000 tonnes, mainly fisheries-derived, and national cleanups recovered 10,285 tonnes of fishing debris from coasts in 2000–2006 [6]. Up to 1,000 tonnes of ALDFG are recovered from the Sea of Japan annually, "mostly bottom gillnets and pots… mainly of non-Japanese origin" [6]. In the North Pacific, a US estimate of 0.06% of driftnets lost per set compounds to "12 miles of net lost each night of the season and 639 miles of net lost in the North Pacific Ocean alone each year" [6]. Hawaii's debris is 83.6% trawl/seine netting, and NOAA and partners have removed "over 600 metric tonnes of ALDFG" from the archipelago [6]. In Australia's Gulf of Carpentaria, a gyre traps nets in "a repetitive cycle of fishing, being washed ashore and being washed back into the water during a storm or spring-tide event" [6]. Pots vanish at industrial rates too: Queensland blue swimmer crabbers averaged ~35 pots lost per year (over 6,000 fishery-wide, under half remaining in the environment); Bristol Bay king crab pot-loss estimates for the same fishery range absurdly — 31,600, 20,000, or 7,000 per year depending on the study — and British Columbia Dungeness traps were lost at 11% per year [6].

The Atlantic's western side shows how thin the record is: Canada's Atlantic gillnet fisheries were estimated at a 2% loss rate — 8,000 nets a year to 1992, 80,000 nets or sheets over 1982–1992 — with one 64 km^2 ground holding an estimated 2,240 lost nets [6]. The Caribbean's 2007 regional workshop could not even agree whether ALDFG was a significant local problem, what caused it, or whether most of it drifted in from outside [6]. That uncertainty is the section's real finding: the global tonnage is an extrapolation from a handful of FANTARED-era European studies, Korean and Japanese surveys, and North American pot counts — most fisheries on Earth have never been measured at all. The next section asks why gear goes over the side in the first place.

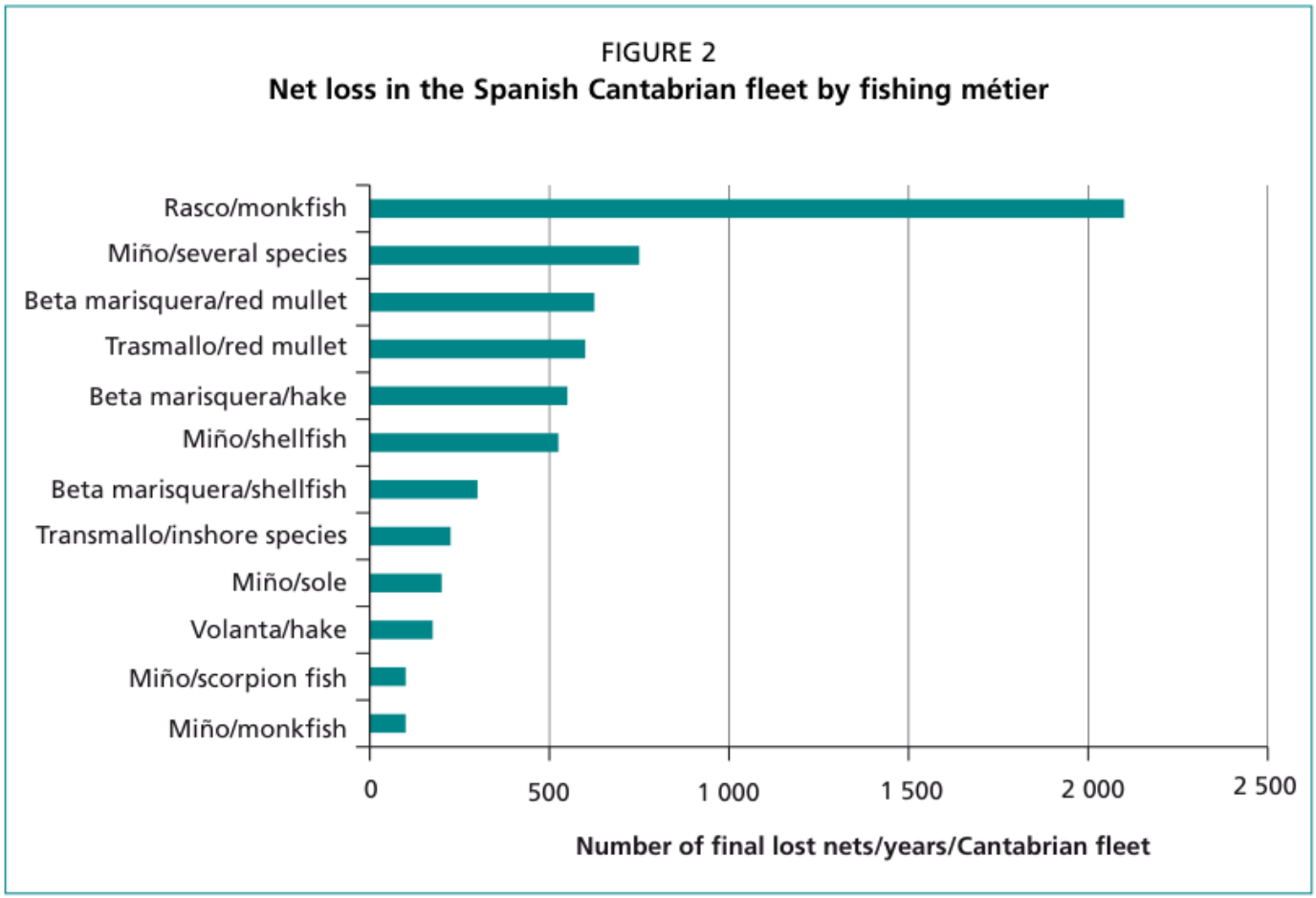


Exhibit. FIGURE 2 Net loss in the Spanish Cantabrian fleet by fishing métier Rasco/monkfish Miño/several species Beta marisquera/red mullet Trasmallo/red mullet Beta marisquera/hake Miño/shellfish Beta marisquera/shellfish Transmallo/insh. Source: S0017.

Reasons gear is abandoned, lost, or otherwise discarded

The FAO's causal chapter starts from the acronym itself: gear is abandoned, lost, or discarded — three different verbs, three different mechanisms — and concedes that "due to the environment in which fishing takes place, and the technology used, some degree of ALDFG is inevitable and unavoidable," with causes varying "between and within fisheries" [6]. Its taxonomy sorts direct pressures into four: enforcement pressure (illegal operators abandon gear to avoid capture), operational pressure and weather, economic pressure (dumping at sea beats paying for disposal ashore), and spatial pressure (gear conflicts), with indirect causes behind them — above all "the unavailability of onshore waste disposal facilities, as well as their accessibility and cost of use" [6].

The best-documented single cause is other fishermen. Gear conflict — "trawled/mobile gear passing through an area in which static gear is positioned" — takes whole fleets at once: in UK tangle-net fishing, "a whole fleet loss would be on average 30 nets," from fleets hauling some 12 km of netting a day; the vessels "patrol their nets at night but are not able to do this while hauling operations are ongoing," leaving gear exposed to any towing vessel [6]. Merchant shipping cuts the dahns and end ropes that mark static gear, "especially in areas of intense activity, such as the English Channel," which "can on occasion be cut leaving the entire fleet without any positional indication on the surface" — gear that is not lost so much as orphaned, unfindable by its own owners [6]. Technology has sharpened the conflict: hake netters in the Channel reported worsening losses because new trawl ground-gear plus "sonar, 3-D mapping software and differential GPS" let trawlers "fish within 25 m of wrecks" once safely inaccessible [6]. Where skippers talk, it works: Baltic losses fell with better communication between sectors, and a formal French–English "gentleman's agreement" allocating blocks to static or mobile gear, swapped every six weeks, "functions well and reduces gear loss considerably" [6]. For the most part, conflict loss is unintentional — one fleet's working day is another's shipwreck.

Weather and bottom do the rest of the accidental work. In the Western Channel, losses "were mainly caused by weather and bottom snagging": rough ground and strong currents hook nets fast; vessels "cut gear adrift for safety reasons (often in very bad weather conditions)" or after snagging obstructions they cannot free [6]. Old gear fails first

— a pattern the report finds from Sri Lanka to the Gulf of Mexico wire-trap blue crab fishery, where causes include “deterioration of buoys/lines/knots, negligence in assembling and maintaining gear, and the use of plastic jugs/bottles as floats”; the deeper cause is economics, since “wherever fishing activity is financially marginal there may be a reluctance or inability of fishers to invest in upgrading the fishing gear they use” [6]. Snagging can kill the vessel, not just the gear: on 10 April 2002 the FV Radiant, fishing 45 miles northwest of the Isle of Lewis with 1,735 m of warp out in 730 m of water, snagged her port trawl door and “effectively became anchored to the seabed”; the load listed her to port, the engine room flooded, and “the vessel capsized while trying to free the fishing gear” — one crew member lost, five rescued [6]. At the small end of the scale, loss is a priced-in risk: Indonesian handliner Renaldi Safriansyah fishes reef edges for high-value grouper knowing “I snag my lines about two times out of ten, but the rewards are good” [6]. And wreck-netting skippers, notably, police themselves: gear is “never abandoned or disposed of on a wreck as this may indicate the location of the wreck to competitors,” and they minimize loss knowing lost gear ghost-fishes their own future grounds [6]. Extreme weather writes off whole coastal fleets at once — hurricanes and tsunamis cause “catastrophic losses in coastal areas,” Katrina’s 2005 debris field prompting NOAA’s Gulf of Mexico Mapping Project [6].

The deliberate pathway runs through the harbour. Shoreside disposal is decisive: gear has “a finite life span” and must go somewhere, but harbourside collection is often absent, leaving municipal facilities that cost time and charges “if indeed such disposal is permitted at all,” so “incentives may be strong to deliberately discard gear at sea, or to illegally dump it at other land-based locations”; even good facilities defeat themselves if “user pays” fees are set too high [6]. A former NAFO senior observer describes the practice plainly: on return journeys in the mid-Atlantic “I do remember seeing old gear being dumped. I think that dumping in the mid-Atlantic was not an uncommon practice… Dumping seldom took place on fishing grounds as this would be self-defeating” — gear went over in open ocean or on deliberately chosen rough ground “where vessels knew fishing conditions to be so bad that no one fished there” [6]. Retrieval economics push the same way: when recovery is “technically too complicated or time consuming” and results uncertain, “it is quicker to discard entangled gear to avoid interfering with hauling and to maximize fishing time” — though the report stresses the counterweight, that fishers’ heavy investment in gear means they usually hunt hard for what they lose [6]. IUU fishing adds its own signature: unmarked gear (or non-marking as itself an IUU act), refusal to communicate with other fleets (raising conflict losses), fishing at night or in bad weather to stay hidden, and gear ditched when inspection looms [6]. Vandalism and theft — buoy lines deliberately cut, documented in Gulf of Mexico crab and English and Scottish pot fisheries — are real but minor [6]. The throughline: almost nothing on this list is malice. It is friction — between fleets, between weather and rope, between a disposal fee and a free ocean. The next section tallies what the abandoned gear does next, alone.

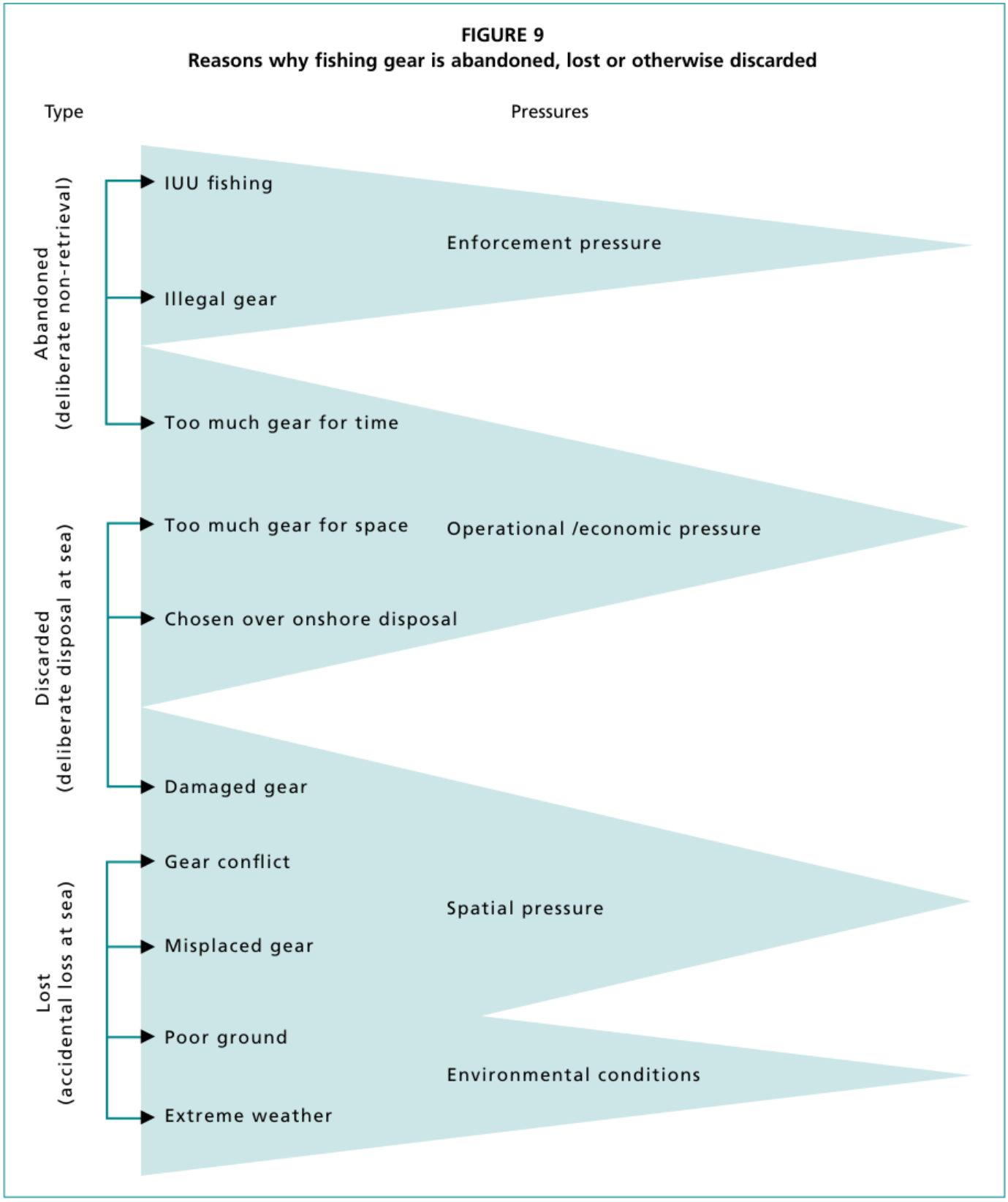


Exhibit. FIGURE 9 Reasons why fishing gear is abandoned, lost or otherwise discarded Type Pressures IUU fishing Enforcement pressure Illegal gear Too much gear for time Too much gear for space Operational /economic pressure Chosen over on. Source: S0017.

Ghost catches: impacts on species, habitats, and propellers

A lost net does not stop fishing; it stops being emptied. The FAO's impacts chapter is built on the few experiments that watched it happen. Off southwest Wales, Kaiser and colleagues set a gillnet and a trammel net 1 km offshore and had divers survey them for nine months: "several hours after both nets had been set, a large number of dogfish were caught, causing the nets to collapse"; fish catches dwindled to near zero by day 70 in the gillnet and day 22 in the trammel, yet the gillnet still took 226 fish and 839 crustaceans over its monitored life, the trammel 78 fish and

754 crustaceans — and "lost nets could continue to catch commercial crustacean species for at least nine months after initial loss," the decline driven by the net rolling up on itself [6]. (Caveat: these nets were deliberately set shallow for diver access, "not necessarily typical of commercial operations.") The self-baiting cycle explains the longevity: "fish dying in nets may also attract scavengers that are then caught in the nets, resulting in cyclical catching by the fishing gear" [6].

The FANTARED 1 experiment — four 100 m monofilament nets cut loose in 15–18 m of water — found the same arc compressed: height and effective fishing area collapsed within weeks as visibility rose; catches, initially "comparable to normally fished gillnets," fell steadily; the catch was 89% fish by number across at least 27 species; "the fishing lifetime of an ALDFG net was found to be between 15 and 20 weeks," and by the following spring the nets were "completely destroyed or heavily colonized by algae and had become incorporated into the reef"; notably, no seabirds, reptiles or mammals were caught in any of the eight experimental nets [6]. A real-world loss ran longer: a 1,200 m trammel net recovered off Tuscany after four months still had one-third fishing at "around 20 percent of normal 'controlled fishing,'" and its hauled biomass was 80% seaweeds and corals against 6% live and 1% dead fish — by the authors' estimate it had taken "some hundreds of kilograms of commercial species" in four months [6]. Scaled to a fishery, Cantabrian tangle nets deployed experimentally caught at commercial rates after 135 days (no monkfish after 224); the modelled ghost catch was 18.1 tonnes, "1.46 percent of the total commercial landings in the area" — flagged as an overestimate — with a worst case of 4.46% (55.3 tonnes) [6]. The report's summary generalizes cautiously: Sancho et al. "considered lost tangle nets to catch around 5 percent of the total commercial catch," while insisting that "overall catch rates of ALDFG vary so greatly that a global estimate would be meaningless" [6].

Depth changes everything. Deepwater gillnets set beyond 500 m in Norway's Greenland halibut fishery settled, after 45 days, at "20 percent to 30 percent" of commercial efficiency — 28 to 100 kg per net per day — "catch rates stabilized at this level and the nets continued to fish for 'long periods of time'" [6]. In Newfoundland's deep water, nets were reported "catching over several years, although at much reduced levels"; pieces of lost salmon gillnet were still catching fish and seabirds after three years despite biofouling; and ten gillnets took about 9,090 kg of cod in Placentia Bay [6]. Pelagic gear, by contrast, mostly self-destructs as a fishing machine: monitored monofilament nets "collapsed soon after deployment" into bundles that caught little, and five 2,000 m pelagic gillnets "formed a large mass of netting within four months" [6]. The summary's rule: ghost-fishing capacity is "highly specific to gear type and the conditions under which it was abandoned, lost or discarded" and to "currents, depth and location" — gillnets and traps ghost-fish on a declining curve; lost trawls "rarely ghost fish but have other impacts such as smothering the benthos and damaging delicate habitats such as coral reefs"; lost longlines rarely fish but leave hooks that "may be embedded in the bodies of seabirds" [6].

Beyond target species, the costs spread. Synthetic gear entangles and is ingested by "sea birds, turtles, seals or cetaceans," with "entanglement generally considered far more likely a cause of mortality than ingestion," and local studies indicating ALDFG "may be a significant cause for mortality for some species at local level," though comprehensive global studies are lacking [6] [6]. Physically, dragged nets and heavy anchoring weights (>100 kg in deep water) scour sponges and corals; some 70% of all marine litter sinks to the seabed where accumulation can smother benthic communities; and the material itself persists — "modern plastics can last up to 600 years in the marine environment" — degrading into microplastic fragments whose toxic-chemical role "is not known" [6]. For people, "a key socio-economic impact is the navigational threat of ALDFG to marine users" — near-invisible nets and lines in the water column are precisely what fouls propellers, shafts and intakes of vessels that never see them — yet the economics remain unbuilt: literature on ALDFG's economic costs "is also very scarce," monitoring, rescue and research costs have never been systematically attempted, and "nor have economic costs been attributed in any meaningful and comprehensive way to ghost fishing catches," leaving advocates "to lobby and work for improvements without sufficient information on costs at their disposal" [6] [6]. APEC's 2004 Honolulu seminar framed the stakes in one sentence — derelict gear causes loss "due to ghost fishing and habitat loss, safety of navigation, and a further decline in endangered and other marine species that are killed or maimed from entanglement or ingestion" [6]. The next section turns from harm to hunt: how lost gear is found, and who goes down to get it.

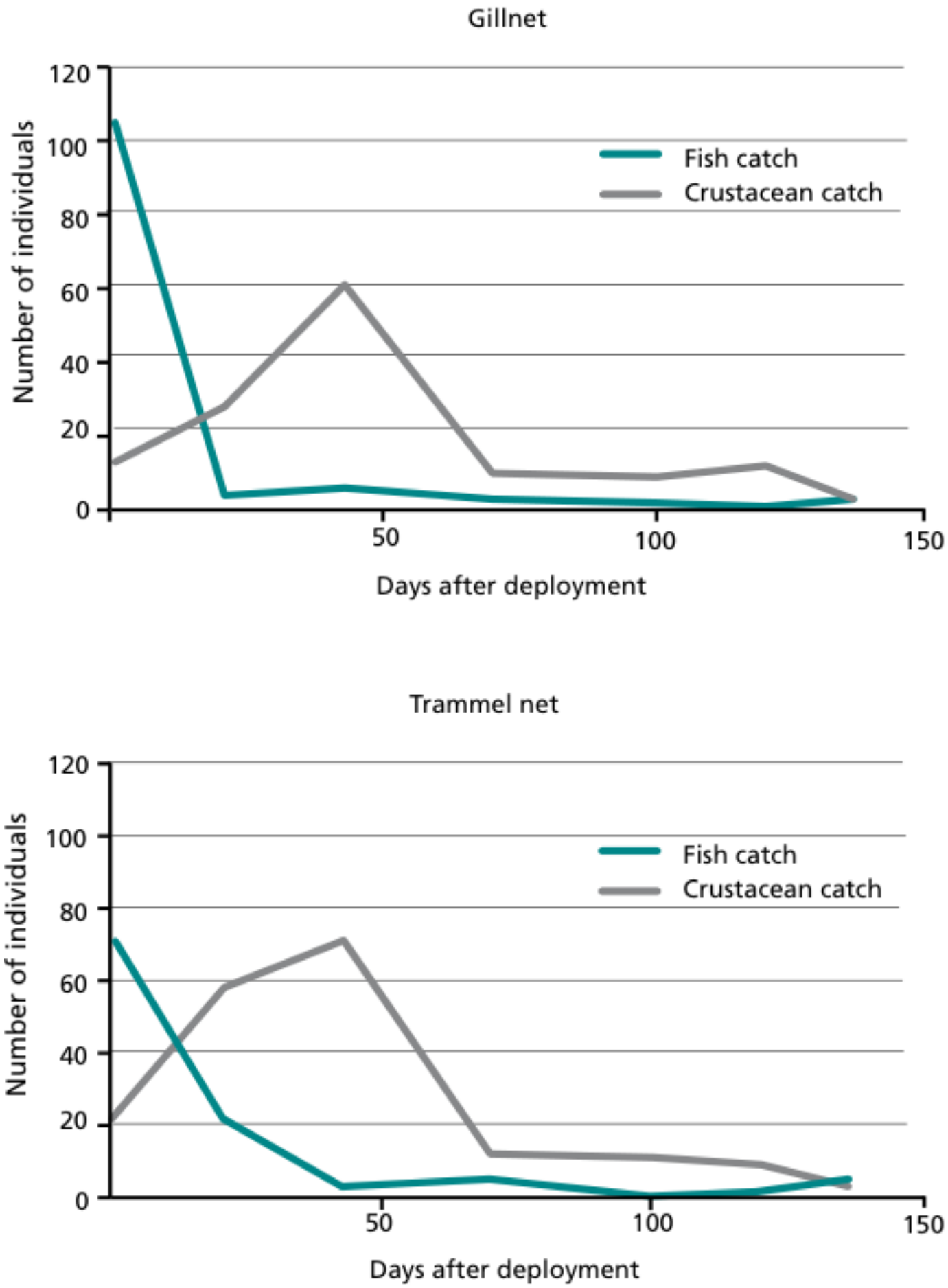


Exhibit. FIGURE 6 Change in catch composition of a “lost” gillnet and a trammel net Gillnet 120 100 Fish catch Crustacean catch 80 60 40 20 0 50 100 150 Days after deployment Trammel net 120 100 Fish catch Crustacean catch 80 60 40 20 0 5. Source: S0017.

Locating lost gear and retrieval programmes

You cannot haul up what you cannot find, and lost gear is engineered by circumstance to be unfindable — unmarked, unlit, often stripped of the very buoys that once announced it. The FAO's review lays out a location ladder, cheapest first. Fishers themselves are the first tier: they "will make every possible attempt to locate and recover their own gear as it has a significant economic cost," weighed against time and fuel [6]. Above that sit surveys, "from those with low costs, such as land-based beach surveys involving volunteers, to those at sea with high costs, using side scan sonar operated from sophisticated marine research vessels" [6]. Beach surveys fit coasts where the harm is onshore entanglement and litter — the Ocean Conservancy's International Coastal Cleanup supplies the standard protocol, and its US numbers give the scale of volunteer effort: from 1986 to 2002 the Cleanup "removed 89 million pounds of debris from more than 130,000 miles of shoreline," with divers adding 2.2 million pounds from over 3,900 miles of underwater habitat [6]. Where location is unknown, "the use of modeling techniques, local knowledge and anecdotal information to identify potential hotspots is essential": in the Northwestern Hawaiian Islands, towed-diver surveys were targeted by mapping high entanglement risk zones (HERZ) — where oceanography concentrates debris — against monk seal nursery areas [6]. Skipper interviews and VMS plots round out the intelligence sources [6].

The top tier is sonar. Side-scan sonar, "more accurate and more affordable in recent years," works best "where relatively large or readily distinguishable items such as pots or traps are to be located" — its images resolve a sport trap as a square, a commercial trap as a circle, the line between them visible [6]. NOAA's Gulf of Mexico project pairs vessel-towed sonar with an autonomous survey vehicle (maximum depth 100 m, worked mostly under 50 m) that maps "submerged wrecks, rock, and other objects that pose a hazard to navigation" — itself so hazard-like that its "deployment must be strictly controlled to ensure it does not itself create a navigational hazard" [6].

Retrieval itself is low-technology and old: programmes "typically entail using a creeper or grapnel to snag nets," run in Sweden and Poland, and routinely by Norway, whose deepwater collaboration with English and Irish authorities swept the northeast Atlantic gillnet grounds [6]. The US variants are more artisanal. The California Lost Fishing Gear Recovery Project (SeaDoc Society, UC Davis, launched July 2005) takes public reports and hires "experienced and certified SCUBA divers to remove gear from near-shore waters in a safe and environmentally sensitive manner," retrieving nearly 11 tonnes by its second year [6]. In the Gulf of Mexico, the Gulf States Marine Fisheries Commission codified trap-removal guidelines and states run annual voluntary closures with fisher-assisted cleanups; between 2004 and 2007 "over 183 boats participated in retrieving nearly 16,000 traps from 1,405,708 acres of coastal waters" [6]. Australia's Carpentaria Ghost Net Programme put AU$2 million into community beach-cleanup networks with coordinated recording of quantities, impacts and origins [6]. Japan runs the steadiest model: in the Sea of Japan "fisheries patrol vessels from the national agency bring any ALDFG identified to shore, as do fishing vessels chartered by fisheries organizations and local government and funded by central government subsidy" [6]. Even the law snags recovery: an NRC report noted US retrieval "may be inhibited by prohibitions against tampering with abandoned gear, the application of cabotage laws and burdensome certification requirements for vessels that transport DFG, and fishery regulations that prohibit vessels from carrying gear that is not a gear type permitted under their license endorsement" [6] — the maritime version of Section 21's authority gaps.

The hard question is whether any of this catches up. Measured against estimated losses, the flagship DeepNet follow-up — an Irish-led retrieval programme — "resulted in the recovery of approximately 35 km to 40 km of lost gear, amounting to around 3 percent of estimated annual gear losses" in that fishery [6]. Port Susan, Washington, shows the funnel within a single survey: side-scan coverage of ~95% of known grounds "identified 403 items," of which "seventy-three percent of those items could be investigated by diver and 174 items or 43 percent of the total items identified were removed" [6]. Effectiveness is mostly expert judgment — baselines rarely exist — and it is fishery-specific: Baltic and Western Channel experts agreed on relevant measures but diverged on acceptability and enforceability, and the report warns a "'one size fits all' approach would not work" [6]. The cost–benefit logic (Brown and Macfadyen) is timing: because ghost catch rates crash within weeks, "by the time a retrieval programme is implemented, ghost nets may only be making very small ghost catches," so retrieval pays mainly "where gear can be located and retrieved quickly," where volumes are huge, or in deep, still water where decline is slow — and in high-trawl areas the fleet itself does the cleanup, picking up or balling up lost nets [6]. The consistent verdict across FANTARED, DeepNet and the EC workshops: "'prevention is better than cure'" — curative programmes are "less effective and more costly than avoidance measures, but they can still be cost-effective when considered against doing nothing," i.e., against the ghost catch itself; and measures only work as suites — "gear marking is only likely to make a significant difference if this is supported by an MCS regime," and waste facilities fail without incentives to use them [6]. The typology maps cause to cure: IUU abandonment answers to port state measures; economic discarding to reception facilities and mandatory return; accidental loss to zoning, marking, reporting, and not setting

gear ahead of forecast storms [6]. One of the few economic instruments in use — Korean authorities paying fishers to deliver unwanted or recovered gear — previews the next section's toolkit: marking, buy-backs, and gear designed to stop fishing when its owner lets go [6].

FIGURE 18
Types of ALDFG, their causes and measures to address

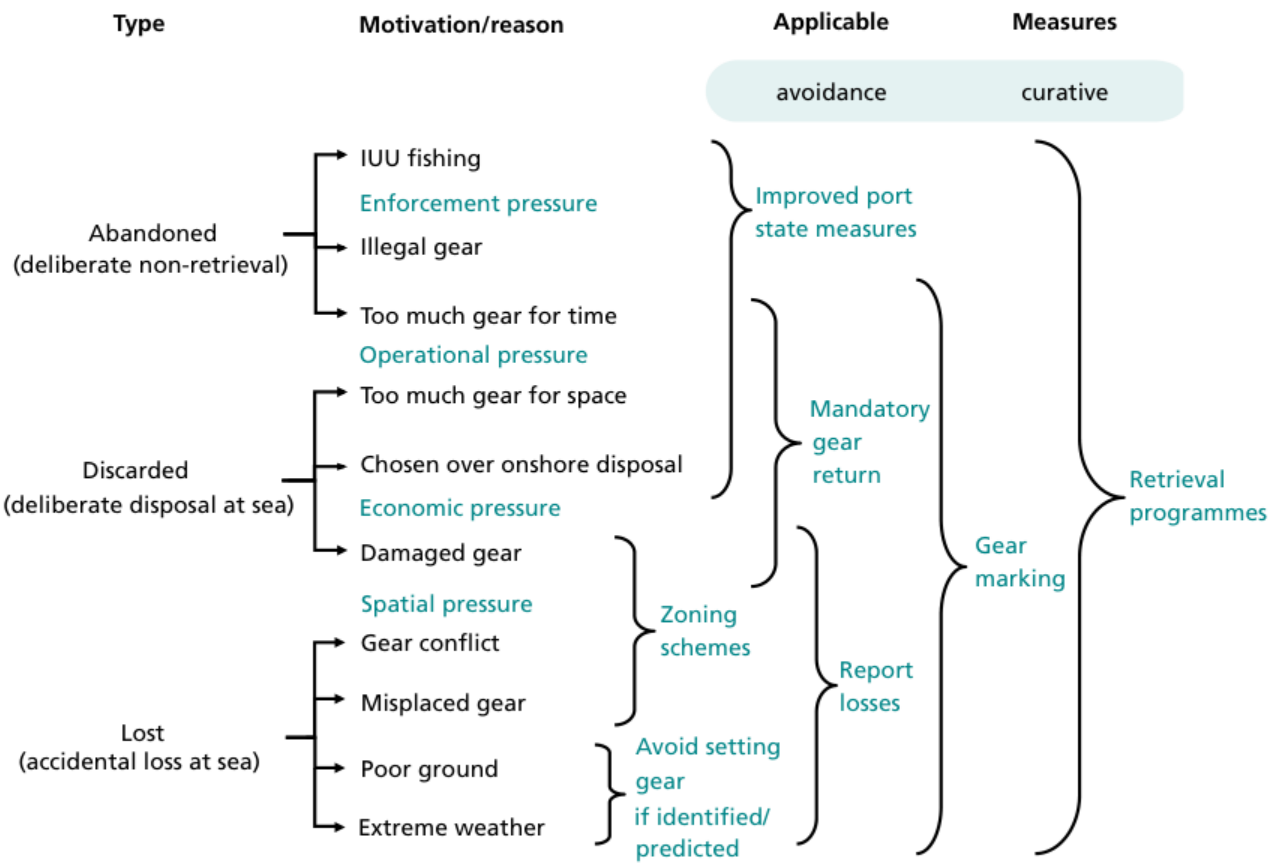


Exhibit. **FIGURE 18 Types of ALDFG, their causes and measures to address Type Motivation/reason Applicable Measures avoidance curative IUU fishing Enforcement pressure Improved port Abandoned state measures Illegal gear (deliberate non-retr. Source: S0017.**

Gear marking, buy-back projects, and biodegradable nets and pots

The FAO organizes every countermeasure into three families — prevent (keep gear out of the sea), mitigate (make lost gear harmless), cure (remove it) — and is blunt about the ranking: preventative measures are "the most effective way to tackle ALDFG," curative ones "inevitably reactive... therefore always less effective," though still cost-effective against leaving gear in place [6]. This section takes the three emblematic tools in turn: marking (prevention through accountability), buy-back (prevention through price), and biodegradable gear (mitigation through design).

Marking. Informal gear marking is "a centuries-old practice to clarify ownership and avoid intra-fishery conflict," but "the mandatory marking of specific gear to enable identification by competent authorities remains far less widespread" [6]. The institutional paper trail is long: FAO's 1991 expert consultation produced the Guidelines for the Application of a System for the Marking of Fishing Gear (Fisheries Report No. 485), covering owner and authority responsibilities, recovery, salvage, liabilities and penalties; a 1993 Supplement standardized tag types and placement per gear type so gear's "presence and extent is obvious to other seafarers" [6]. The Code of Conduct for Responsible Fisheries (§8.2.4) states that gear "should be marked in accordance with national legislation in order that the owner of the gear can be identified," with "uniform and internationally recognizable gear marking systems" [6]. Practice lags the paper: most national requirements label the marker buoy rather than the gear, "to support enforcement of licences or for reasons of navigational safety" — not to trace ALDFG [6]. The exceptions are specific: CCAMLR's Conservation Measure 10-01 requires pots, buoys and floats marked with "the vessel name, call sign and flag state" in

the Antarctic; Canadian rules require shrimp traps and ring nets marked with the operator's name; Korea launched a gear-marking initiative in 2006 under its National Integrated Management Strategy for Marine Litter and promoted it through NOWPAP; and EC rules from 2006 require passive gears and beam trawls to carry the vessel's port licence number — but only outside territorial waters, with most member states not extending the requirement inshore [6]. The structural flaw is physical: where tagging exists, as in Australia's Northern prawn fleet, "tags tend to be attached to headropes and groundropes rather than directly to sections of net or line," while most ALDFG "is predominantly made up of nets and lines" — the fragment that washes up carries no return address [6]. Embedded fixes exist — coded wire tags implanted in netting and scanned on recovery, or a "rogue yarn" of different twist or colour worked into twine, used in Japan to distinguish management areas [6]. The chapter's verdict stands: "there are few examples of national requirements for gear marking intended to address the problem of ALDFG, i.e. marking to prohibit the deliberate abandonment of gear through enabling identification of ownership" [6].

Buy-back and reception. The cheapest gear disposal is a harbour that takes gear free. MARPOL Annex V Regulation 7 obliges governments to ensure port reception facilities "without causing undue delay to ships," yet small fishing harbours chronically lack them — for Pacific Island States, SPREP identified solid waste management as the number one issue, with 90% of fishing operations foreign [6]. Certification can be theatre: the US Coast Guard's Certificate of Adequacy programme certifies ports on capability or on-call providers, not on actually accepting garbage, so crews "ready and willing to pay for disposal services… are not always able to secure these services, even from those ports with COAs" [6]. Where disposal is folded into port fees it works — Baltic cod gillnetters drop end-of-life nets in harbour containers with costs "already contained as part of port fees, so there appears little economic incentive for fishers to deliberately discard nets at sea"; Greek net fishers strip old netting to the municipal tip at no cost [6]. Korea goes further and pays: its Waste Fishing Gear Buy-back project (since 2003, run by the Ministry of Maritime Affairs and Fisheries) purchases waste gear returned to port — designed because fishers who hauled litter up while fishing "used to… throw it back into the sea" — supplying boats with large hardwearing bags for quayside deposit; it is "reported to be highly effective in terms of recovery and disposal of gear, but there is no evidence that cost–benefit analysis has been undertaken for a scheme that is dependent on significant public funding" [6]. (A related Korean scheme was partly prompted by the sinking of a passenger ferry entangled in discarded gear.) The North Sea's Fishing-for-Litter project (Dutch, from March 2000) pays in kind — free disposal of trawled-up litter — and by 2004 had 54 boats in four countries landing 450 tonnes [6]. The FAO's own recommendation list ties the knot: revise MARPOL Annex V around ALDFG with "a particular focus on the marking of fishing gear to identify ownership, defining what constitutes reasonable losses of gear, providing port reception facilities, and lowering the limit of gross tonnage (GT) that requires fishing vessels to carry garbage record books" [6].

Biodegradables. Mitigation is the weakest leg, and the report says why: such measures "are limited in their extent and application as many may increase costs through reduced effectiveness of gear or higher gear prices. Consequently, the development of innovative materials has been slow and the return to biodegradable netting by the industry has been very limited" [6]. Trials continue — including net materials with increased sound reflectivity to cut cetacean bycatch, and designs encouraged through WWF's International Smart Gear Competition [6] — but the acceptance problem is documented: expert groups for both the Baltic and the Western Channel rated biodegradable nets among the measures fishers consider unacceptable [6]. The design dilemma is intrinsic: a net that degrades on schedule after loss must not degrade during use, and the fishery will not pay more for gear that catches less. So the toolkit's honest ordering is the report's own — mark gear so owners can be found, make harbours take it back free or buy it back, and treat biodegradability as a research programme, not a solution. The next section zooms out to the treaty layer: MARPOL Annex V and the slow international recognition of ghost gear as a named problem.



Exhibit. FIGURE 11 Crab trap with rot cord Source: Fisheries and Oceans, Canada. Source: S0017.

MARPOL Annex V and international recognition of the ALDFG problem

Ghost gear is a transboundary problem wearing national clothes: nets lost off one coast fish on another's, which is why "regional and international cooperation to prevent ALDFG is vital" [6]. The recognition stack begins with the UN General Assembly. Resolution A/RES/59/25 (2004) called on states, FAO, IMO, UNEP's Regional Seas Programme, and regional fisheries bodies "to take action to address the issue of lost or abandoned fishing gear and related marine debris through the collection of data on gear loss, economic costs to fisheries and other sectors, and the impact on marine ecosystems" [6]. A/RES/60/30 (Oceans and the Law of the Sea) noted the lack of information on marine debris, urged states to integrate debris into national waste-management strategies, and invited IMO to review MARPOL Annex V; A/RES/60/31 pressed for data plus "both preventative and curative measures"; A/RES/61/222 repeated the waste-strategy call and welcomed the Annex V review; A/RES/61/105 reaffirmed ALDFG's importance and pushed it onto the agenda of FAO's Committee on Fisheries (COFI) in 2007 [6]. The 6th UNICPOLOS meeting (New York, June 2005) took up marine litter and abandoned fishing gear at the Assembly's request, and UNCLOS itself supplies the general duty — Articles 1, 192, 194, 197, 207, 211 and 216–218 oblige nations against marine debris broadly [6].

The operative treaty is MARPOL Annex V, the garbage annex of the IMO's pollution convention, in force since 31 December 1988 and twice amended: it "completely prohibits certain discharges of ship-generated garbage (e.g. plastics)" and zones the rest by distance from land — plastics are illegal to discharge at any distance [6]. Its fishing-gear clause is the hinge of this whole block: "the prohibition of the discharge of plastics specifically prohibits the discharge of synthetic fishing nets; however, the Annex does not apply to the accidental loss of such nets, provided that all reasonable precautions have been taken to prevent such loss" [6]. Deliberate dumping is banned; genuine loss is excused — and the line between them is a judgement call made by the only witnesses, the crew. Annex V "is applicable to all vessel types including fishing vessels of all sizes," and Regulation 9 requires ships of 400 GT and over to keep garbage records including "the discharge, escape or accidental loss (of garbage that includes synthetic fishing material)... and to record the circumstances of and reasons for the loss" [6] — the threshold that ex-

empts most of the world's fishing fleet, and whose lowering the FAO recommends [6]. Regulation 7 commits governments to port reception facilities "without causing undue delay," the duty whose patchy fulfilment Section 29 documented [6]. The Annex V Guidelines, meanwhile, "call for fisheries managers to utilize fishing gear identification systems that provide information such as vessel name, registration number and nationality, and they encourage governments to consider the development of technology for more effective fishing gear identification" [6].

Around the treaty sits the institutional division of labour the FAO's stakeholder analysis maps with unusual candour [6]. UNGA supplies the mandate; FAO leads on fisheries through COFI; IMO's leverage is "high, especially due to ongoing MARPOL Annex V review"; UNEP's Regional Seas work is "considered the only systematic work on the marine litter problem at the regional and supra-regional level" — 12 Regional Seas programmes running since 2005, each with status reviews, action plans and cleanup days [6] [6]. APEC's 2004 Honolulu seminar demanded action "at the national, regional, and global levels" and a standing body on derelict gear, and the 2005 Bali Plan of Action extended the remit to "derelict fishing gear and derelict vessels" alike [6]. The EU grounded measures in the Common Fisheries Policy and, in Regulation 356/2005, laid down marking rules for passive gear [6]. National governments "often bear the costs of clean-up" and hold the legislative pen; the catching sector's role is decisive in both directions — it causes ALDFG and suffers from it, and because "enforcement problems with any legislation" are endemic, its "buy in" is what makes measures real [6]. Two quiet conflicts are named: processors have little leverage "unless some sort of certification scheme were introduced," and gear manufacturers — key to marking and technical fixes — "may have an interest in continued ALDFG as a means of increasing sales of new gear" [6]. The review's summary judgment is the block's epitaph: "sound international policy, legislative and regulatory regimes have been developed and are in place (e.g. MARPOL Annex V). However, there are significant deficiencies in the implementation and enforcement of these regimes," and progress "depends to a significant degree on changing human behaviour in addition to providing the relatively straightforward technological fixes" [6].

The modern layer tightens the same screws: Breaking the Plastic Wave calls for "harmonization of reporting gear production and losses, codes of practices, and communication protocols, such as the newly adopted FAO guidelines in the PSMA and voluntary guidelines on the marking of fishing gear," plus "inclusion of gear loss in sustainability criteria by fishery certification bodies" [15]. Its shipping-litter chapter underlines the enforcement gap at sea: dumping from vessels is "illegal under international law, with some exemptions (MARPOL Annex V). Nevertheless, the practice is believed to be widespread," with EU shipping litter alone estimated at 54,000–67,000 tonnes of plastic a year — 35% of maritime-source plastic, "the other 65 per cent is ALDFG" [15]. Best-practice regulation now means indirect fees that make port disposal free at the point of use, refunds on docking fees when waste is delivered, and digital waste reporting shared between governments [15]. The pattern across forty years of instruments is consistent: the law learned to name ghost gear long before it learned to see it. Identification — marked gear, recorded losses, reported positions — is the bottleneck everywhere. The paper now scales that problem up from nets to entire fleets: vessels that switch off their transponders, fish invisibly, and make ghost gear look easy to find.

FIGURE 3
Examples of ALDFG in northern Australia



A 6 tonne Taiwanese gillnet with large entangled shark washed ashore in Arnhem Land.

Aboriginal rangers loading an ALDFG fishing net collected from the shore onto a truck for recycling/ disposal, Arnhem Land, Australia.



Exhibit. FIGURE 3 Examples of ALDFG in northern Australia Aboriginal rangers loading an ALDFG fishing net collected from the shore onto a truck for recycling/ disposal, Arnhem Land, Australia. Source: S0017.

Dark Fleets: Vessels That Hide

Satellite mapping reveals extensive industrial activity at sea

The modern ghost ship is not empty — it is unreported. Its crew fishes, its holds fill, and it simply never appears on the public map. Paolo and colleagues (Global Fishing Watch and partners, Nature, January 2024) built the counter-map. The stakes frame the paper: more than a billion people depend on the ocean for their primary food source, 260 million are employed by marine fisheries, about 80% of traded goods move by sea and nearly 30% of world oil comes from offshore fields — a "1.5–2.5 trillion dollar 'blue economy'" growing faster than the global economy, while a third of fish stocks are fished beyond sustainable levels and 30–50% of critical marine habitats have been lost [20]. Yet the observational base is feeble: "on land, maps exist for almost every road," while at sea the vessel monitoring systems used in fisheries are proprietary, and the public system — AIS, the transponder network built for collision avoidance — is full of holes [20]. Not all vessels are required to carry AIS; rules vary by country, size and activity; "vessels engaged in illicit activities often turn off their AIS transponders or manipulate the locations they broadcast"; the largest recent illegal-fishing and forced-labour cases involved fleets that "mostly did not use AIS devices"; and coastal blind spots and government restrictions on terrestrial AIS widen the dark [20]. The paper's term for the invisible: vessels "not visible on publicly accessible AIS data" — 'dark vessels' [20].

The method replaces trust with radar. The team analysed "2 petabytes of satellite imagery spanning the years 2017–2021," covering more than 15% of the ocean — the coastal belt where over 75% of industrial activity concentrates — and trained three convolutional neural networks: object detection at >97% accuracy (length estimation R^2 = 0.84), offshore-infrastructure classification at >98%, and fishing/non-fishing vessel classification at >90% [20]. They classified "more than 67 million image tiles" of Sentinel-1 synthetic-aperture radar (which sees metal hulls through cloud and darkness) and Sentinel-2 optical imagery; SAR catches most objects over 15 m — detection >70% for 25-

m vessels and >90% for vessels of 50 m and up — and they matched detections against "53 billion vessel GPS positions from the AIS" to label each contact publicly tracked or not [20] [20].

The headline: darkness is the norm in fishing. Across 23.1 million vessel detections, about 63,300 vessels were present at any given moment, 42–49% of them fishing vessels — and "about three-quarters (72–76%) of globally mapped industrial fishing did not appear in public monitoring systems, compared with one-quarter (21–30%) for other vessel activities" [20]. Activity is widespread yet concentrated: a vessel was detected at least once in 84% of 0.1° grid cells, but "half of all vessel activity was concentrated in less than 3% of the cells"; 86% of fishing and 75% of non-fishing activity sat in waters shallower than 200 m — just 7% of the ocean; and Asia accounted for ~67% of all vessel activity, Europe 12%, North America and Africa 7% each [20]. The correction to the AIS worldview is severe: AIS alone "wrongly suggest[s] that Europe and Asia have comparable fishing activity," while the satellite map shows Asia taking 70% of all fishing detections and "nearly 30% of all mapped fishing vessels… concentrated in the exclusive economic zone of China alone"; in the Mediterranean, AIS implies European states fish more than ten times the hours of African states, while the imagery shows detections "fairly balanced between the northern and southern parts" [20]. Untracked density clustered in Indonesia, South and Southeast Asia, and the northern and western African coasts [20]. Most non-fishing vessels — transport and energy — do broadcast: only about a quarter were missing, many explainable by poor satellite AIS reception rather than intent [20].

The map also catches behaviour. The densest fishing on Earth in 2017–2019 was not on any public chart: the western Korean Peninsula, at "about 40 vessels per 1,000 km²," its dark fleet swelling each May during China's domestic fishing moratorium and collapsing 85% when North Korea shut its borders for COVID — the fleet the next section illuminates in detail [20]. Even the world's flagship protected areas leak: the Galápagos Marine Reserve averaged "more than 5" untracked vessels per week and the Great Barrier Reef Marine Park "more than 20" [20]. At fine scale the imagery resolves tactics — mixed tracked and untracked trawlers working banks and canyon edges between Tunisia and Sicily; off Bangladesh, where "almost no vessels are publicly tracked," fleets following bathymetric contours radiating from the Ganges Delta [20]. And the time series works: the map measured a global fishing decline of 12 ± 1% through the pandemic while transport and energy held steady or grew [20]. The implication for this paper's theme is direct: we spent three sections learning that ghost gear cannot be governed because it cannot be identified. Paolo et al. show the same was true of most of the world's industrial fishing — until radar satellites made the ocean's industrial layer visible whether or not it chooses to transmit. The following sections look into three kinds of darkness the map exposed: state-sanctioned dark fleets, deliberate transponder silence, and the sanctions-era shadow tanker fleet.

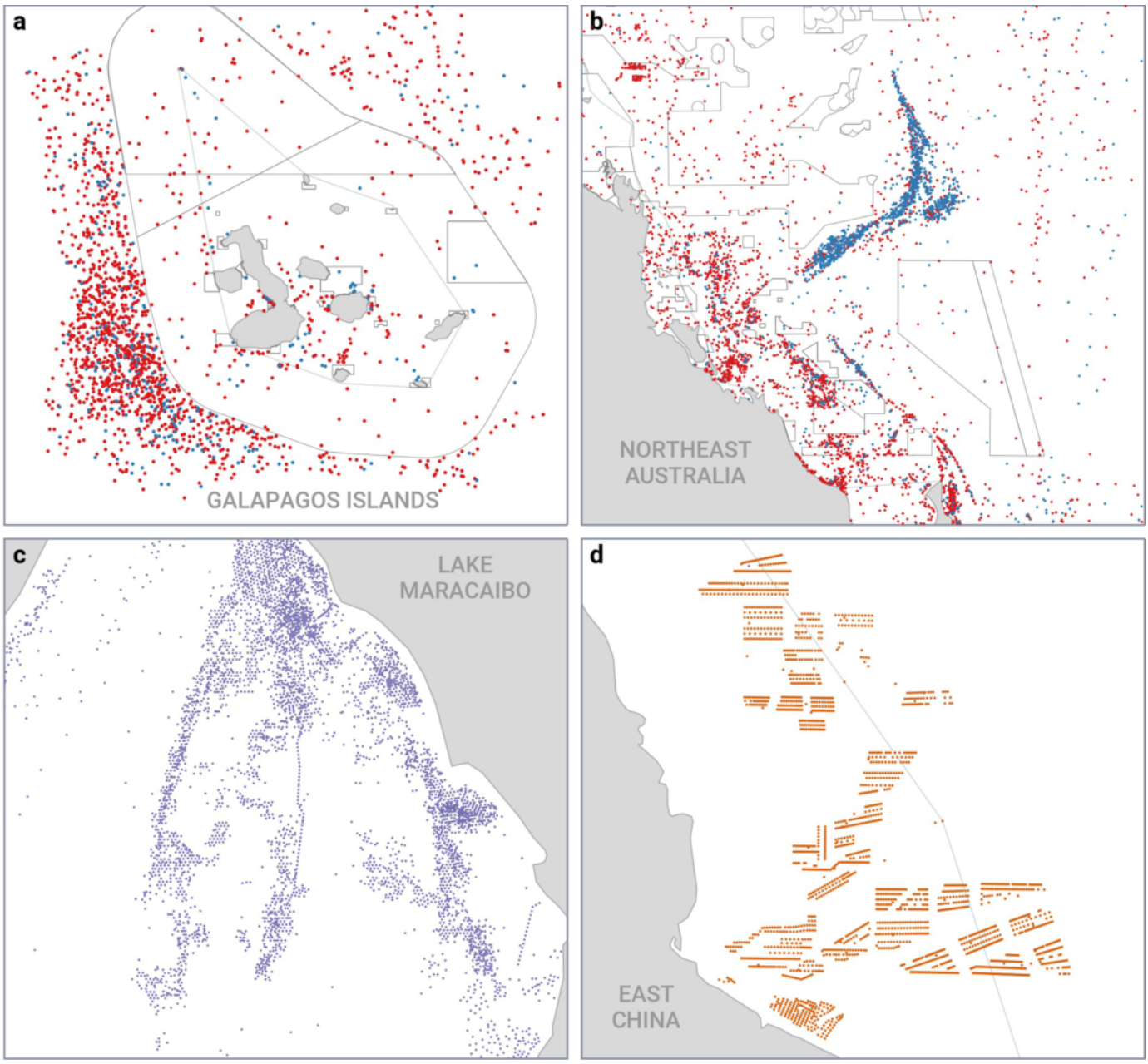


Exhibit. Fig. 7 Satellite imagery-based detection allows monitoring at local scale. Source: S0021.

Illuminating dark fishing fleets in North Korea

This is where the paper's two meanings of "ghost ship" collide. In the waters between North Korea, South Korea, Japan and Russia, geopolitical tension and disputed boundaries have created "a vacuum of shared data and management" over a single shared stock — Todarodes pacificus, the Japanese flying squid [21]. The fishery matters to everyone: squid is South Korea's top seafood by production value, a top-five seafood in Japan, and was North Korea's third-largest export before sanctions; yet reported catches have fallen 80% in South Korean waters and 82% in Japanese waters since 2003, the Chinese distant-water fleet has fished North Korean waters since 2004, and its activity and catches have gone unpublished since 2016 — making stock assessment impossible [21]. After North Korea's 2017 ballistic-missile tests, UN Security Council resolutions (2371, 2375, 2397) prohibited procuring seafood from North Korea, joint ventures without UN approval, and North Korea's sale or transfer of fishing rights — binding Chapter VII sanctions, so violations "would constitute a violation of public international law and domestic Chinese law" [21] [22]. The South Korean Coast Guard nevertheless watched hundreds of vessels cross into North Korean waters. Park et al. (2020) decided to count them from orbit.

The team fused four satellite technologies, each compensating for the others' blindness: AIS (identity and movement, but used by "only a fraction of vessels"), SAR (finds all large metal vessels through cloud, but with irregular coverage), VIIRS (global daily revisit, detects bright fishing lights, cloud-limited), and high-resolution optical imagery (best confirmation, weather-bound) [21]. A convolutional neural network trained on Planet's 3-m PlanetScope imagery

picked out pair trawlers — the dominant foreign type — verified by tasked 0.72-m SkySat imagery and by SAR from three satellites; the fraction of vessels that did broadcast AIS betrayed the fleet's origin: signals matched to 140 vessels in the imagery showed tracks home to Chinese ports such as Shandong [21]. The counts: "at least 796 distinct pair trawlers operated in North Korean waters in 2017 and at least 588 did so in 2018," plus lighting vessels — 55–60 m ships carrying "up to 700 incandescent bulbs and generating over 1000 lux, equivalent to the light of some football stadiums," their glare separable in VIIRS from every other fleet's, with minima of 108 (2017) and 130 (2018) [21]. In total, "over 900 industrial vessels originating from China in 2017 and over 700 in 2018 were detected fishing intensely in North Korean waters," logging an estimated 91,400 fishing days in 2017 and 67,300 in 2018, and taking ">160,000 metric tons of Pacific flying squid worth US $440 million in 2017-2018" [22] [21].

The legal geometry is a fork with violations on both prongs. The vessels "often do not carry appropriate papers" and "may plausibly be so-called 'three-no boats' operating outside of official Chinese authority, with no registration, no flag, and no license to operate" — in which case they fish illegally under Chinese law (which requires ministerial approval for foreign waters) and UNCLOS coastal-state rights; alternatively, if approved by either government, "those state(s) are in violation of UN Security Council sanctions" — a scenario China denies [21]. Either way, the authors conclude, the fleet is "the largest known case of illegal fishing perpetrated by a single distant-water fleet" [21]. It also fishes against a neighbour playing by opposite rules: South Korea caps squid with a total allowable catch, limits lighting power, bans pair trawling and permits fewer than 40 small trawlers, while the revealed fleet uses brighter lights, pair trawls, and hundreds of hulls on the same declining stock [21].

The squid war's other navy is the tragedy's core. Displaced from their own waters by the industrial fleet's competition, North Korea's small-scale fishers — in 10–20 m wooden boats lit by only 5–20 bulbs — pushed into the Russian EEZ: "about 3000 North Korean vessels fished in the Russian EEZ during 2018," their fishing days there rising from 39,000 in 2015 to 222,000 in 2018 [21] [21]. Russia "has authorized fewer than 100 North Korean boats since 2014, and in 2017, no permits were granted," so most of the 3,000 were unauthorized [21]. These boats are "severely underequipped for the long-distance travel required to reach Russian fishing grounds," and the cost is written on Japanese beaches: "between 2014 and 2018, 505 North Korean boats washed ashore on Japanese coasts. These incidents frequently involve starvation and deaths, and local media reports that many fishing villages on the eastern coast of North Korea have now been coined 'widows' villages'" [21]. GFW's own summary names them plainly: "'ghost boats' — battered vessels containing no crew or, in some cases, only human remains" [22]. The 2019 follow-up found the system stable: close to 800 Chinese vessels detected, a 2019 catch of ~76,700 tonnes worth ~US$240 million (satellite-estimated 82,200 fishing days against the Coast Guard's 134,000), cumulative post-sanctions catch value ~US$560 million, North Korean vessels in Russian waters back above 2,000, and the count of reported "ghost boats" highly correlated with the number of North Korean boats in Russian waters [22] [22]. Park et al. close on method, not outrage: combining satellite sources "can reveal the activities of dark fleets," and what was once a state surveillance capability "can now be performed… by small, independent groups of researchers" — transparency as the enforcement mechanism of first resort [21]. The next section generalizes from one sea to the planet: where vessels go dark, and how much fishing vanishes with them.

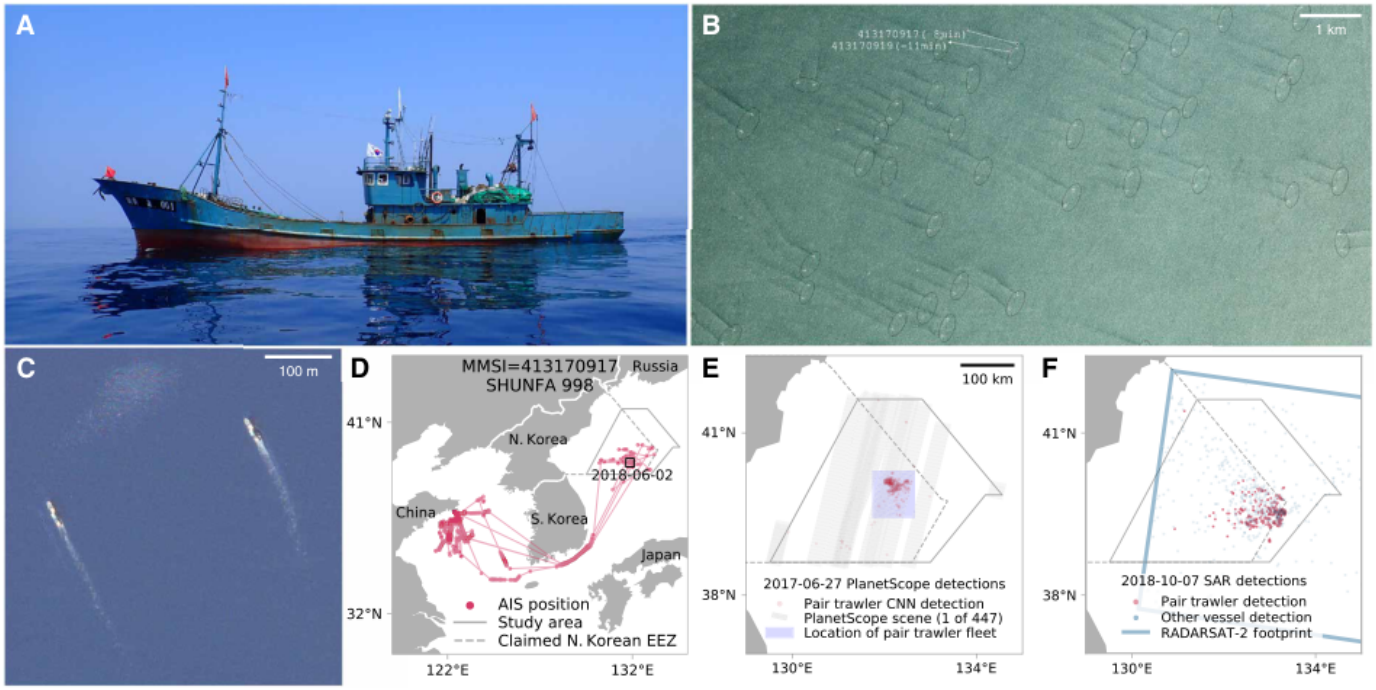


Exhibit. Detecting the dark fleet: (A) a Chinese pair trawler; (B) trawler wakes in SAR imagery; (D) the AIS track of SHUNFA 998 entering the North Korean EEZ; (E-F) PlanetScope and RADARSAT-2 detections inside North Korean waters (Park et al. 2020) [S0018]. Source: S0018.

Hot spots of unseen fishing vessels

Welch et al. (2022) start from the gap inside the tracking system: not vessels that never transmit, but vessels that transmit and then, deliberately, stop. The context is the economics of illegal fishing — global IUU losses of "US$10 billion and US$25 billion annually, with one in every five wild-caught fish harvested illegally or unreported," plus documented links to forced labour and trafficking [23]. AIS, built as "a collision avoidance tool," has been repurposed into the public's window on fishing; intentional disabling closes that window, and the act is ambiguous by design — AIS "are not universally mandated, nor are vessels always required to keep their devices on," and disabling can be legal (confidential fishing locations, piracy-prone waters) or criminal, obscuring "unauthorized fishing activity in exclusive economic zones (EEZs) and MPAs, or unauthorized transshipments" that launder illegal catch and have been linked to forced labour [23]. The paper's move is to treat the silence itself as data: "we can use these events as a data source to direct research and improve management" [23]. Using a rule-based classifier to separate likely intentional disabling from innocent signal gaps in Global Fishing Watch's archive — more than 3.7 billion AIS messages from 2017–2019 — they found "more than 55,000 suspected intentional disabling events in waters greater than 50 nautical miles from shore," involving "more than 40% of fishing vessels in these waters," and "obscuring up to 6% (>4.9 M hours) of fishing vessel activity" [23]. Four gear types accounted for 96% of the time lost [23].

The geography is the finding: "more than 40% of the total hours lost to suspected disabling occurred across four hot spots: the Northwest Pacific (13%), adjacent to the EEZs of Argentina (16%) and West African nations (8%), and near Alaska, USA (3%)" [23]. The first three share a profile the authors name: all "have been previously deemed to be areas of IUU concern… rich fishing grounds and limited management oversight — a formula that may embolden vessels to engage in IUU activities" [23]. In the Northwest Pacific, one cited study found "more than half of the vessels flagged to China operating in the Northwest Pacific had suspected disabling events and estimated that more than half of their chub mackerel catch was unreported" [23]. Off Argentina, the hypothesis is a dark-fleet relay: unauthorized fishing inside Argentine waters with catch off-loaded "to high seas refrigerated cargo vessels, thus avoiding visiting ports to off-load illegal catch" [23]. Off West Africa — where illegal fishing costs up to US$2.3 billion a year — disabling strips monitoring from waters that include authorized incursions by industrial fleets [23]. A structural point follows: vessels with government VMS remain trackable by their flag state, "although access to VMS data is often tightly restricted," so disabling blinds everyone except the flag state — and the three big hot spots are distant-water fleets in foreign or international waters whose VMS data local authorities cannot see [23].

Alaska, the fourth hot spot, inverts the story and proves the method can distinguish motives. Its disabling fleet is US trawlers, monitored by US authorities via VMS in "one of the most intensely managed regions in the world" — so "disabling in this region likely indicates location hiding of high-quality fishing grounds from competitors (unlike AIS,

VMS cannot be legally disabled, nor can VMS messages be received by competitors)" [23]. Darkness here is not crime but trade secrecy, and the driver models agree: chlorophyll — a proxy for productive water — "was an important driver of disabling in trawlers" [23]. Two behavioural patterns cut across the map. First, transshipment: "disabling events were common in areas with high transshipment activity, while fishing activity occurred in areas with lower transshipment activity," indicating "a desire to have these possibly unauthorized transfers obscured from public view" — most pronounced for squid jiggers, which loiter heavily and face "relatively little management oversight" compared with tuna fleets, whose transshipment rules and observer requirements are stricter [23]. Second, boundary-hugging: "disabling events were concentrated in waters adjacent to EEZ boundaries… most likely to occur adjacent to foreign and disputed EEZs, suggesting that some of these events may indicate vessels disabling their transponders before entering unauthorized locations to fish illegally," with contested boundaries creating "managerial blind spots, which could make IUU activities more profitable and less likely to be prosecuted" — though opaque access agreements mean some of those dark crossings were licensed all along [23].

Set against Section 31's census, the scale needs care: an estimated 50–80% of fishing operations more than 100 nautical miles offshore are by AIS-carrying vessels and "a majority of the effort on the high seas is by vessels broadcasting AIS," so this dataset illuminates "an additional 4.9 M hours (6%) of fishing vessel activity by vessels with AIS" — the deliberate dark, layered on top of the never-tracked majority Paolo et al. mapped [23] [20]. The management read is precise: disabling maps show exactly "where stricter AIS requirements and enforcement may support improved fisheries management" [23]. A switched-off transponder, in other words, is not an absence of evidence; analysed at scale, it is evidence — of where the fishing is rich, the oversight thin, and the boundary worth crossing in the dark. The next section widens the lens from where disabling happens to why: a global model of its drivers.

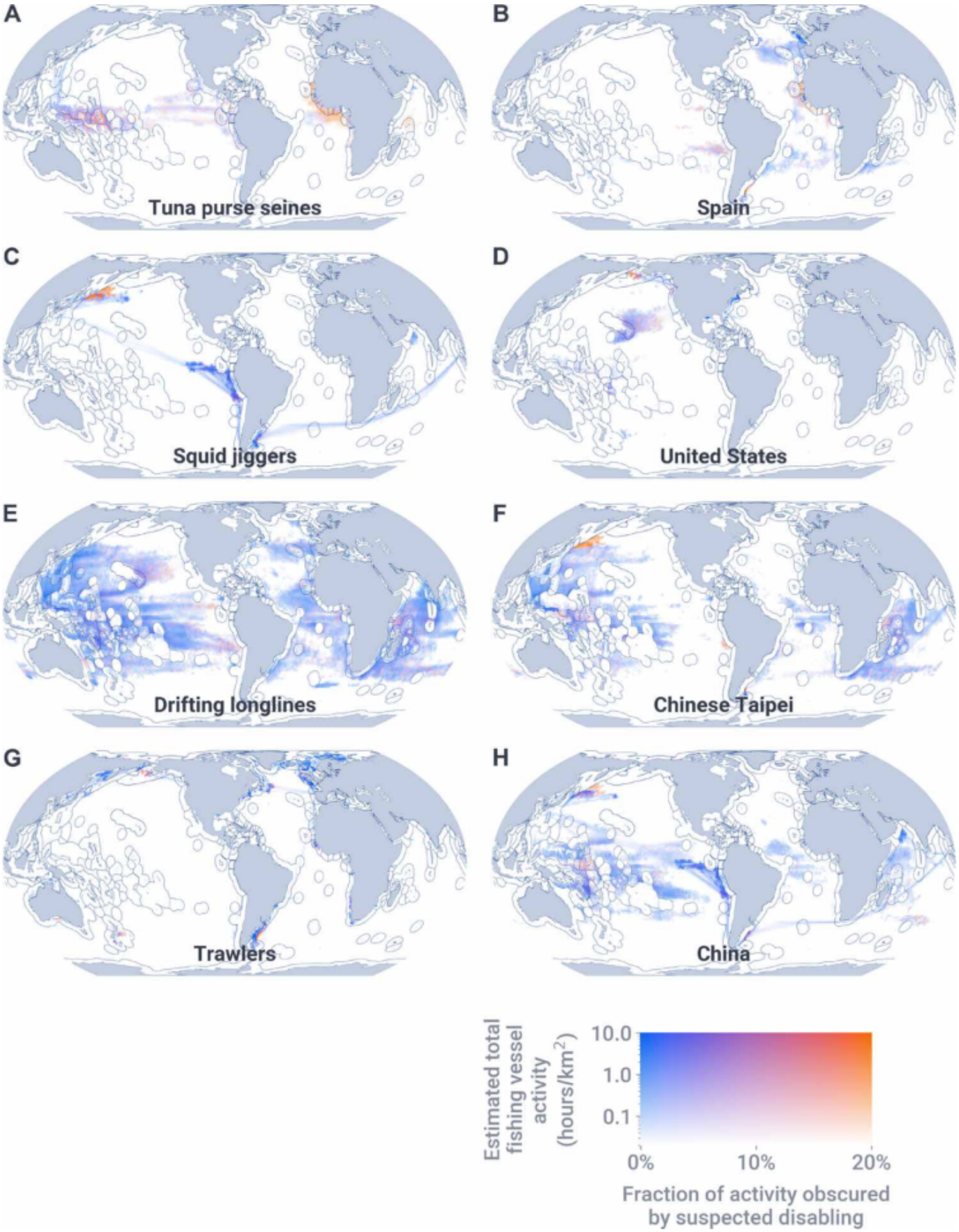


Exhibit. Fig. 2. Estimated total fishing vessel activity and the fraction of this activity obscured by suspected disabling events. Source: S0020.

A global picture of AIS disabling and its drivers

Section 33 mapped where the signal dies; Welch et al.'s driver analysis asks who kills it, and why. The dataset is industrial in scale — "more than 28 billion AIS messages from 2017 to 2019" from Spire and Orbcomm via Global Fishing Watch, machine-classified into fishing events and gear types [23]. By gear, the fractions of activity obscured in-

vert the absolute numbers: “tuna purse seines had the highest fraction of vessel activity obscured by disabling events (up to 21%), followed by squid jiggers (up to 7%), and drifting longlines and trawlers (both up to 5%),” while drifting longlines contributed the most total hours lost [23]. By flag, “vessels flagged to four key nations made up 82% of time lost to suspected disabling events” beyond 50 nautical miles: Spain had the highest fraction obscured (up to 14%), then the United States (up to 8%), Chinese Taipei (up to 6%) and China (up to 5%), with China largest in absolute hours [23]. The authors flag the selection effect themselves: “AIS usage is biased toward vessels flagged to upper- and middle-income countries; thus, these results, in part, reflect who uses AIS (higher AIS usage leads to high opportunity to disable AIS)” [23] — the map of disabling is a map of the tracked world misbehaving, not of all darkness.

To find drivers, the team fitted boosted regression trees contrasting locations with suspected disabling against locations with fishing but no disabling, for the four dominant gears and overall: the models achieved 32.7% mean explained deviance and 0.83 mean AUC, “a strong capacity to discriminate” — and, tested on unseen data, strong predictive power [23] [23]. Driver one is the rendezvous. Loitering by refrigerated cargo vessels — the proxy for transshipment — “was the most important driver in the full and squid jigger models (relative importance of 49 and 37%, respectively)”; disabling climbed where loitering ran 1,000–10,000 hours while ordinary fishing sat in areas of 100 hours or less, and “more than half of the disabling events by squid jiggers were close enough to loitering refrigerated cargo vessels to be able to undertake a transshipment” [23]. Driver two is the border. Distance to shore dominated the drifting-longline model (33%) and ranked third overall (13%): disabling occurrence “peaked at or just outside 200 nautical miles from shore,” the EEZ limit, “with occurrence decreasing further offshore,” while fishing itself was bimodal, inshore and beyond 200 miles [23]. The excess is not explained by fishing density: more than 96% of EEZ-adjacent disabling events occurred next to foreign EEZs; waters adjacent to Argentina, Russia and Peru showed disabling shares above their fishing shares — “more than 47% of all disabling events adjacent to the Argentinean EEZ” — and although only 11% of EEZs have contested boundaries, “disabling events were twice as likely within or adjacent to these EEZs compared to fishing activity” [23].

Drivers three and four are innocent, and the model separates them cleanly. Chlorophyll — productive water — “was the most important driver in the trawler model (50%),” and the pattern for trawlers and tuna purse seines “suggests that these gear types may typically disable their AIS devices to hide the locations of productive fishing grounds from competitors,” competition being fiercest for trawlers because trawling is confined to continental shelves [23] [23]. For tuna purse seines, “piracy was the most important driver of… disabling, indicating that one of the reasons tuna purse seines disable their AIS devices is likely to hide their positions in dangerous waters” [23]. The case files show the criminal end of the spectrum: the Oyang 77 “disabled its AIS device nine times adjacent to the Argentinean EEZ before being apprehended by the coast guard and charged with illegally fishing within Argentina’s waters,” and the Lu Rong Yuan Yu 668 disabled twice beyond the EEZ “and was eventually captured by the coast guard for illegally fishing within Argentina’s waters” [23]. The most elegant evidence is accidental: one vessel disabled its shipboard AIS but forgot its gear-mounted unit — fitted “to reduce loss of expensive fishing gear” — which kept broadcasting through “an encounter with a refrigerated cargo vessel and a transshipment event confirmed by the vessel’s flag state”; that particular transshipment was properly authorized, proving the mechanism (transfers happen inside the silences) without proving every silence criminal [23].

That restraint is the section’s intellectual core. “The legality of disabling events is often difficult to discern because policies on AIS usage vary by region, regional fisheries management organization (RFMO), flag state, and vessel size,” an “intricate patchwork” that makes individual illegality hard to establish — yet the study “demonstrated strong links between suspected disabling and behaviors of IUU concern, specifically potentially unauthorized border crossings and transshipments” [23]. The application is predictive enforcement: models that forecast disabling can “position at-sea or airborne enforcement and surveillance” and, combined with risky-port intelligence, focus the port inspections the Port State Measures Agreement mandates [23]. The authors’ epitaph for the SDG and CBD targets to eliminate IUU by 2020 — missed, in part for want of exactly this data — is the block’s thesis in one line: “it is the absence of AIS data itself that contains a wealth of information” [23]. Fishing fleets go dark for advantage; the next two sections meet a fleet that is dark by design and by statecraft — the sanctions-era shadow tanker fleet, where the cargo is oil, the flags are rented, and the stakes are measured in billions.

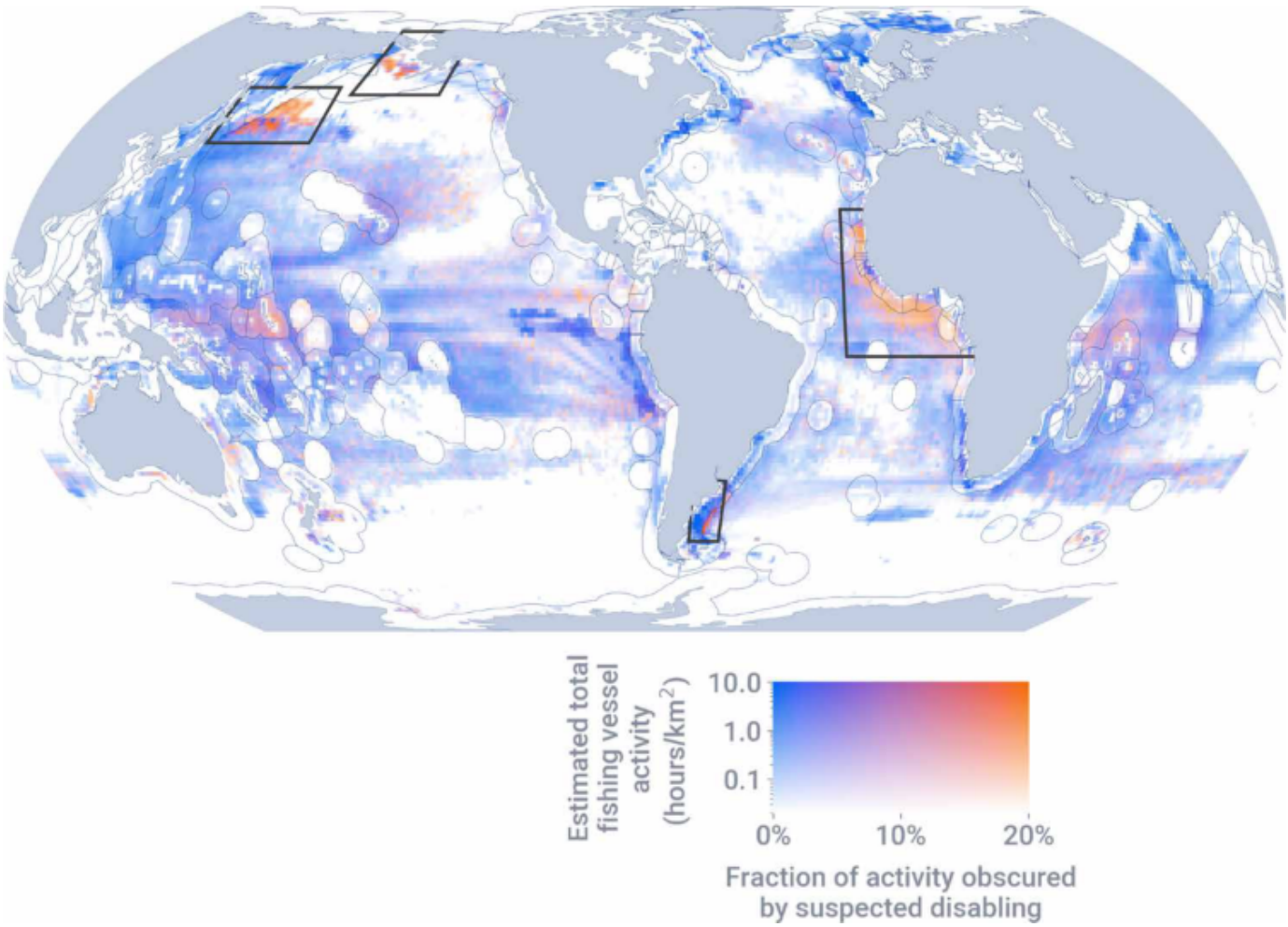


Exhibit. Fig. 1. Estimated total fishing vessel activity and the fraction of this activity obscured by suspected disabling events in areas with sufficient satellite reception quality (10 positions/day). Source: S0020.

The shadow fleet ecosystem: tankers, sellers, managers, and flag states

The fishing fleets of the previous sections go dark one vessel at a time. The shadow fleet is darkness as an industrial system — a parallel shipping industry assembled to move Russian oil outside Western insurance, finance and price-cap enforcement, and tracked monthly by the KSE Institute's Russian Shadow Fleet Tracker. Its September 2026 edition (data for August 2026) gives the ecosystem's anatomy. Volumes first: Russian seaborne oil exports fell to 4.9 million barrels per day (−6.1% month-on-month, −12.4% year-on-year), but shadow-fleet exports rose to 3.0 mb/d (+1.8% m-o-m) — the dark share grows as the total shrinks [24]. The fleet that month comprised "258 unique tankers to export oil, including 166 (64%) shadow ones, of which 120 were EU-designated (47% of the total fleet)"; the EU-designated subset — tankers individually listed by the EU — alone carried 53% of all exports and 86% of shadow volumes [24]. The Baltic is the densest theatre: 128 unique tankers, 65 (51%) shadow, exports of 2.5 mb/d with shadow volumes up 16% m-o-m [24].

The hulls and their paper. A shadow tanker is defined less by its steel than by its documentation stack — owner, manager, flag, insurer — each layer chosen for opacity. Flags concentrate remarkably: "Russia and Sierra Leone were the leading flag states for both the shadow and EU-designated fleets in August 2026, jointly accounting for 56% and 59% of exports, respectively," and in the Baltic the pair reached 63% (shadow) and 69% (EU-designated), while "the share of Equatorial Guinea flag stood at record 8% of total" [24]. Around them orbits a long tail of micro-registries — Oman, Cameroon, Panama, São Tomé and Príncipe, Barbados, Guinea-Bissau — and, in the tracker's own chart category, "False/Unknown" flags [24] [24]. Insurance, the layer that once tied tankers to the International Group of P&I Clubs, is being re-domesticated: "52 of 166 (31%) shadow fleet tankers were identified as ones operating under Russian insurance. Of these 52, 48 (92%) were sanctioned by at least one jurisdiction: EU, US, UK, Canada, Australia, and New Zealand" [24].

The operators. Management is concentrated, not atomized. "Chinese and Russian ship managers remained the largest operators of both the shadow and EU-designated fleets in August 2026, jointly handling 64% of exports for both," rising to 75% (shadow) and 78% (EU-designated) from Baltic ports [24]. The top 10 ship managers moved 881

kb/d — 29% of shadow volumes — and 889 kb/d (34%) of the EU-designated fleet; several of the biggest are themselves sanctioned, including South Fleet Ltd and Invest Fleet Ltd (Russia) and Dreamer Shipmanagement LLC-FZ and Albatross Shipmanagement (UAE) [24]. In the Baltic the top 10 managers handled 371 kb/d (31%) of shadow and 355 kb/d (34%) of EU-designated volumes [24].

The sellers. Above the ships sit the trading desks that actually sell the oil. In June–August 2026 "the top 10 sellers accounted for 85% of crude oil and 56% of oil product among shadow fleet" (84% and 61% for the EU-designated fleet); the leading crude seller for both was UAE-based Redwood Global Supply FZE LLC — itself an EU-designated company — at 21% and 22% respectively, while Rosneft led oil products at 16% and 19% [24] [24]. In the Baltic book, the concentration is tighter still — top-10 sellers at 97% of crude and 75% of products for the shadow fleet, with Rusexport the top identified crude seller — and the sanctioned names recur: Redwood and Surgutneftegas sit in the top-10 seller lists of both fleets [24].

The buyers. The demand end is the least hidden layer: "China and India remained the main destinations for both the shadow and EU-designated fleets in August 2026, jointly accounting for 64% and 63% of exports, respectively" [24]. From the Baltic, India and Singapore led identified destinations at about half of volumes — while "47% and 48% had unknown destinations," a reminder that even the tracker's cargo-level view dissolves at the edges [24].

The enforcement lag. The tracker's most damning statistic is about the designation regime itself: "during August 2024–August 2026, 444 of 625 (71%) EU-designated oil tankers loaded at Russian ports at least once after their designation, transporting a total of 1.2 billion barrels, equivalent to 28% of Russia's total seaborne oil exports during the period" [24]. Designation, in other words, names the fleet faster than it stops it. The ecosystem's resilience is structural: substitute registries when flags are pressured, substitute insurers when clubs withdraw, substitute sellers when traders are listed, and a buyer base outside the sanctioning coalition. It is Section 21's lesson inverted — there, derelicts persisted because no authority could act; here, a fleet persists because every actor in its stack can be swapped. The next section reads the same tracker for the trade's overall shape: export routes, the designated fleet's share over time, and what two years of designations have and have not changed.

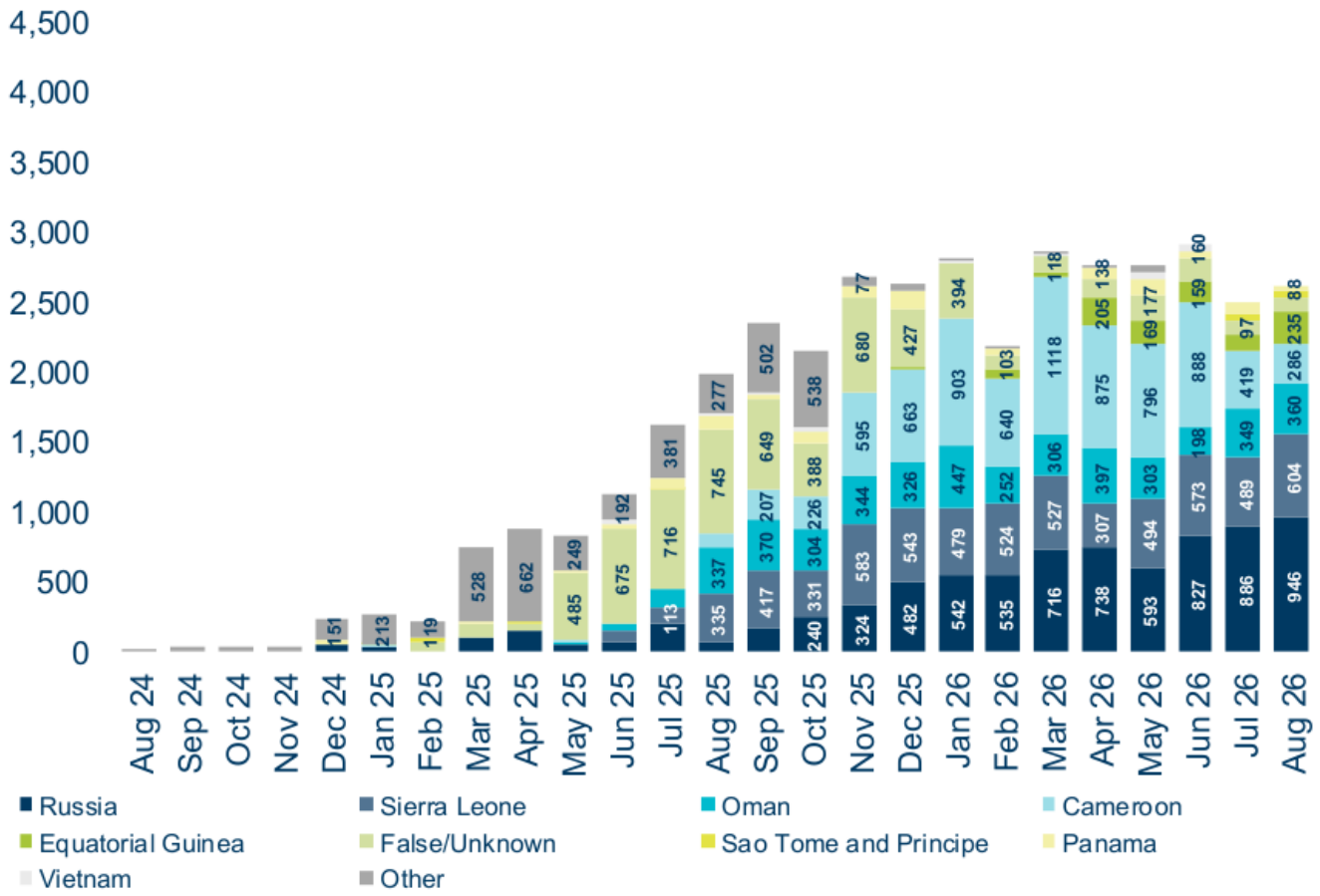


Exhibit. EU-designated fleet volume by flag state, kb/d 4,500 4,000 3,500 3,000 2,500 2,000 1,500 1,000 500 0 Russia Sierra Leone Oman Cameroon Equatorial Guinea False/Unknown Sao Tome and Principe Panama Vietnam Other Aug 24 Sep. Source: S0026.

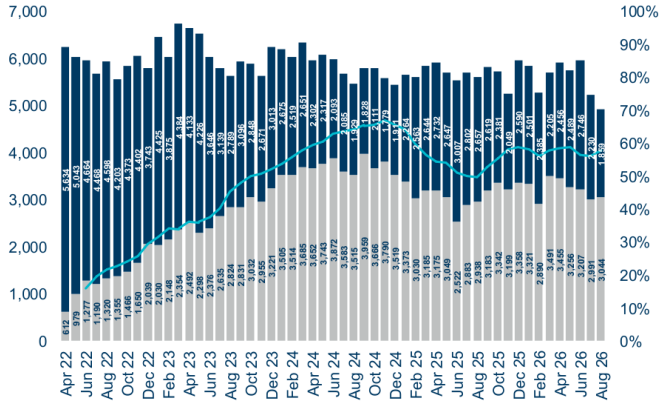
Russian seaborne oil exports and the EU-designated fleet

The tracker's categories deserve precise statement, because everything in this block turns on them. Its definitions: the **shadow fleet** comprises "tankers without links to G7+ jurisdictions for registered ownership, management, and flagging, as well as without P&I insurance by the International Group of P&I Clubs"; the **EU-designated fleet** comprises "tankers that conducted loadings in the respective month after being designated by the EU, including ones that may also be designated by other jurisdictions"; everything else is the "white fleet" [24]. Note what the middle category measures — not tankers stopped, but tankers listed and still loading. That is the story of Russian oil exports in August 2026: a shrinking trade carried increasingly by named ships.

The trend lines run in opposite directions. Total seaborne exports have slid from above 6 mb/d in 2022 to 4.9 mb/d in August 2026 (−12.4% year-on-year), while the shadow fleet's share of what remains has climbed steadily across the tracker's April 2022–August 2026 series, and shadow volumes rose in absolute terms in August (+1.8% m-o-m to 3.0 mb/d) even as the total fell [24] [24]. The designated fleet's share has grown the same way since designations began in earnest: EU-listed tankers carried 53% of all August exports [24]. Regionally, the Pacific and Baltic dominate the dark trade: "shadow tankers transported the largest oil volumes from Pacific Ocean (45%) and Baltic Sea ports (40%) in August," with EU-designated tankers nearly identical at 43% and 39% [24]. From the Baltic specifically — 2.5 mb/d of exports — India took 48% of shadow-fleet volumes and Singapore 2%, while "shipments with an unknown destination account for 47% for shadow fleet and 48% for EU-designated fleet" [24] [24]. A trade in which half the cargoes sail for "unknown" is a trade whose bills of lading are part of the camouflage.

The fleet itself is old — old enough to reconnect this block to the derelicts of Sections 19–24. "Tankers older than 15 years accounted for 92% of the shadow fleet" in August 2026; in the Baltic the figure was 94% [24] [24]. These are hulls the mainstream market was retiring: bought up as the white fleet renewed itself, reflagged to the registries of Section 35, insured outside the International Group system (31% of shadow tankers identified under Russian insurance alone), and worked on the longest hauls in the trade — Baltic and Pacific loadings to India and China [24] [24]. The parallel to the ADV problem is structural: vessels whose value has fallen below the cost of compliant operation do not disappear; they migrate to the jurisdiction — or the non-jurisdiction — where they can still trade, and the casualty risk migrates with them, now attached to cargoes of crude oil rather than abandoned yachts, and to insurance arrangements whose ability to pay for a major spill is untested.

And the designations? The tracker's two-year audit is the block's verdict: "444 of 625 (71%) EU-designated oil tankers loaded at Russian ports at least once after their designation, transporting a total of 1.2 billion barrels, equivalent to 28% of Russia's total seaborne oil exports during the period" [24]. The August snapshot shows the same behaviour in miniature: "among 166 shadow fleet tankers, 120 (72% of shadow fleet or 47% of the total fleet) made voyages while under EU sanctions" [24]; in the Baltic, 49 of 65 shadow tankers (75%) did so [24]. Designation has become a registry of the trade rather than a barrier to it — the EU list and the KSE tracker between them now provide the best public census of the fleet, much as GFW's disabling dataset turned transponder silence into evidence. What designation changes is the cost stack (insurance, finance, port access in coalition states) rather than the physical flow; what it has not produced is interdiction on the water. The exports data close the dark-fleet block with the same asymmetry the fishing sections found: visibility is achievable — satellites, cargo data and corporate registries can reconstruct the system in near-real time — but visibility is not enforcement. The paper now changes register entirely, from watching ships to owning them after they sink: salvage law, sunken warships, and the legal afterlife of wrecks.



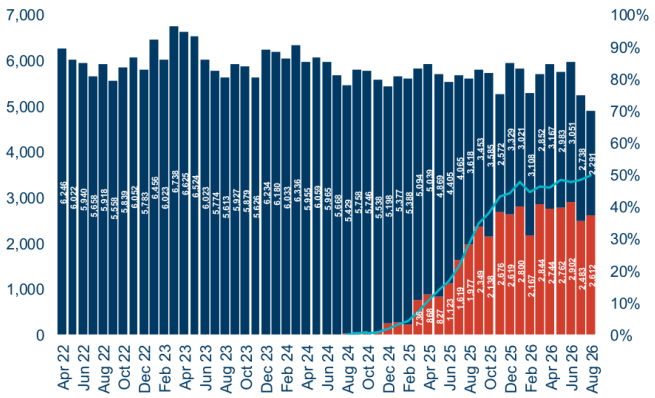


Exhibit. Russian seaborne oil exports (EU-designated vs. not EU-designated), kb/d 7,000 100% 7,000 100% 90% 90% 6,000 6,000 80% 80% 5,000 70% 5,000 70% 60% 60% 4,000 4,000 50% 50% 3,000 3,000 40% 40% 2,000 30% 2,000 30% 20% 20% 1. Source: S0026.

Law of the Lost Ship: Salvage, Abandonment, and Sunken Warships

Immunity of sunken warships and salvage law

A warship, in law, does not stop being a warship when it sinks. Aznar-Gómez's survey begins from the afloat rule: warships attract sovereign immunity, "historically accepted both by tribunals and doctrine" — the 1926 Brussels Convention immunizes government vessels in public service from arrest and seizure, UNCLOS Articles 32, 95 and 236 "plainly recognise that warships enjoy immunity" (a warship being defined in Article 29 by flag, command and discipline), and the ILC's draft Article 16 confirms immunity for warships, naval auxiliaries and state vessels in non-commercial service [18]. For sunken warships, "as a principle, nothing in the legal texts suggests that they lose their immunity by the simple fact of their sinking." The functional counter-argument — a wreck is no longer a "ship," so immunity lapses — "has been not only criticised by doctrine but rejected by State practice — domestic and international" [18]. The European Parliamentary Assembly's Recommendation 1486 (2000) implicitly concedes the rule by urging states to negotiate agreements mitigating the immunity they retain over sunken war vessels, "with particular regard to the high proportion of such vessels which are of significant historic value and to the high proportion which contain human remains and especially those which are war graves" [18].

Salvage law is doubly excluded. First, its premise is peril: as O'Keefe and Nafziger put it, "the law of salvages relates solely to the recovery of items endangered by the sea… For underwater cultural heritage, the danger has passed; either a vessel has sunk or an object has been lost overboard. Indeed, the heritage may be in greater danger from salvage operations than from being allowed to remain where it is" [18]. The 1989 International Convention on Salvage, read on its terms, defines salvage as assisting property "in danger" and a vessel as a structure "capable of navigation" — language that "plainly excludes wrecks from salvage operations" [18]. Second, warships are carved out expressly: the 1910 Brussels salvage convention (Art. 14) "does not apply to ships of war or to Government ships appropriated exclusively to a public service"; the 1967 Protocol's partial extension still confined claims against a state to that state's own courts; and the 1989 Convention's Article 4(1) reaffirmed that it "shall not apply to warships or other non-commercial vessels owned or operated by a State and entitled, at the time of salvage operations, to sovereign immunity under generally recognised principles of international law unless that State decides otherwise" [18]. Article 29 adds that state non-commercial cargoes may not be seized, arrested or proceeded against in rem without the owner state's consent [18]. States could also reserve historic wrecks out of the Convention under Article 30(1)(d) — nearly all major maritime states did — and IMO's position was that the exclusion existed even without a reservation, given the Convention's private-law character; IMO's draft Wreck Removal Convention likewise "expressly leaves aside the removal of sunken warships otherwise decided by the flag State" [18]. Aznar-Gómez submits the resulting customary rule: salvage law does not apply to warships "unless otherwise expressly decided by the flag State" [18]. Domestic law agrees — the US Salvage Act excludes "ships of war," and Britain's Protection of Wrecks Act 1973 and Protection of Military Remains Act 1986 remove protected wrecks from salvage [18].

State practice is consistent, and it is possessive. The US State Department's 1965 formulation: "where ownership to vessels or cargoes resided in US Government at time of sinking, the US retains title thereto subject to explicit transfer or abandonment. In the absence of transfer or abandonment of US interests, therefore, salvage of such cargoes or hulks requires US consent" [18]. When Japan paid $2,214,007.36 in 1938 for sinking the USS Panay in the Yangtze and claimed the wrecks as its property by indemnity, Washington accepted transfer for the private vessels but refused the gunboat: it found "no authority in law for acceding in any case to such a request" of salvage [18]. The Soviet Union told Japan in 1980, over Japanese salvage of the Admiral Nakhimov, that "a sunken warship is completely immune from the jurisdiction of any State other than the flag State"; Japan's rejoinder — that capture in 1905 had transferred all rights to the captor — is the exception that marks the rule's boundary (capture, not sinking) [18]. Britain's position, restated in the 1989 Exchange of Notes with South Africa over HMS Birkenhead (lost 1852 with 445 people and reputedly 240,000 gold sovereigns), is that "the Crown maintains rights and interest in ships of the Royal Navy which have sunk, wherever they may be and without time limit" — the Notes regulated salvage terms while safeguarding the wreck's status as a military grave and expressly not prejudicing UK ownership [18]. Even Germany's sunken U-boats kept their owner through total defeat: Norwegian courts affirmed title to U-76 (sunk 1917) "rested with Germany until 1945," and Singapore's High Court held of U-859 (sunk 1944 with a mercury cargo) that "the German State has never ceased to exist despite Germany's unconditional surrender in 1945 and whatever was the

property of the German State… still remains the property of the German State" [18]. The cooperative model exists too: Britain and Canada's 1997 Memorandum on HMS Erebus and HMS Terror — Franklin's ships, sunk 1848 — has Britain, "as owner of the wrecks," assign "custody and control" to Canada while expressly not waiving "ownership or sovereign immunity… while they are on the seabed" [18].

What salvors are left with is the law of finds, and it turns entirely on abandonment: salvage gives compensation, not title, whereas finds can give title against all but the rightful owner — so "where there is no intention to abandon and no doubt as to who is the owner such claims to acquire title (as opposed to salvage rights) appear unfounded" [18]. The Sea Hunt litigation operationalized this: Spain's frigates La Galga (1750) and Juno (1802) were held not abandoned — a 1763 peace treaty did not do the work — under the Columbus-America standard that abandonment "must be proved by clear and convincing evidence, such as an owner's express declaration," an inference a court may draw for ancient wrecks only until "a previous owner appear[s] and assert[s] his ownership interest" [18]. The next section takes up the other frame states use for these wrecks — not immunity, but heritage.

Sunken warships as underwater cultural heritage

Section 37 kept sunken warships inside the law of state property and immunity. The heritage frame starts from the same wrecks and asks a different question — not "whose is it?" but "what is it for?" Aznar-Gómez's bridge between the two: even on the controversial assumption that a wreck is a ship no more, "the wrecks have [not] ceased to be the public property of the flag State," and, as Anastasia Strati puts it, sunken state vessels "retain their status as public State property so that their recovery may require the consent of the flag State" [18]. Quéneudec's comment on the Alabama case generalizes it: the transformation of a ship into a wreck "never has the effect of making the owner's right disappear ipso facto… this holds for wrecks in general and a fortiori still more for wrecks of public ships"; a state's property in its warship wreck "subsists as long as that State has not expressly renounced its right or voluntarily transferred it, save in the case of capture by another State in exercise of belligerent right" [18]. The wreck's contents split legally: objects "still of a military character (documents, instruments, weapons, etc.)" keep sovereign immunity by their jure imperii calling, while objects whose character has become "historical, spiritual or archaeological (old guns, human graves, charts, etc.)" merit different treatment — but treatment, not transfer, since "States are still owners of these artefacts, precluding the reversion of title to the finder since a rightful owner always exists unless an express abandonment act has been proven" [18]. US admiralty's vintage rule says the same of loose artefacts: when "articles are lost at sea the title of the owner in them remains," and where property "was involuntarily taken from his control, the law is hesitant to find an abandonment" [18].

The heritage regime proper is mostly national, and it typically displaces admiralty rather than decorating it. The US Abandoned Shipwreck Act was passed because Congress judged that "admiralty principles are not well-suited to the preservation of historic and other shipwrecks to which this Act applies" — such wrecks are "not considered… to be in marine peril, necessitating their recovery by salvage companies"; the ASA vests title to abandoned wrecks embedded in state submerged lands in the state, and §2106(a) provides that "the law of salvage and the law of finds shall not apply" to them [18]. Scovazzi notes salvage and finds are anyway parochial: bodies of law "typical of a few common law systems but… complete strangers to other domestic legal systems" [18]. The civil-law and patrimony statutes surveyed instead assert state ownership or permit control directly: Spain's Ley 16/1985 makes the civil-code finds rule inapplicable to historic patrimony; China's 1989 Regulations (Art. 3) "grant ownership of underwater cultural relics to the State"; Tunisia's 1994 code (Art. 73) vests archaeological objects in inland and territorial waters in the state; Italy places cultural objects in the public domain; Australia's Historic Shipwrecks Act 1976 (§13) subordinates any action on wrecks to a ministerial permit; South Africa's National Heritage Resources Act 1999 treats heritage resources as part of the national estate, requiring permits for any activity directed at them; Israel's Antiquities Law 1978 does the same [18] [18]. At the multilateral level, the 2001 UNESCO Convention on the Protection of the Underwater Cultural Heritage supplies the safeguard framework the IMO itself points to (its Articles 3, 5, 9 and 10), and the Council of Europe's Parliamentary Assembly has recommended avoiding commercial recovery operations altogether [18] [18].

The friction point is which state. Coastal and flag states collide most sharply where empire left the wrecks: Latin American and Caribbean governments declared at Santo Domingo (16 June 1998) that underwater cultural heritage "is the property of the State in which it is found and through this it is the heritage of the Humanity" [18]. Spain and Uruguay are litigating the principle over the frigate San Salvador, sunk with her crew on 31 August 1812 in Maldonado Bay: Uruguay, under decrees of 1976 and 1986, "considers that title over the wreck reverted to Uruguay," while Spain reiterated that it "had not abandoned the vessel," insisted that in the San Salvador's case "human remains should be treated with particular respect," and proposed a general cooperation agreement on "common underwater

cultural heritage" [18]. The UNESCO Convention's compromise — visible already in Section 21's authority discussion — is procedural rather than proprietary: a state with a "verifiable link, especially a cultural, historical or archaeological link" to a wreck is an "interested State" entitled to be informed and consulted (Articles 7(3), 9(5), 11(4)), a formula built precisely for the Etruscan problem — flag states that no longer exist [18]. The moral centre of the heritage frame is the grave: the Parliamentary Assembly's Recommendation 1486 (2000) pressed states to mitigate retained immunity "with particular regard to… the high proportion which contain human remains and especially those which are war graves" [18], and the Birkenhead settlement shows the practice — salvage terms negotiated under an agreement that "safeguarded the wreck's status as a military grave" [18]. Heritage law thus neither frees the wreck nor freezes it: title stays with the flag state, meaning accrues with time, coastal states gain voice and sometimes custody (the Erebus/Terror model of Section 37), and commercial recovery — the salvor's business model — is the one activity every instrument in this section is designed to exclude. The next section returns to the salvor's side of the courtroom: a US case about not a ship at all, but a Navy aircraft, and what "abandonment" requires when the owner is the federal government.

Federal salvage rights and the Navy aircraft case

The cleanest statement of the federal position comes from a case about an airplane. The aircraft was a Navy TBD-1 Devastator torpedo bomber, built in 1938, which flew neutrality patrols, joined the carrier Yorktown in 1941, and fought at the Coral Sea — where TBD-1s sank the Japanese carrier Shoho and badly damaged Shokaku — and at Midway. On 1 July 1943, flying a torpedo-attack training mission from Miami, it suffered mechanical failure; "the pilot and crew parachuted safely from the plane, which crashed in deep international waters approximately eight miles east of Miami Beach" [4]. The Navy "did not know exactly where the TBD-1 crashed, and it 'struck' the plane from the inventory of active aircraft in September 1943. Since that time, the Navy has taken no steps to locate or salvage the plane" [4]. In 1990, searchers hunting Spanish galleons found it; the National Museum of Naval Aviation declined the location for want of acquisition budget; the coordinates were sold to collector Douglas Champlin's Windward Aviation, which filmed the wreck in 1994 and 1998, recovering a canopy portion and the radio mast, investing "over $130,000" [4]. Both sides agreed the aircraft was uniquely valuable: a Midway and Coral Sea veteran, and "no TBD-1 planes have been preserved for display or study. In fact, the only other known TBD-1 also lies submerged in deep water" [4]. After negotiations failed, Champlin's International Aircraft Recovery sued in rem for exclusive salvage rights, a liberal salvage award, or title under the law of finds. The district court sided with IAR. The Eleventh Circuit (July 17, 2000) reversed in the opinion's first paragraph: "the United States, as owner of the plane, can prohibit IAR's salvage efforts" [4].

Abandonment. The Navy's paperwork did not give the plane away. "Striking" an aircraft from the active inventory is an accounting act, not a disposition: the court leaned on the analogous Kern Copters case, where the Ninth Circuit held the Army had not abandoned a helicopter crashed in Guatemala by "'dropping [it] from accountability records'" [4]. Nor did the Surplus Property Act of 1944 help IAR: the Act ordered that surplus property "shall be disposed of," but its focus "was on the sale of surplus of World War II matériel, including salvage and scrap" — it "did not mention the abandonment of property" [4]. When the Surplus Property Administration's Regulation §8319 (1945) first let agencies abandon matériel, it required "an affirmative finding made by a 'responsible officer, approved by a reviewing authority,' and reduced to writing" — typically plus factual findings on value and handling costs and thirty days' published notice offering the property for sale or donation; "the Navy neither made such findings nor compiled the written report" for the TBD-1, and the Board's case-by-case abandonment permissions (one covered surplus submarine and torpedo netting) included no crashed Navy planes [4]. Behind the regulations stands the Property Clause point from the opinion's first section: federal property is disposed of only as Congress provides, and IAR's invitation to apply admiralty's common-law abandonment test instead failed because its authority, Deep Sea Research, "involved a privately-owned steamship with privately-insured cargo," and IAR cited "no other cases for the proposition that the common law of admiralty supercedes the Constitution" [4] [4].

Salvage. Nobody disputed that salvage law can reach a submerged airplane — "it is settled that all manner of objects other than vessels and their cargo are subject to salvage," and raising sunken craft is a recognized salvage service [4]. The fight was over consent. The district court had reasoned that a salvor holds an absolute right to aid a vessel in marine peril whenever a prudent owner would have accepted help. The Eleventh Circuit accepted the distress rule's core — a deserted vessel in peril may be salvaged without prior assent (The Laura; The Barque Island City), and its owner "cannot contest the salvor's right to attempt a rescue by claiming after the fact that the assistance was unwanted" — but held that an owner who does object, timely and while only its own property is at stake, prevails [4]. The authority is Merritt & Chapman (1927): "salvage cannot be exacted for assistance forced upon a ship"

[4]. The contrary glosses collapsed on inspection: Tidewater Salvage's suggestion that rejection must itself be prudent was dicta that "appears never to have been put to the test," and Spreckels, quoted whole, actually affirms that "the owner of a vessel disabled or in distress does not thereby lose the control of his property. He has the right to refuse or accept any offers of assistance that may be made, or to adopt his own measures for the preservation of his vessel" [4]. A volunteer who forces services on an unwilling owner is, in the doctrine's older phrase echoed in the opinion's authorities, a gratuitous intermeddler entitled to no award — the same principle that lets a wreck's title-holder bar salvors outright, as Spain did in Sea Hunt [4].

The case is the domestic twin of Section 37's international rule: for state craft, time, depth and neglect do not transfer rights; only an authorized, affirmative act of the sovereign owner does. The Devastator stayed on the seabed — the last of its kind, owned by a Navy that would neither raise it nor let anyone else. The next section generalizes the point into the paper's legal synthesis: abandonment in law versus abandonment in fact, and why the gap between them is where ghost ships live.

Abandonment in law versus abandonment in fact

Every ghost in this paper lives in the same gap: between a thing the sea has taken and a legal act that says so. The law, across all four of the paper's domains, refuses to infer abandonment from circumstances. For a derelict vessel in Washington, "abandoned" is not a description but a decision — "the legal ability of a governing agency to declare a vessel 'abandoned' and thus remove and dispose of it" — and the Blue-Ribbon glossary splits the world cleanly: "'Abandoned' — the legal condition of the vessel, in terms of ownership. 'Derelict' — the physical condition of the vessel, in terms of seaworthiness" [5] [5]. A boat can rot at anchor for a decade — abandoned in fact to any observer — and remain, in law, simply owned, because the declaration that would move her into a removal programme has never been signed; the Florida turn-in programme polices the same boundary from the other side, accepting only vessels that "do not yet satisfy the definition of 'derelict' or 'abandoned'" [5]. For fishing gear, the legal line runs through intent no observer can see: MARPOL bans discharging synthetic nets yet "does not apply to the accidental loss of such nets, provided that all reasonable precautions have been taken" — so the identical net is lawful flotsam or illegal dumping depending on a state of mind aboard a vessel that has sailed away [6].

For state craft, the requirement hardens into formality. US doctrine since 1965: the government "retains title thereto subject to explicit transfer or abandonment," and salvage "requires US consent" [18]. Admiralty's Columbus-America standard demands abandonment "proved by clear and convincing evidence, such as an owner's express declaration," and bars the inference precisely when it matters — "should a previous owner appear and assert his ownership interest" [18]. The IAR litigation shows the machinery: the Navy's striking of the TBD-1 from its inventory was an accounting entry, not a disposition (the Army's crashed helicopter in Kern Copters was likewise not abandoned by being "dropp[ed] from accountability records"); the Surplus Property Administration permitted agency abandonment only on written findings by a responsible officer, approved by a reviewing authority — findings never made [4]. The Eleventh Circuit conceded the plane — "submerged in a corrosive environment and slowly disintegrating" — "is in a state of marine peril," yet reversed because the district court "underappreciated the authority of a vessel's owner to prevent others from interfering with its property" [4]. The owner's right to refuse is ancient and absolute in form: "if the master of a burning vessel prefers to allow her to burn rather than to permit outside parties to extinguish the flames, he may do so... he should not be assisted against his will" (Chouteau, 1881); and Merritt & Chapman's rule that "salvage cannot be exacted for assistance forced upon a ship" [4] [4]. Even the refusal has a factual dimension the court remanded to explore: statutes governing public property may amount to a "constructive rejection," but the Navy's years of negotiation might have been acquiescence — the remand asks "when the United States effectively rejected the salvage efforts," with an award for past efforts still possible [4]. Law and fact contaminate each other to the end.

The heritage regime resolves the tension by demoting the question. UNESCO's Convention, Article 4, provides that activities directed at underwater cultural heritage "shall not be subject to the law of salvage or law of finds, unless it: (a) is authorized by the competent authorities, and (b) is in full conformity with this Convention, and (c) ensures that any recovery of the underwater cultural heritage achieves its maximum protection" — three cumulative conditions, so that "even with an express abandonment act of the flag State," salvage on a sunken warship as heritage would need authorization and full archaeological method [18]. Article 2(5) makes "the preservation in situ of underwater cultural heritage... the first option before allowing or engaging in any activities directed at this heritage" [18]. The reason is salvage's engine: "no cure no pay," under which "salvage is motivated by economic considerations; the salvor is often seeking items of value as fast as possible rather than undertaking the painstaking excavation and treatment of all aspects of the site" — salvors' sonar targets wrecks with precious metals and untargets those con-

taining “spices, charts, human beings” [18]. Forrest’s conclusion, quoted by Aznar-Gómez: salvage law “does not promote a uniform system of law applicable to [underwater cultural heritage] and is therefore inappropriate as the bases for an international agreement”; the replacement regime turns on historical importance rather than marine peril, with peril surviving only as the factor deciding “whether the [heritage] should be recovered or preserved in situ” [18]. Dromgoole and Gaskell’s dictum — “there comes a time when salvage law must give way to archaeological law” — is the heritage block in one sentence [18].

Read back across the paper, the pattern is exact. The Baychimo was abandoned in fact in November 1931 — crew flown out, hull vanished in a storm — yet her ethnographic cargo moved through the world on a chain of custody, paperwork and a misspelled donor name, not on the law of finds [1] [1]. The shadow fleet inverts the derelict: legally impeccable on paper — flagged, managed, insured after a fashion — while factually outside every system the paperwork invokes [24]. The dark fishing fleets are legally present and factually invisible until a satellite supplies the missing fact [20]. And the ADV statutes exist because legislatures learned that without a legal act of abandonment, factual abandonment is unactionable — the hulk stays, the owner is gone, and no one may touch it [5]. Ghost ships are not, in the end, a nautical phenomenon. They are what falls between a fact the sea has already decided and a formality no one has performed. The paper’s final movement turns from law back to physics and narrative: what investigators can reconstruct after the fact, and the sea’s own last word on hulls that sail themselves.

The Deadliest Crossing: Forensic Oceanography and the Mediterranean

Forensic oceanography: blaming the rescuers

The paper has spent forty sections learning to read the sea’s evidence — drift models, sonar, satellite constellations. Forensic Oceanography, the research project of Charles Heller and Lorenzo Pezzani, turns those same tools to a courtroom purpose: reconstructing events in the Mediterranean to assign responsibility for deaths that official narratives prefer to leave causeless. Their report Blaming the Rescuers (2017) is a self-described “counter analysis” of claims by Frontex, the European Border and Coast Guard Agency [25]. The backdrop is the disaster of April 2015: shipwrecks on 12 and 18 April killed “more than 1,200 people… the largest to have been documented in recent Mediterranean history” [25]. Their earlier report, Death by Rescue (2016), attributed those deaths to the termination of Italy’s Mare Nostrum operation, which “had patrolled close to the Libyan coast to rescue migrants in distress”; its end “left a huge gap in Search and Rescue (SAR) capabilities” that merchant vessels partially filled, and deaths spiked — a policy failure the European Commission’s President conceded in terms: ending Mare Nostrum “was a serious mistake… It cost human lives” [25]. The SAR gap was filled by NGOs — and by 2016–17 the NGOs themselves were under a “de-legitimisation and criminalisation campaign” involving Frontex, politicians, media, and exploratory inquiries by Italian prosecutors [25].

The substantive accusation, as the report summarizes Frontex’s argument, has three limbs: SAR NGOs are “(1) constituting a ‘pull-factor’ leading to more migrants attempting the dangerous crossing; (2) ‘unintentionally helping criminals’ by encouraging smugglers to use even poorer quality boats and more dangerous tactics; (3) in turn making the crossing more dangerous for migrants” [25]. Catania prosecutor Carmelo Zuccaro later adopted the line in a fact-finding mission on the NGOs [25]. Forensic Oceanography’s method is to concede the trends and contest the causality. The trends are real and agreed: crossings rose from 153,842 (2015) to 181,436 (2016); average loading of rubber boats rose “from an average of 103 people per boat in 2015 to 122 in 2016”; deaths rose from 2,892 to 4,581 and the mortality rate “from 1.84 in 2015 to 2.5 in 2016” — making 2016 “the deadliest year on record for Mediterranean crossings” [25] [25]. What they challenge is “the way” Frontex connected them, which “occlud[ed] much more important processes and actors — including the regional political and economic contexts in Africa, EU policies and operations at sea, the intervention of the Libyan Coast Guard and the organisation of smuggling in Libya” [25].

The counter-causality runs through the EU’s own anti-smuggling operation. By early 2016, “the effects of the EUNAVFOR MED operation and the dynamics within Libya had combined to make the conditions of crossing increasingly dangerous”: more people, in worse boats, in worse weather, “in distress closer to the Libyan coast” [25]. The mechanism was interdiction economics — as naval operations destroyed captured wooden vessels, smugglers shifted to expendable rubber boats: SAR events “leaped from 906 in 2015 to 1,424 in 2016,” smugglers “increasingly sent out many boats at once,” with “a staggering peak of 53 SAR events in one day reached on 29 August” 2016; loadings reached 200 people per rubber boat; the rough-cut floor planks “broke under the migrants’ weight” (Frontex’s own AFIC report); suffocation deaths, once a horror of wooden holds, were “reported by NGOs even on rubber boats” [25].

A new tactic appeared in EUNAVFOR MED's own reporting: "a skiff towing a rubber boat without an engine, which is then left adrift" — lethal in the 26 May 2016 case documented by the Watch The Med Alarm Phone [25]. The naval command itself supplied the legal conclusion: migrant vessels, given "the limited supply and the degree of overloading," are "to be considered in distress according to international conventions from the moment they launch," and "the majority of migrants still die inside or very close to Libyan territorial waters" [25]. NGOs, the report argues, moved toward the Libyan coast in response to distress moving shoreward; their presence "may in turn have contributed to consolidating specific shifts in smugglers' practices — such as no longer providing migrants with a satellite phone — but was not the cause of the worsening conditions of crossing" [25].

The statistical limb is presented with its caveat attached — a model of forensic honesty: monthly mortality in 2016 "had risen in early 2016 before NGO SAR assets returned to the central Mediterranean following their winter break, and declined in parallel to their redeployment," rising again "only when SAR NGOs' presence decreased at the end of the autumn" — a "striking negative correlation" (−0.314 Pearson's R) between NGO vessel numbers and mortality, which the authors note "is not statistically significant" and only reliable for 2016, though 2015 showed −0.532, borderline significant [25]. The scale of the accused operation is itself part of the rebuttal: at maximum deployment the NGOs fielded 12 vessels and had become "the largest SAR operator in the central Mediterranean" — because no state would — and over Easter 2017 they "took the lead in the rescue of more than 9,000 migrants," with ships like Jugend Rettet's Iuventa, nominal capacity 100, taking aboard hundreds [25] [25]. The report's conclusion is political: the "toxic narrative" diverts attention from "the continuing SAR gap that has made SAR NGOs essential in the first place," continues deterrence-by-danger by other means, and steers policy toward reliance on the Libyan Coast Guard — interceptions that return migrants to conditions the UN human rights office documented as "arbitrary detention, torture, sexual violence, forced labour and trafficking" [25]. For this paper's purposes, the deeper point is methodological: forensic oceanography treats the sea's data exhaust — drift, timing, position, correlation — as testimony. The next section stays in the same waters, on the smugglers' side of the evidence problem.

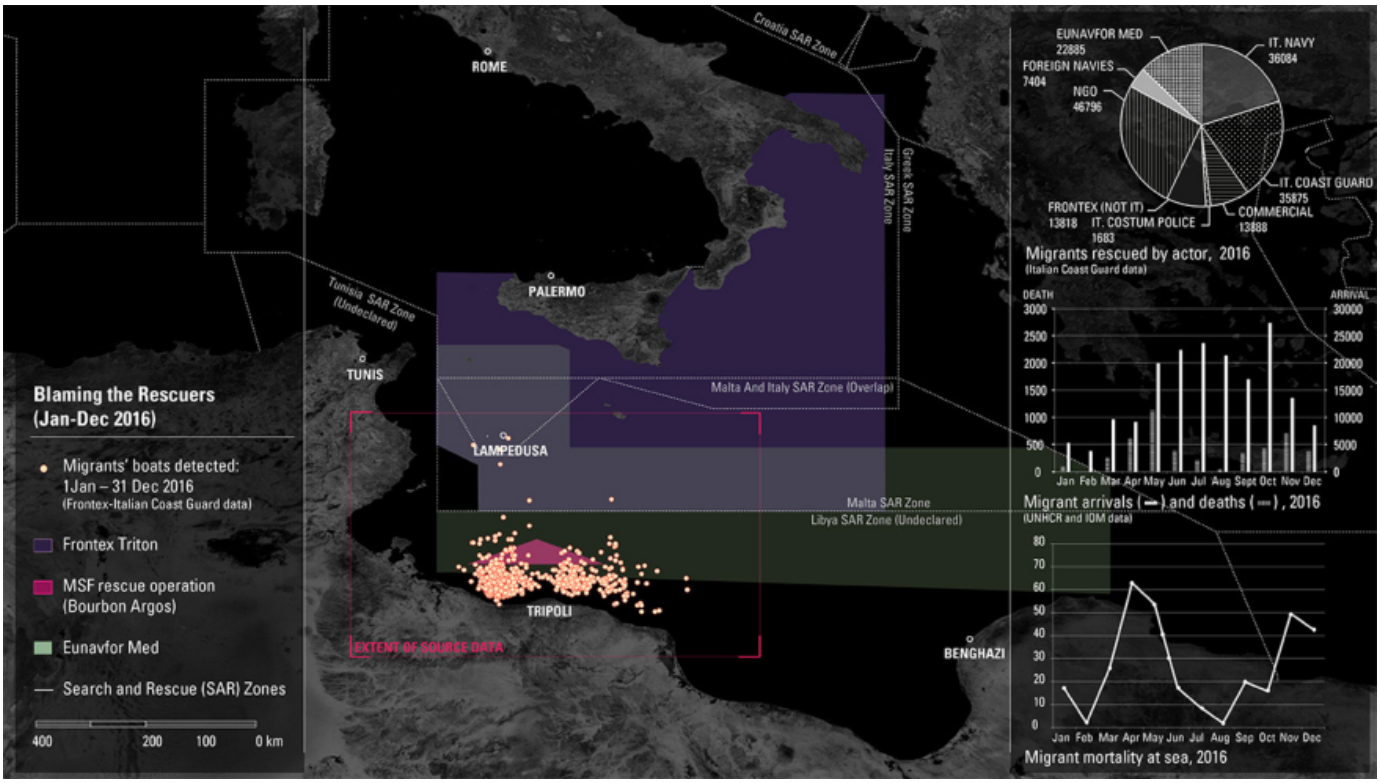


Exhibit. Map and figures of the situation in the central Mediterranean, January-December 2016: SAR zones, detected migrants' boats off Tripoli, and rescues by actor (Forensic Oceanography) [S0023]. Source: S0023.

Smugglers' tactics and the Central Mediterranean route

The Central Mediterranean route runs from the Libyan coast to Malta and Italy — Lampedusa and Sicily — and it is crossed, in the European Parliamentary Research Service's summary of Frontex data, "in old, often unseaworthy fishing boats or small rubber dinghies, usually with insufficient fuel and without proper navigation systems" [26]. It is one corridor in a system whose volumes swing violently: route detections show the Central Mediterranean carrying 40,000 people in 2013, 170,760 in 2014 and 153,964 in 2015, while the Eastern Mediterranean route exploded from

50,830 (2014) to 885,386 (2015) — seventeen times higher in a single year [26] [26]. The system's deadliness is unmatched: the IOM estimates the Mediterranean accounted for 75% of all border-related fatalities worldwide in 2014, with 22,400 deaths between 2000 and 2014 and more than 3,770 people lost at sea in 2015 alone [26]. And every number in this paragraph carries the evidence problem the next section takes up: given irregular entry, "figures can only be estimates" — some entrants are never detected, others double-counted, Greece arrivals "again reported when entering the EU for the second time through Hungary or Croatia" [26].

Against that system, the pull-factor theory of §41 fails its natural experiments. Frontex's own 2017 Annual Risk Analysis recorded 181,459 arrivals from sub-Saharan, West and Horn Africa in 2016, up 18%, a trend "consistent with previous year-on-year increases" that began in 2014–15, when NGO presence was still minimal [25]. The control case is the Western Mediterranean route from Morocco, which registered "a 46% increase… from 2015 to 2016, in the absence of any SAR NGO assets" [25]. And the reverse experiment had already been run: ending Mare Nostrum did not reduce crossings in early 2015 — it only increased deaths [25]. What actually moved the Central Mediterranean numbers was composition and origin-country pressure. Frontex's early-2016 analysis shows the 2015 headline decline (153,946 detections, −10%) was a Syrian story — Syrians fell from about 40,000 (2014) to fewer than 7,500 (2015) as they shifted to the eastern route — while East and West Africans "steadily increased from below 80,000 in 2014 to more than 108,000 in 2015 (+42%)" [25]. EUNAVFOR MED nonetheless claimed its first-year 9% flow reduction partly for its own "deterrence effect," a reading Forensic Oceanography treats as mistaking a route substitution for a policy success [25]. The drivers the report credits are "worsening economic and political crises" across Africa and the turmoil in Libya itself, which left migrants with "little choice but to attempt the sea crossing, with or without proactive SAR" [25].

The tactics that made the crossing lethal were manufactured onshore, in two phases. After the fall of the Qaddafi regime in 2011, a previously stable smuggling business fragmented: new entrants offered lower prices without "the willingness or knowhow to organise safe crossings," competition brought subpar navigation and heavier loading, and the resulting "low cost" model's casualties — the October 2013 shipwrecks — are what prompted Italy to launch Mare Nostrum [25]. Then, from late 2014 and consolidated by late 2015, came what Mark Micallef calls "resource predation": militias first taxed smuggling, then ran it, treating migrants as "simply another commodity to be exploited in the broader resource predation carried out by armed groups" [25]. Frontex's intelligence reporting concedes the fusion — on the Libyan coast "the militia's 'commanding officer' in the region is the head of the network" — and journalist Nancy Porsia's reporting describes migrants "taken hostage by militias which sell them to smugglers from one leg to the other across the journey"; the main hub shifted from Zuwara to Sabratha as smuggler and militia fused, and the Libyan Coast Guard's own interventions grew alongside [25]. The downward spiral in crossing conditions, the report stresses, "has been documented since 2013" — before any NGO flotilla existed [25].

The naval operation then selected for the worst boat. EUNAVFOR MED's own 2015 report recorded, from the operation's start, that "smugglers provide migrant vessels with less fuel, food and water and launch them in more difficult weather conditions" — at a time when NGOs performed only 13% of rescues and operated mostly outside the 24-nautical-mile contiguous zone [25]. (Frontex's AFIC 2015 dissented, blaming rescue-vessel proximity — citing a rubber boat found 22 miles off Libya with 112 people and fuel for only about 12 hours; the attribution was contested from the start [25].) The destruction campaign scaled: 67 boats destroyed in the operation's first six months of 2015 became 269 in the next ten months of 2016 — 225 rubber boats, 40 wooden, 4 speed boats — and military sources put 75–80% of all intercepted vessels destroyed in 2016, excluding the Italian Navy's parallel destructions [25]. The logic was explicit: smugglers "can no longer recover [wooden] smuggling vessels on the High seas, effectively rendering them a less economic option for the smuggling business" [25]. It worked — and the market's answer was the disposable dinghy. By February 2016 wooden boats were about 3% of intercepted migrant vessels, down from around 25% in 2015, and rubber boats came to figure in 70% of SAR events; smaller boats plus more crossers multiplied SAR events from 906 to 1,424 [25]. The system's later behaviour confirms the hydraulics: Frontex's Risk Analysis for 2018 records the EU–Turkey statement suppressing Aegean departures while pressure diverted — a temporary Black Sea route (537 detected, August–November 2017, the most since 2009), a Western Mediterranean flow that "more than doubling the previous record" in 2017 with smuggler-supplied rubber dinghies appearing east of Gibraltar, and a Western African route down to 421 detections under a Spain–Senegal–Mauritania surveillance arrangement [27]. Squeeze one corridor and the flow, the boats, and the dying move — which is precisely why attributing any of it to the rescuers requires the evidence discipline the next section examines.

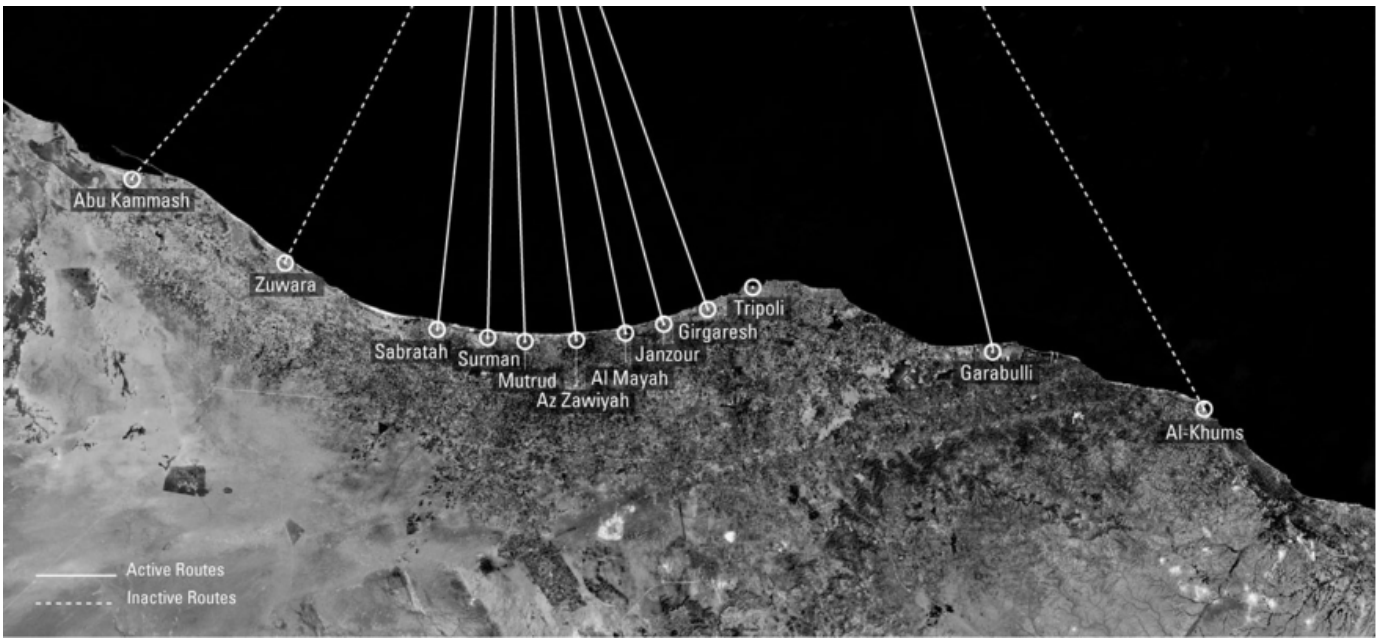


Exhibit. Map of main smuggling routes along the Libyan west coast. Source: S0023.

Deterrence, danger, and the evidence problem at sea

Frontex describes its own Risk Analysis reports as produced in the "pursuit of factual exactness, truth and exhaustive analysis" — a striking self-image for the document at the centre of the rescuer controversy [25]. The 2017 Annual Risk Analysis's core evidentiary exhibit is a genuine pattern: through 2015 and early 2016, smugglers instructed migrants to call MRCC Rome by satellite phone, "more than half of all rescue operations were initiated in this manner," and NGOs figured in under 5% of incidents; from June to October 2016 "the pattern was reversed" — satellite-phone-initiated rescues fell to 10% and NGO operations rose above 40% [25]. From this correlation the report generalizes: Libyan smugglers "heavily relied on the International Convention for the Safety of Life at Sea (SOLAS), and associated SAR as well as humanitarian assistance efforts, turning it into a distinct tactical advantage," and "all parties involved in SAR operations in the Central Mediterranean unintentionally help criminals achieve their objectives at minimum cost, strengthen their business model by increasing the chances of success" [25]. Forensic Oceanography names the method: isolate one actor "from the web of interactions with other actors which together shape the dynamics of migration across the sea," align parallel trends, and let proximity do the causal work — the same argument, they note, that Frontex ran against Mare Nostrum in 2014, when its executive director told the European Parliament that smugglers had "abused of the proximity of the operation to the Libyan coast… they are putting less fuel, less food, less water on the vessel," an argument that "contributed to delegitimizing and ultimately terminating the operation" [25].

The aggregation trap is the cleanest illustration of the evidence problem. On annual totals, 2016 presents "the paradoxical impression… that the increase in both the mortality rate and the share of SAR operations carried out by NGOs were simultaneous" — the attackers' Exhibit A [25]. Disaggregate by month and the picture inverts: mortality spiked in April–May 2016, when the NGO flotilla was still redeploying from its winter break — one vessel (SOS Méditerranée) present most of April, five in May, accounting for just 4% and 12% of rescues — and "the mortality rate was brought down over the summer months, just as the NGO flotilla reached its peak deployment" of eleven vessels and a 35% rescue share in June [25]. The statistical statement of §41 (−0.314 Pearson, not significant; −0.532 for 2015, borderline) is what the data will bear, and the report prints the caveat rather than the headline [25]. November supplies the counter-case: NGO assets fell to seven yet performed their highest-ever share of rescues, 47%, because EUNAVFOR MED and Frontex shares decreased — while record October–November crossings and high winds drove mortality up anyway [25] [25].

Frontex's handling of that November surge is the report's specimen of evidence-free attribution: its Triton update asserted smugglers encouraged rough-weather sailing "because they will be rescued soon after their departure," a claim made, the authors note, "without providing any evidence of this causal connection" [25]. The rival explanations were on the record: NGO proximity had been constant for months (why would it detonate in October?); West and Central African migrants — a rising share — cross through winter; and Libya was convulsed by fighting for the port of Zawiya and in Tripoli, while the coastal road from Tripoli toward Zawiya, closed since October 2015, reopened for

September to mid-November, easing the movement of migrants to the coast [25]. The criminal-collusion variant fared no better under scrutiny: prosecutor Zuccaro's "proven facts" — phone calls from Libya, floodlights, transponders cut — dissolve into standard practice (distress calls to rescuers are the system's design; SAR vessels light themselves at night to be found) and into the known poverty of AIS coverage off Libya, where transmission gaps are routine; "the claims of criminal activity have been increasingly retracted by several prosecutors as they failed to find conclusive evidence" [25]. What the evidence does support is adaptation, conceded in §41: EUNAVFOR's own report ties shorter rescue distances (75 → 35 → 20 nautical miles) and smugglers' declining satellite-phone provision to their tracking of NGO positions broadcast via AIS — consolidation of a deterioration made elsewhere, not its cause [25].

Deterrence, finally, has its own ledger, and it does not balance. The Mare Nostrum precedent stands: withdraw proximate rescue and crossings continue while deaths multiply (§§41–42). When Central Mediterranean detections did fall, Frontex's Risk Analysis for 2018 dates the break precisely — "a sudden reversal took place in July 2017" — in the era of Libyan-coastguard interdiction, not NGO withdrawal, while aggregate detections hit their lowest since 2013 even as Frontex conceded that migratory "pressure on the external borders remained high" and was diverting to the Western Mediterranean [27]. The same document quietly concedes the denominator problem: detections ran below asylum applications, implying entrants who arrived undetected, and "it is difficult to quantify these phenomena" [27]. That is the evidence problem at sea in one line: the events that matter most — the unrescued boat, the undetected landing, the death unwitnessed — are exactly the ones the instruments miss, and every actor's preferred statistic is a shadow cast by its own sensors. This paper has met that problem in every register: drift models that cannot nowcast a hull, sighting logs that map human passage rather than debris, AIS archives whose gaps are the evidence. The Mediterranean controversy is what happens when those shadows are mistaken, by an agency claiming factual exactness, for the sea itself. The final sections return to the sea's own uninstrumented violence: the waves that arrive unannounced, and the storms that scatter everything.

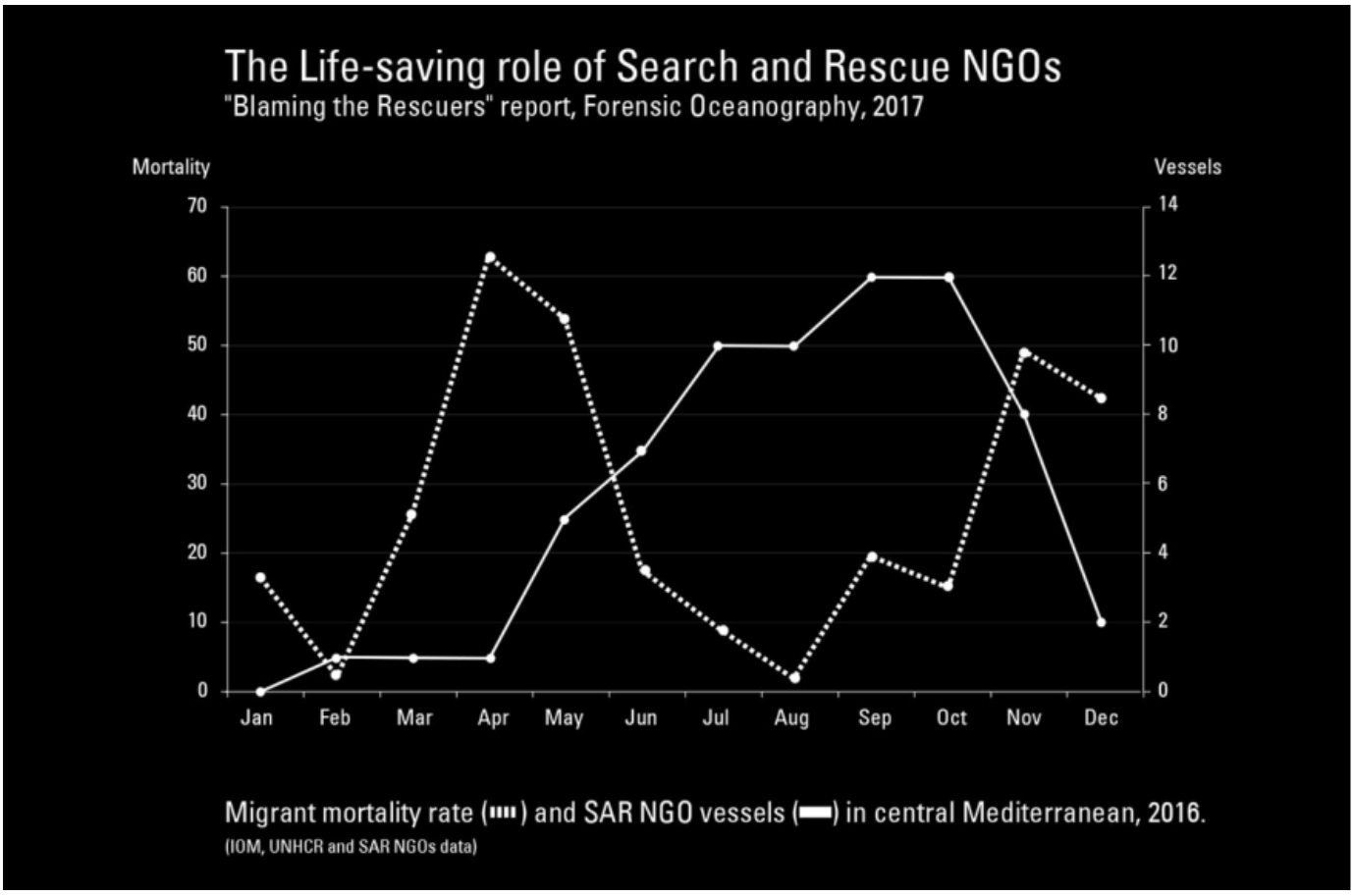


Exhibit. Migrant mortality rate (dotted) and SAR NGO vessels on station (solid) in the central Mediterranean, 2016 (Forensic Oceanography; IOM and UNHCR data) [S0023]. Source: S0023.

When the Sea Itself Is the Hazard

Rogue waves measured and simulated: the Andrea wave

Every ghost ship in this paper implies a sea state nobody recorded. Rogue waves are the extreme case of that evidence problem: for most of maritime history they existed only as survivor testimony — the mariner's version of reportage without paperwork — because the instruments almost never happened to be watching. Field data "are usually recorded in 17–30 min every 3rd hour and therefore are affected by sampling variability," with conditions between measurements simply "assumed to be stationary due to lack of information"; worse, the higher-order statistics that betray rogue-wave risk, skewness and kurtosis, are so unstable that "at least 250–350 repetitions of a 17 min wave record will often be required" for reliable estimates [28]. The Andrea wave is precious because it beat those odds. During the Andrea storm, which crossed the central North Sea on 8–9 November 2007, a rogue wave was recorded at the Ekofisk field (Magnusson and Donelan, 2013) — "comparable in characteristics, both with respect to the wave height and wave crest criterion, to the well-known New Year wave (called also the Draupner wave) measured by Statoil at the Draupner platform the 1 January 1995" [28]. Two instrumented monsters, a decade apart: enough to interrogate.

The interrogation couples two model families whose complementarity is the study's methodological point. A spectral model like WAM "provides sea state description only in a form of the two-dimensional wave spectrum but does not give any information about the instantaneous position of the sea surface"; phase-resolving nonlinear models — here the Higher Order Spectral Method solving the Euler equations — "give us the water surface elevation out of which statistical properties of individual waves can be extracted," and can be re-run to manufacture the repetitions the ocean never provides [28]. The hindcast backbone is ECMWF archive data at a grid point about 50 km from Ekofisk, in 74 m of water, sampled every 6 hours: significant wave height peaked at 9.8 m at 06:00 UTC on 9 November 2007, with the storm's highest steepness, 0.14 — but the Andrea wave itself was recorded at 00:40 that night, between samples, so "because the hindcast data are sampled every 6 h it is not possible to detect the steepness at [the moment] when the Andrea wave is recorded" [28]. The event sits in the instrument's blind interval, exactly as §43's unrescued boats sit in theirs.

Mechanism attribution proceeds by elimination from the recognised catalogue: "linear Fourier superposition (frequency or angular linear focussing); wave–current interactions; crossing seas; quasi-resonance nonlinear interactions (modulational instability); shallow water effects" [28]. Linear focusing is set aside as rare; no strong current was reported at Ekofisk, an intermediate-depth site, eliminating wave–current interaction and shallow-water effects [28]. Crossing seas — the crowd's favourite — fails its own test: it requires two wave trains of similar energy and frequency crossing at an angle β where kurtosis peaks ($40 < \beta < 60$), but the hindcast shows a storm dominated throughout by wind sea, its significant height reaching about 10 m while swell stayed "only slightly above 1 m" for most of the storm, the two comparable only at the storm's start and end; "none such conditions... have been identified in the ECMWF hindcast data" [28]. What remains is modulational instability: the directional spectrum during the critical window (18:00 UTC 8 November to 06:00 UTC 9 November) shows one wave system, and "the Andrea wave might have occurred in the sea state more prone to extreme waves as a result of modulational instability (i.e. the sea state with relative high steepness and not particularly broad spectrum)" [28].

The simulations then test whether such a sea state can actually throw an Andrea-class crest. The hindcast spectra seed HOSM ensembles — the initial surface evolved for 70 peak periods, 150 repetitions with randomized amplitudes and phases [28]. The ensemble's maximum crest ratio reached $C_{max}/H_s = 1.4$ at kurtosis 3.35, "comparable to the one characterising the Andrea wave $C_{max}/H_s = 1.63$ recorded at the Ekofisk field... however, not as high" [28]. The shortfall is reported without cosmetic, alongside its likely partial explanation: the study location sits ~50 km from Ekofisk and the simulation window does not cover 00:40 UTC, so the modelled maxima "could reach their maximum prior to the time of the occurrence of the Andrea wave at Ekofisk" [28]. Two further findings widen the lens: rogue-capable sea states "occur several times during the Andrea storm history" at the study location, and North Sea scatter diagrams suggest similar-steepness states are not particularly rare — though scatter diagrams omit the directional spectrum that decides the mechanism [28]. The conclusions are correspondingly careful: the coupled WAM+HOSM approach yields "sea surface elevation, the maximum wave crest, skewness and kurtosis when rogue waves are present" from real hindcast spectra and agrees satisfactorily with the Ekofisk record, but HOSM "cannot be applied for prediction of rogue waves for very steep sea states because it does not include wave breaking," the 6-hour sampling may miss decisive information, model uncertainties were not investigated — and, most strikingly, the storm's rogue waves were deduced during its build-up and decay at several locations, "not... when the storm

reaches the largest Hs" [28]. The biggest sea is not the most dangerous one; the danger lives in the steep, narrow-spectrum shoulders of the storm. That asymmetry — peak violence decoupled from peak size — is also how storms treat everything floating in them, which is the paper's final subject: what high-windage debris does when the weather that made the Andrea wave arrives.

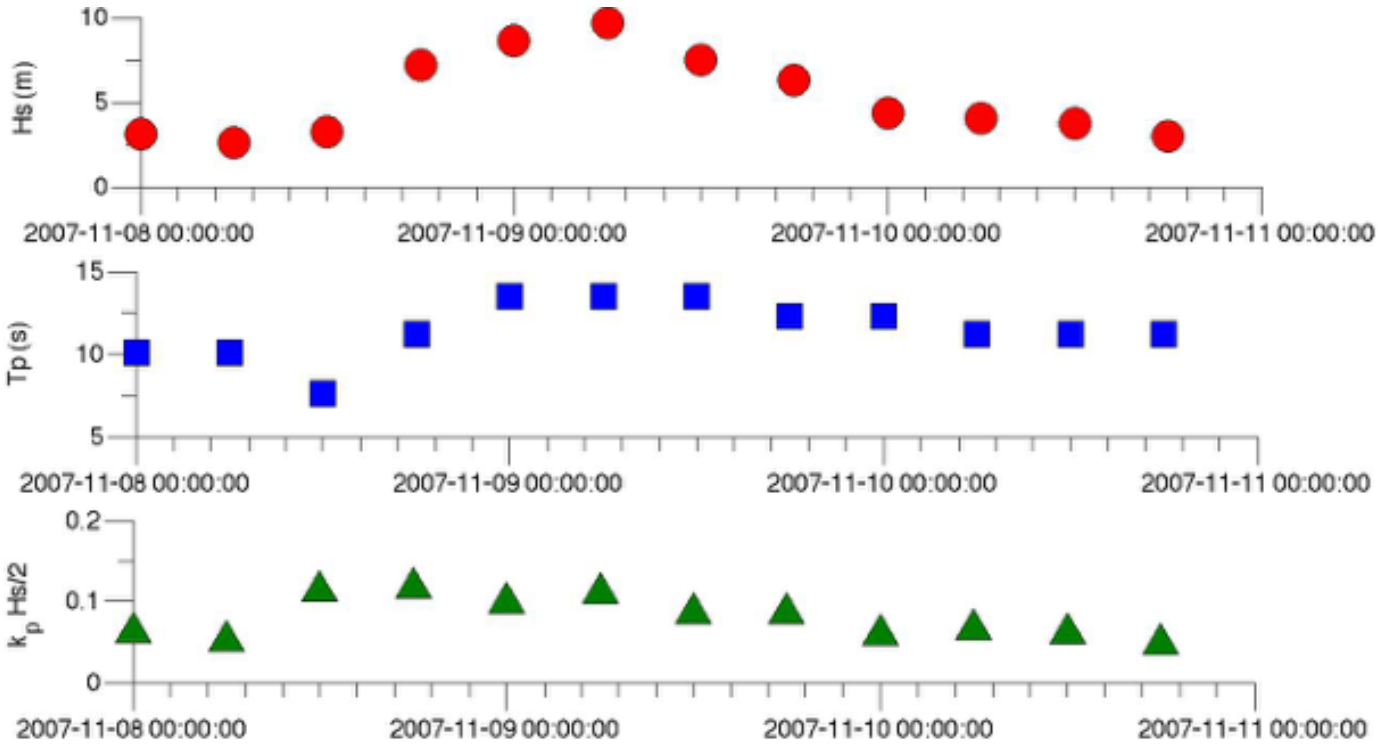


Exhibit. Fig. 2. History of significant wave height, spectral wave period and sea state steepness for the total sea during the Andrea storm. Source: S0025.

High-windage debris and storm-driven dispersal

Everything in this paper has been sorted, all along, by a single physical variable: how much of a floating object stands above the waterline. Windage is governed by a simple drag balance — an object with equal surface area above and below the waterline (A/B = 1) "will drift relative to the water at a rate of 2.5% of the 10-metre wind speed," in the wind's direction, which "may be aligned somewhat differently than the direction of the surface current"; in general, "objects with significant windage will be transported across the ocean more rapidly than zero-windage objects" [2]. Total velocity decomposes as U = U_water + U_wind, and the second term is destiny [2]. The Lagrangian analysis behind the global mass budget quantifies the fork: of 2,510,918 modelled particles, with no wind influence 96% had spent at least two consecutive days near the shoreline within a year — but with a 2% windage coefficient the figure rises to 98%, while the probability of transport to offshore waters falls; "probability of stranding increases with windage coefficient," converging in the fitted budget at a stranding probability of 97% against offshore transport of 33% [12]. High-windage debris is therefore almost certain to meet a coast, quickly; what lingers offshore is the low-windage residue — "in good agreement with observations reporting mainly low-windage debris accumulating in oceanic gyres" [12].

The Great Pacific Garbage Patch is that residue, and its inventory testifies to the sorting. The objects collected there "exhibited no or very little air draft when placed in seawater," many aerially sighted objects "seemed fully submerged," and "ghostnets, which were the main contributors to the total mass of GPGP plastic, generally have a draft of several metres, and therefore are unlikely to be influenced by wind transport"; the null-windage model scenario best reproduced the measured concentration gradients [10]. Modelled higher-windage particles behave oppositely — "transported over larger areas, with a higher likelihood of beaching, as well as exiting oceanic 'garbage patches'" — with 1%-windage North Pacific particles spread from the patch to "the Hawaiian archipelago and the North American coastline" [10]. The material evidence aligns: foam, the highest-buoyancy fraction, is negligible inside the GPGP yet abundant in Alaska beach clean-ups, and the early shoreline sightings of Tohoku tsunami debris were precisely its high-windage vanguard — "floating docks, boats and large buoys" [10]. The corollary is a permanent measurement bias: patch surveys "underestimated amounts of higher windage debris that may be passing through the GPGP," be-

cause on monthly averages its concentration is minimal even though its throughput over time "may be more substantial" [10]. The patch is what storms and winds leave behind, not what the ocean carries.

Delivery to shore is itself a storm-season transaction. Ocean currents "can only bring debris close to the shore. Debris is then deposited on the shoreline by the action of retreating tides, onshore winds, and waves" [2]. On the British Columbia coast the calendar is explicit: summer's northerly winds drive upwelling and offshore transport that "help to protect the coast," while winter's southerlies "create downwelling conditions and onshore transport of surface waters which favours deposition of debris along the coast" [2]. Meanwhile turbulence smears every compact release into weather of its own: with eddy diffusivity of 5,000–10,000 m^2/s in the Kuroshio Extension, a debris cluster disperses over roughly 400 km — some 500,000 km^2 — within six months, not uniformly but in "patchy clumps and elongated filaments" [2]. A storm like Andrea's does all of this at once: it is at once the hazard of §44, the engine of the crossing speeds of §2 (high-windage debris making the Pacific transit in 9–12 months at 4–5% windage, the Ryou-Un Maru at ~3.5%), and the sorting machine that decides what beaches, what drifts on, and what joins the gyre.

That is the thread on which the whole paper hangs. The Baychimo and the Ryou-Un Maru were high-windage hulls: fast, far-travelling, forever fetching up against coasts to be sighted, boarded, and storied. The Friendly Floatees rode lower and slower, numerous enough to become instruments. The drifter buoys were engineered to eliminate windage altogether, drogued at 15 metres, and so revealed the gyres that trap their un-drogued kin for centuries. The microplastic smog rides lowest of all, beneath notice until the nets came. Even the paper's human fleets obey the geometry: a shadow tanker is a hull that maximizes its presence above the legal waterline while minimizing its draft in the registries, and a migrant dinghy is the opposite — all exposure, no freeboard, its fate decided by the same wind that decides a dock's. Ghost ships, in the end, are simply what the wind has not yet delivered to a shore — and the sea keeps its own ledger of every arrival, written in the only archive that never closes: the beach.

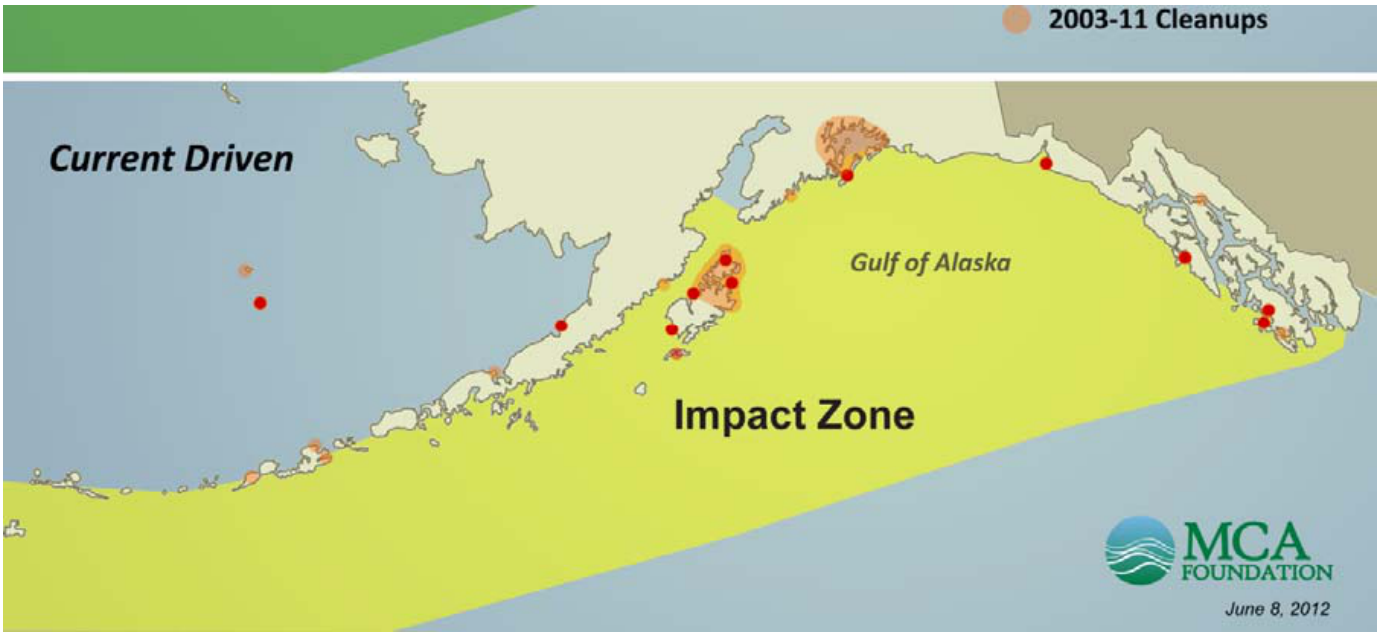


Exhibit. Figure 3. Expected distribution of high windage and current-driven debris. Source: S0013.

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